

Bulk universality of random regular graphs

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Abstract

We consider the adjacency matrix of a uniformly random simple d -regular graph on N vertices. For every fixed $d \geq 3$ and every fixed bulk energy, we prove that the rescaled eigenvalue point process converges to the Sine_1 process with intensity $1/\pi$. We also establish universality of consecutive gaps at deterministic bulk labels and of fixed-energy correlation measures. The eigenvector input consists of polynomial estimates for mixed fourth moments on a uniformly sampled star. These estimates prevent loss of variance in the qualitative joint Gaussian limits of Backhausz–Szegedy, yielding independent variance-one Gaussian waves for fixed tuples at nearby energies. Exact switching identities then give a stationary marked flow and the microscopic loop hierarchy, which identifies the point-process limit. The appendices prove the stronger correlation and gap statements using counting-moment bounds and conditional log-gas laws for tagged limits.

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1 Introduction

We study a uniformly random simple, undirected d -regular graph $\mathcal{G} = \mathcal{G}_{N,d}$ on the labelled vertex set $\llbracket N \rrbracket = \{1, \dots, N\}$, where $d \geq 3$ is fixed. Throughout, $N \rightarrow \infty$ along integers for which dN is even and the graph exists. Its adjacency matrix A is real symmetric and has zero diagonal. We order its eigenvalues, including multiplicities, as

$$d = \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_N.$$

We choose an orthonormal eigenbasis with $\mathbf{u}_1 = N^{-1/2} \mathbf{1}$, where $\mathbf{1}$ is the all-ones vector. This vector has eigenvalue d ; every other basis vector is orthogonal to it, including when the graph is disconnected. All bulk intervals considered below are compact subsets of $(-2\sqrt{d-1}, 2\sqrt{d-1})$, and hence exclude this eigenvalue.

One of the most prominent questions for a random matrix model is its local eigenvalue statistics. For Wigner matrices, the universality of bulk eigenvalue gaps and correlation functions has been established under general assumptions [8, 16, 17]. The results were later extended to sparse matrices in [14, 15, 21], including adjacency matrices of Erdős-Rényi graphs in regimes with polynomially growing average degree. Very recently, Henheik and Knowles [20] proved convergence of the bulk point process at a fixed energy for $(\log N)^{2+\kappa} \leq Np \leq N^{1-\kappa}$, for every fixed $\kappa > 0$.

The same question becomes much more difficult for random regular graphs, where the adjacency matrix is highly structured and its entries are no longer independent. A local resampling technique was developed in [5], and it was proved that when $N^\varepsilon \leq d \leq N^{2/3-\varepsilon}$ for a fixed $\varepsilon > 0$, the eigenvalue density of $A/\sqrt{d-1}$ matches with the semicircle law on all mesoscopic scales. Together with a comparison between switching dynamics and Dyson Brownian motion, [2] proved the bulk universality in the same degree regime. More recently, Bourgade and Huang [9] proved convergence of the bulk point process at a fixed energy for $(\log N)^{24} \ll d \leq N^{1/2}$ using microscopic loop equations.

When d is fixed, the empirical eigenvalue distribution converges to the Kesten-McKay law [26, 27]. Local laws and eigenvector delocalization were proved for sufficiently large fixed degree by Bauerschmidt, Huang and Yau [4], and for every fixed $d \geq 3$ by Huang and Yau [23]. Numerical simulations have long suggested that the rescaled local eigenvalue statistics in the bulk agree with those of the Gaussian Orthogonal Ensemble (GOE) [25, 28].

In this paper, we prove bulk universality for every fixed $d \geq 3$. We first state convergence of the local point process, and then the stronger gap and correlation conclusions. Write

$$\rho_d(x) = \frac{d\sqrt{4(d-1)-x^2}}{2\pi(d^2-x^2)} \mathbf{1}_{|x| \leq 2\sqrt{d-1}}, \quad \rho_{\text{sc}}(x) = \frac{\sqrt{4-x^2}}{2\pi} \mathbf{1}_{|x| \leq 2}$$

for the Kesten-McKay and semicircle densities, respectively. Our GOE convention is the real symmetric matrix whose upper-triangular entries are independent, with off-diagonal entries $N(0, N^{-1})$ and diagonal entries $N(0, 2N^{-1})$. Its limiting density is ρ_{sc} ; write $\mu_1 \geq \dots \geq \mu_N$ for its ordered eigenvalues.

We use the normalization of Bourgade and Huang [9, Section 1.1]: Θ_1 denotes the translation-invariant Sine₁ point process with intensity $1/\pi$, so

$$\mathbb{E}\Theta_1(I) = \frac{|I|}{\pi} \quad \text{for every bounded interval } I \subset \mathbb{R}.$$

Thus the mean microscopic spacing is π . All eigenvalue and gap scalings below use this convention. Here δ_x denotes a unit point mass at x , and point measures count eigenvalues with their multiplicities.

Main theorem (Fixed-energy point-process convergence). *Fix $d \geq 3$ and a deterministic energy $E \in (-2\sqrt{d-1}, 2\sqrt{d-1})$. Then, as $N \rightarrow \infty$,*

$$\sum_{j=1}^N \delta_{N\pi\rho_d(E)(\lambda_j - E)} \implies \Theta_1$$

in distribution for the vague topology on locally finite counting measures on $\mathbb{R}$. The randomness on the left is only that of the uniformly random simple d -regular graph.

The vague topology tests a measure against continuous compactly supported functions. Equivalently, the conclusion can be written in terms of Laplace functionals: for every nonnegative $f \in C_c(\mathbb{R})$,

$$\lim_{N \rightarrow \infty} \mathbb{E} \exp \left\{ - \sum_{j=1}^N f(N\pi\rho_d(E)(\lambda_j - E)) \right\} = \mathbb{E} \exp \left\{ - \int_{\mathbb{R}} f(x) d\Theta_1(x) \right\}.$$

The expectations on the two sides are under the graph law and the Sine₁ law, respectively. The distinguished eigenvalue $\lambda_1 = d$ escapes every bounded microscopic interval, so omitting it gives the same limit.

The main text proves this theorem in Section 3.4. Convergence in law does not by itself justify expectations of unbounded local counting observables, and a fixed-energy window does not keep track of a deterministic eigenvalue label. The following stronger theorem treats both issues. Its correlation statement is proved in Appendix A; its gap statement is proved in Appendix B, using the comparison established in Appendix C.

For the gap normalization, define the upper-tail quantiles by

$$\int_{\gamma_i}^{2\sqrt{d-1}} \rho_d(x) dx = \frac{i}{N}, \quad \int_{\gamma_i^{\text{sc}}}^2 \rho_{\text{sc}}(x) dx = \frac{i}{N}, \quad 1 \leq i \leq N.$$

These quantiles follow the decreasing eigenvalue order. We use this convention throughout the main text and the graph gap argument; Appendix C explicitly reverses the order for its auxiliary GOE comparison.

Theorem 1.1 (Gap and fixed-energy correlation universality). *Fix $d \geq 3$ and $k \geq 1$.*

(i) *For every $0 < \delta < 1/2$ and $F \in C_b(\mathbb{R}^k)$,*

$$\lim_{N \rightarrow \infty} \sup_{\delta N \leq i \leq (1-\delta)N} \left| \mathbb{E} F(N\pi\rho_d(\gamma_i)(\lambda_i - \lambda_{i+1}, \dots, \lambda_{i+k-1} - \lambda_{i+k})) \right. \\ \left. - \mathbb{E} F(N\pi\rho_{\text{sc}}(\gamma_i^{\text{sc}})(\mu_i - \mu_{i+1}, \dots, \mu_{i+k-1} - \mu_{i+k})) \right| = 0.$$

The supremum is over integer indices; all displayed labels exist for sufficiently large N .

(ii) *Fix $\tau > 0$, a deterministic energy $E \in [-2\sqrt{d-1} + \tau, 2\sqrt{d-1} - \tau]$, and $\varphi \in C_c(\mathbb{R}^k)$. Put $\tilde{E} = E/\sqrt{d-1}$. Then*

$$\lim_{N \rightarrow \infty} \left(\mathbb{E} \sum_{i_1, \dots, i_k}^{\neq} \varphi(N\pi\rho_d(E)(\lambda_{i_1} - E), \dots, N\pi\rho_d(E)(\lambda_{i_k} - E)) \right. \\ \left. - \mathbb{E} \sum_{i_1, \dots, i_k}^{\neq} \varphi(N\pi\rho_{\text{sc}}(\tilde{E})(\mu_{i_1} - \tilde{E}), \dots, N\pi\rho_{\text{sc}}(\tilde{E})(\mu_{i_k} - \tilde{E})) \right) = 0. \quad (1.1)$$

The sums run over ordered tuples of distinct indices in $\{1, \dots, N\}$. In both parts, the graph and GOE expectations are separate; no coupling of the two ensembles is required.

Part (i) concerns a deterministic eigenvalue label and the consecutive gaps following that label. Part (ii) instead observes the spectrum near a deterministic energy, without averaging that energy. The first window moves with the random spectrum, whereas the second does not; in particular, the two statements require different additional arguments beyond the main point-process theorem. The sums in (ii) are factorial-moment observables. We assert convergence of correlation measures tested against $C_c(\mathbb{R}^k)$, not pointwise convergence of finite- N correlation densities: the graph model is discrete, and such densities are not assumed to exist.

The common input to all three eigenvalue conclusions is the analysis of bulk eigenvectors. It allows us to pass to the microscopic limit in exact switching identities. In the main text these identities yield loop equations for the fixed-energy process; in the gap appendix they also yield conditional log-gas laws for a process viewed from a fixed label.

A Gaussian wave on the infinite d -regular tree is a centered Gaussian process, invariant under tree automorphisms, which satisfies the eigenvalue equation at every vertex [13]. We use the normalization in which every coordinate has variance one, and call its restriction to a root and its d neighbors the Gaussian star law. Gaussian waves are the tree analogue of Berry's random wave model [6].

For an orthonormal selection $\mathbf{u}_1, \dots, \mathbf{u}_k$ with eigenvalues $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_k)$, let $\nu_{\mathbf{X}}$ be the law, conditional on the graph, of $(\sqrt{N}\mathbf{u}_a(v_j))_{a,j}$ at a uniform root v_0 with uniformly ordered neighbors $v_1, \dots, v_d$. In this empirical law the graph and the selected eigenvectors are fixed; the only randomness is the choice of root and neighbor ordering. Let $\nu_{\mathbf{Z}_{\boldsymbol{\lambda}}}$ be the product of the centered variance-one Gaussian star laws at these eigenvalues. Here the subscripts enumerate the selected eigenvectors. For a Gaussian wave with eigenvalue λ , the covariance between the root and a neighbor is λ/d , and the covariance between distinct neighbors is $(\lambda^2 - d)/(d(d-1))$. The following theorem is the quantitative eigenvector input used throughout the paper.

Theorem 1.2 (Polynomial mixed fourth moments). *Fix a compact bulk interval K . There are $b, \chi, C > 0$ and graph events of probability at least $1 - CN^{-b}$ such that, on each fixed graph in these events, simultaneously for $1 \leq k \leq 4$, every orthonormal selection of k real eigenvectors with eigenvalues in K , and every coordinate choice $(a_\ell, j_\ell) \in \{1, \dots, k\} \times \{0, \dots, d\}$, $1 \leq \ell \leq 4$,*

$$\left| \int \prod_{\ell=1}^4 x_{a_\ell, j_\ell} d\nu_{\mathbf{X}} - \int \prod_{\ell=1}^4 x_{a_\ell, j_\ell} d\nu_{\mathbf{Z}_{\boldsymbol{\lambda}}} \right| \leq CN^{-\chi}. \quad (1.2)$$

Repetitions are allowed in the coordinate choices. The selected eigenvalues may lie anywhere in K , and the constants depend only on d and K .

Corollary 1.3 (Qualitative joint Gaussian stars). *Fix a compact bulk interval K and $k \geq 1$. For every deterministic sequence $\delta_N \downarrow 0$,*

$$\sup_{\substack{\text{orthonormal selections } \mathbf{u}_1, \dots, \mathbf{u}_k \\ \lambda_a \in K, \max_{a,b} |\lambda_a - \lambda_b| \leq \delta_N}} W_2(\nu_{\mathbf{X}}, \nu_{\mathbf{Z}_{\boldsymbol{\lambda}}}) \xrightarrow{\mathbb{P}} 0. \quad (1.3)$$

The convergence in probability is with respect to the random graph; the measures inside W_2 are its conditional empirical sampling laws and their Gaussian comparison laws. The supremum over an empty set is zero. The cost is Euclidean in the $k(d+1)$ physical star coordinates. In particular, every fixed tuple selected measurably from the graph whose eigenvalues converge in probability to a common bulk energy has independent variance-one Gaussian star limits, conditionally on the graph. No convergence rate is asserted. For $k = 1$, the restriction on the eigenvalue diameter is vacuous.

The uniformity in the preceding statements is quenched: one first chooses a graph in a common good event, and then may choose any of the allowed eigenvectors on that graph. It is stronger than a separate high-probability statement for each predetermined eigenvector. Within a multiple eigenspace, any real orthonormal choice is allowed. When random choices of eigenvectors are needed, we fix a measurable orthonormal eigenbasis, for example by applying Gram–Schmidt in a deterministic order to spectral projections of the coordinate vectors. The results do not depend on this convention, and we do not insert independent random signs in the empirical star laws.

We use the convention $W_2(\mu, \nu)^2 = \inf \mathbb{E}|\mathbf{X} - \mathbf{Y}|^2$, where the infimum is over couplings. The two statements have distinct roles in the proof. The polynomial mixed fourth moments control covariances of quadratic eigenvector observables, including sums over eigenvalues throughout K . Qualitative convergence identifies the joint law of each fixed collection of eigenvector marks whose eigenvalues approach the same energy.

Previously, for fixed $d \geq 3$, [1] proved that almost eigenvectors of random regular graphs converge locally to Gaussian waves with variance in $[0, 1]$. The paper also obtained joint Gaussian limits for fixed tuples at nearby energies. For edge eigenvectors, [19] proved convergence to independent Gaussian waves of variance one, jointly with and independently of the edge eigenvalues. The result holds on every fixed-radius ball, with independent random signs attached to the eigenvectors. For growing degrees $N^\varepsilon \leq d \leq N^{2/3-\varepsilon}$ with a fixed $\varepsilon > 0$, asymptotic normality of bulk eigenvectors was established by Bourgade, Huang and Yau [10].

This paper considers bulk eigenvalue statistics. Near the edge of the spectrum, Bauerschmidt, Huang, Knowles and Yau [3] proved universality for $N^{2/9+o(1)} \leq d \leq N^{1/3-o(1)}$. Huang and Yau [24] extended this range to $N^{o(1)} \leq d \leq N^{1/3-o(1)}$, while He [18] established edge universality for $N^{2/3+o(1)} \leq d \leq N/2$. In [22], Huang, McKenzie and Yau proved optimal eigenvalue rigidity up to subpolynomial factors and edge universality for every fixed $d \geq 3$.

1.1. Proof ideas

The point-process proof and the appendix upgrades use the polynomial mixed fourth moments in Theorem 1.2 and the qualitative Gaussian limit in Corollary 1.3. To obtain the moment estimates, we develop a quantitative fourth-moment refinement of the Gaussian maximality argument of Backhausz–Szegedy [1, Theorem 11.2]; see Proposition 2.8. The heat-flow comparison follows [1, Proposition 11.1], while fourth-chaos coercivity and a high-noise entropy expansion yield the explicit moment bound. For an eigenvector with eigenvalue λ , we add correlated Gaussian noise with covariance proportional to $\text{Im}(A - \lambda - i\eta)^{-1}$, where $\eta = N^{-c}$ for a small fixed $c > 0$. The local laws of Huang and Yau [23] control both the local covariance and the total entropy of this perturbation. The resulting local and global entropy contributions cancel up to a polynomial error. We need this deficit estimate only at one fixed noise level, as stated in Proposition 2.13. Ornstein–Uhlenbeck monotonicity transfers the bound to the high-noise regime, where the abstract fourth-moment estimate applies.

For the full law, we use the qualitative joint neighborhood conclusion of Backhausz–Szegedy for fixed tuples at nearby energies. The fourth-moment bound makes all quadratic coordinate products uniformly integrable. The exact empirical covariance therefore passes to every Gaussian limit, fixing each variance at one and making the channels independent. This also gives qualitative convergence in W_2 ; a quantitative full-law stability estimate is unnecessary for the switching argument.

At fixed degree, a simple switching changes adjacency matrix entries by quantities of order one. We therefore use the exact resolvent identity for the corresponding rank-two perturbation. In this identity, products of eigenvector coordinates are summed over the spectrum. The qualitative joint Gaussian limit identifies each fixed collection with eigenvalues approaching the same energy; the polynomial mixed fourth-moment estimate controls the

contribution of the remaining eigenvectors throughout the fixed bulk interval. Together with the switching identities, they give tightness at the microscopic scale.

Fix a bulk energy E , set $\sigma = \pi\rho_d(E)$ and $a_E = \operatorname{Re} m_d(E + i0)$, and, for $z, w \in \mathbb{C}^+$, define

$$m_N(z) = \frac{1}{N} \operatorname{Tr}(A - z)^{-1}, \quad m_d(z) = \int_{\mathbb{R}} \frac{\rho_d(t)}{t - z} dt,$$

$$s_N(w) = \frac{m_N(E + w/(N\sigma)) - a_E}{\sigma}.$$

Write s for a subsequential limit of s_N . The switching identity also involves the normalized 2×2 compression of the resolvent to the two switching directions. Its limit, denoted by S , has a spectral representation with Gaussian vectors attached to the eigenvalues. We call these vectors the eigenvector marks. By the quenched uniformity in Corollary 1.3, the limiting marks are independent standard Gaussian vectors conditionally on the scalar spectral data.

Simple switching is reversible: the original and switched graphs form an exchangeable pair. Thus the limiting law of (s, S) is invariant under the switching involution $\mathcal{T}$. It is also invariant under rotations $\mathcal{R}_\theta$ of the Gaussian marks. Conjugating these rotations by switching gives the flow Φ_θ below, which preserves the same law. Although the switching is discrete, we can differentiate this flow. If $\mathcal{V}$ is the infinitesimal generator and F is a polynomial observable of the scalar and marked resolvents, then

$$\Phi_\theta = \mathcal{T}\mathcal{R}_\theta\mathcal{T}, \quad \mathbb{E}\mathcal{V}F = 0.$$

Averaging the resulting identities over the Gaussian marks gives the microscopic loop equations, starting with

$$\mathbb{E}[s'(w) + s(w)^2 + 1] = 0.$$

The full hierarchy is proved in Proposition 3.9. The characterization of Bourgade and Huang [9, Theorem 1.5] identifies the limiting point process as Θ_1 , in the same coordinates as the main point-process theorem. This proves that theorem and is the endpoint of Section 3. Appendix A then establishes uniform integrability of every fixed local counting moment. It upgrades point-process convergence to the correlation statement in Theorem 1.1(ii).

For Theorem 1.1(i), Appendix B keeps the eigenvalue at a deterministic bulk label as the origin when taking the microscopic limit. With this choice of origin, the switching argument gives loop equations and integration-by-parts identities for the labelled eigenvalues. The loop equations yield rigidity and control the force exerted by distant eigenvalues. To use the integration-by-parts identities for a conditional density, we first integrate the rooted law over translations, freeing the tagged eigenvalue from the origin. The identities then show that, given the exterior eigenvalues, the density in a finite block has the form of a $\beta = 1$ log gas; see Lemma B.7.

Proposition B.8 isolates the final implication: rigidity and these conditional Gibbs laws identify the gaps with those of the Palm process Θ_1^0 associated with Θ_1 , with the same mean spacing π . Its proof, given in Appendix C, completes the distant exterior by Gaussian classical locations and applies the local log-gas comparison of Erdős and Yau [16, Proposition 5.3], which uses Dyson Brownian motion [12]. GOE kernel asymptotics [11] and Campbell's formula identify the reference gap law. The microscopic graph limit is taken first, and the conditional block size is then sent to infinity.

Logical dependencies of the proof. The main-text proof has three stages. First, the entropy tools, the weighted Gaussian-star frame, and fourth-chaos coercivity convert a small entropy deficit into polynomial control of mixed fourth moments. Second, the random-graph local law supplies this deficit uniformly over fixed finite collections of eigenvectors, and the qualitative theorem of Backhausz–Szegedy identifies variance-one joint Gaussian waves at

nearby energies. Third, exact switchings and Gaussian marks yield the microscopic loop hierarchy and identify the fixed-energy point process. None of these stages uses the appendix upgrades.

The two parts of Theorem 1.1 are proved separately after this identification. The correlation argument adds counting-moment bounds to the point-process limit. The gap argument instead adapts the marked construction to a deterministic eigenvalue label; rooted loop equations, rigidity, and coordinate integration by parts give conditional $\beta = 1$ log-gas laws, whose gap statistics are identified by Gaussian completion. The graph limit is taken before the auxiliary spectral and conditional-block cutoffs are removed.

Organization and notation. The main text consists of Sections 1–3. Section 2 proves the polynomial mixed fourth moments and the qualitative joint Gaussian limit at nearby energies. Section 3 constructs the microscopic marked limit and its stationary switching flow, derives the loop hierarchy, and ends in Section 3.4 with the proof of the main point-process theorem. Appendix A proves Theorem 1.1(ii) by controlling local counting moments. Appendix B constructs the tagged limit, proves rigidity and conditional Gibbs laws, and deduces Theorem 1.1(i) from Proposition B.8. Appendix C proves that abstract identification proposition by Gaussian completion and a GOE comparison.

Unless stated otherwise, the large parameter is N , and dependence on N is suppressed. Constants $c, C > 0$ may change from line to line; their allowed dependence is specified in the relevant statement. The notation $O_\alpha(\cdot)$ records additional dependence on α . Bold symbols denote vectors, and $\mathbf{u}(v)$ is the scalar coordinate of $\mathbf{u}$ at vertex v . For a star, $X_{a,j}$ denotes channel a at position j : $j = 0$ is the root and $1 \leq j \leq d$ are its neighbors. An index a on a selected eigenpair labels that selection, rather than the global eigenvalue order.

We write $|\mathbf{v}|$ for Euclidean length, $\|M\|$ for the operator norm of a matrix, M^T for transpose, and M^* for conjugate transpose. Thus $\text{Im } M = (M - M^*)/(2i)$ for a matrix-valued resolvent. For a scalar random variable and $1 \leq p < \infty$, $\|X\|_p = (\mathbb{E}|X|^p)^{1/p}$; the probability law is specified locally. For Gaussian function spaces it is always stated explicitly. We use $[M_1, M_2] = M_1 M_2 - M_2 M_1$ and the total variation convention $\|\mu - \nu\|_{\text{TV}} = \sup_A |\mu(A) - \nu(A)|$. For graph-dependent matrices and transforms, a tilde denotes the switched graph; in particular G and $\tilde{G}$ are the original and switched Green functions.

Probability spaces and order of limits. The probability law in use must be distinguished from any additional sampling on a fixed graph. Unsubscripted $\mathbb{P}$ and $\mathbb{E}$ refer to the law currently under discussion: the graph law in a graph-only statement, the joint law when a proposal or noise is included, and the law of the limiting state after passage to a limit. A superscript, such as $\mathbb{E}^\mu$, specifies the probability law explicitly.

After the graph and its eigenvectors have been fixed, $\mathbb{E}_\star$ averages only a uniform root and a uniform ordering of its neighbors. The notation $\mathbb{E}_\bullet$ instead averages the two independent uniform oriented edges in a switching proposal, before acceptance is imposed. The corresponding conditional variances are $\text{Var}_\star$ and $\text{Var}_\bullet$. For the Gaussian channel, $\mathbb{E}_\mathbf{Y}$ averages only the auxiliary Gaussian vector, with its signal and covariance fixed. An empirical profile already contains the finite averages over roots and neighbor orderings; those samples are not averaged a second time in $\mathbb{E}_\mathbf{Y}$. In a limiting marked state, conditioning on the scalar state leaves only the Gaussian marks random. Independence is used only where it is specified in the construction.

A compact bulk interval K and the degree d are fixed before N tends to infinity. A finite number of selected eigenvectors is also fixed before that limit. In marked resolvent sums, we first retain a fixed microscopic window and a fixed number of marks, then take $N \rightarrow \infty$, and only afterwards remove the cutoffs. In the gap argument, the limiting tagged process is constructed first; the conditional block size is then sent to infinity. Keeping these orders

separate is what allows qualitative Gaussian convergence to be sufficient.

Resolvent conventions and elementary estimates. For a real symmetric matrix A and $z = E + i\eta$ with $\eta > 0$, the spectral theorem gives

$$G(z) = \sum_j \frac{\mathbf{u}_j \mathbf{u}_j^T}{\lambda_j - z}, \quad \text{Im } G(z) = \eta G(z) G(z)^*, \quad \|G(z)\| \leq \eta^{-1}.$$

Here $G(z)^*$ denotes the conjugate transpose. In particular, $\text{Im } G(z)$ is positive definite and

$$\frac{1}{N} \sum_j \frac{1}{|\lambda_j - z|^2} = \frac{\text{Im } m_N(z)}{\eta}.$$

This last equality is the Ward identity in the form used below. Our Stieltjes convention is $(x - z)^{-1}$, so the imaginary part is positive in the upper half-plane. An atom of mass r at x contributes $r/(x - z)$: its *pole weight* is r , whereas its complex-analytic residue at $z = x$ is $-r$.

We use repeatedly the following elementary implication. If $f(w) = a + \int ((x - w)^{-1} - x/(1 + x^2)) d\mu(x)$, with $a \in \mathbb{R}$ and μ a positive measure, then on any compact $B \subset \mathbb{C} \setminus \mathbb{R}$ contained in one half-plane,

$$\sup_{w \in B} |f(w) - f(i)| \leq C_B \text{Im } f(i) \quad \text{in the upper half-plane.}$$

Indeed, $(x - w)^{-1} - (x - i)^{-1} = (w - i)/((x - w)(x - i))$, and the absolute value of this kernel is at most $C_B/(1 + x^2)$. The lower half-plane follows by complex conjugation. Cauchy's integral formula gives the corresponding bounds for derivatives on smaller compact sets. Thus a moment estimate at one height provides local analytic tightness; it does not, by itself, identify the poles.

Finally, our microscopic coordinate is $x = N\sigma(\lambda - E)$, where $\sigma = \pi\rho_d(E)$. This is the coordinate used in both the main point-process theorem and Theorem 1.1, as well as in the resolvent arguments and the appendices. The limiting point process is Θ_1 with intensity $1/\pi$; no further dilation is made when identifying its law.

Acknowledgment

YH is supported by NSFC No.12322121 and the National Key R&D Program of China No.2023YFA1010400. The research of JH is partially supported by the Sloan Research Award.

LLM disclosure

GPT-6 assisted the proof development at the level of a coauthor. The authors are responsible for the entire content of the paper.

2 Quantitative Gaussian wave limit

We collect the general entropy and Gaussian tools and apply them to prove polynomial mixed fourth-moment estimates and qualitative joint Gaussian limits for empirical eigenvector laws. An empirical star is random only through the uniform root and the uniform ordering of its neighbors. Auxiliary Gaussian noise is introduced explicitly when needed; probabilities of good graph events refer to the random graph.

2.1. General entropy and Gaussian tools

We collect statements about fixed probability laws, covariance matrices, and finite-dimensional Gaussian spaces. Each expectation in this subsection is taken under the law specified in the corresponding statement. These tools will later be applied after the graph and the selected eigenvectors have been fixed.

2.1.1. Entropy and Gaussian identities. For a probability vector $\mathbf{p} = (p_j)_j$, write $H(\mathbf{p}) = -\sum_j p_j \log p_j$, with $0 \log 0 = 0$ and natural logarithms. For a discrete random variable X , the abbreviation $H(X)$ means $H(\text{Law}(X))$. A finite-valued random probability vector needs more care: $H(\hat{\mathbf{p}})$ is the entropy of its realized entries and is itself a random variable, whereas $H(\text{Law}(\hat{\mathbf{p}}))$ is the entropy of the random profile as an outcome. We never use the first notation for the second quantity. In particular, $\mathbb{E}H(\hat{\mathbf{p}})$ and $H(\mathbb{E}\hat{\mathbf{p}})$ are generally different.

For a density p on a Euclidean space, write $h(p) = -\int p \log p$. The standard Gaussian law on $\mathbb{R}^m$ is denoted by $\gamma_m = N(0, I_m)$. On a subspace, the reference measure is its induced Euclidean volume. We also write $h(X) = h(\text{Law}(X))$ when that law has a density on the specified space. Conditional entropy is an average of entropies of conditional laws; for example,

$$H(X | Z) = \mathbb{E}[H(\text{Law}(X | Z))], \quad h(X | Z) = \mathbb{E}[h(\text{Law}(X | Z))].$$

In the discrete expression the inner H is the entropy of the conditional probability vector, evaluated at the realized Z . Finally, put $H_2(t) = -t \log t - (1-t) \log(1-t)$. All uses below have finite displayed entropies.

If a finite-valued label J selects a component law P_J with probability q_J , write $\mathbf{q} = (q_j)_j$. The entropy chain rule gives

$$\sum_j q_j h(P_j) \leq h\left(\sum_j q_j P_j\right) \leq \sum_j q_j h(P_j) + H(\mathbf{q}). \quad (2.1)$$

Indeed the difference is the mutual information between the sample and its component label, which lies between zero and $H(\mathbf{q})$. The same bounds hold for discrete entropy.

More explicitly, if X is the mixture sample, then $h(X | J) = \sum_j q_j h(P_j)$ and the chain rule gives

$$h(X) - \sum_j q_j h(P_j) = H(J) - H(J | X).$$

The right side lies between zero and $H(J)$. This calculation also explains why the upper bound is independent of overlap between the components: complete overlap gives zero mutual information, while perfectly distinguishable components give $H(J)$. For probability laws $\mathbf{p}, \mathbf{q}$ on an alphabet of size M , with total variation distance v ,

$$|H(\mathbf{p}) - H(\mathbf{q})| \leq H_2(v) + v \log M. \quad (2.2)$$

This follows by coupling the two samples to disagree with probability v and applying the entropy chain rule to the disagreement indicator.

For completeness, let (X, Y) be such a coupling and $B = \mathbf{1}_{X \neq Y}$. Given Y and $B = 0$, the value of X is known; given Y and $B = 1$, it has at most M possibilities. Therefore $H(X | Y) \leq H_2(v) + v \log M$. The inequality $H(X) \leq H(Y) + H(X | Y)$ gives one direction, and exchanging X, Y gives the other. This proof uses total variation in the convention $\frac{1}{2} \sum_j |p_j - q_j|$.

For $\mathbf{Z} \sim N(\mathbf{m}, Q)$ in $\mathbb{R}^n$, with Q positive definite,

$$h(\mathbf{Z}) = \frac{1}{2} \log \det(2\pi e Q). \quad (2.3)$$

Every density with covariance Q has entropy at most the right-hand side, by nonnegativity of its relative entropy with respect to the Gaussian law with the same mean and covariance.

Indeed, writing g for that Gaussian density, its logarithm is a constant minus a quadratic form. Matching means and covariances implies $\int p \log g = \int g \log g$, and therefore $\int p \log(p/g) = h(g) - h(p) \geq 0$. This also proves the maximum-entropy bound when the competing law is a non-Gaussian mixture.

If a Gaussian vector $(\mathbf{Z}_1, \mathbf{Z}_2)$ has covariance blocks Q_{11} , Q_{12} and Q_{22} , with Q_{11} invertible, then

$$\text{Cov}(\mathbf{Z}_2 | \mathbf{Z}_1) = Q_{22} - Q_{12}^T Q_{11}^{-1} Q_{12}. \quad (2.4)$$

This is obtained by subtracting the linear regression on $\mathbf{Z}_1$; the residual is Gaussian and independent of $\mathbf{Z}_1$.

The residual is $\mathbf{Z}_2 - \mathbb{E}\mathbf{Z}_2 - Q_{12}^T Q_{11}^{-1}(\mathbf{Z}_1 - \mathbb{E}\mathbf{Z}_1)$. Its covariance with $\mathbf{Z}_1$ is zero; for jointly Gaussian vectors, zero covariance implies independence. Computing its covariance gives the Schur complement in (2.4).

We also use the Gaussian Poincaré inequality in the form

$$\text{Var}(f(\mathbf{Z})) \leq \|Q\| \mathbb{E}|\nabla f(\mathbf{Z})|^2, \quad (2.5)$$

for locally Lipschitz f with square-integrable gradient. It follows from the standard Gaussian inequality by the change of variables $\mathbf{Z} = \mathbf{m} + Q^{1/2}\mathbf{G}$; smooth approximation gives the locally Lipschitz version used for the softened quantizer below.

2.1.2. Gaussian mixtures and discretization.

Lemma 2.1 (Entropy stability of Gaussian mixtures). *Let $w_1, \dots, w_r \geq 0$ satisfy $\sum_{j=1}^r w_j = 1$, and let $\mathbf{m}_1, \dots, \mathbf{m}_r \in \mathbb{R}^m$ be fixed. For $0 \leq t \leq 1$, set*

$$p_t = \sum_{j=1}^r p_{t,j}, \quad p_{t,j} = w_j \mathcal{N}(\mathbf{m}_j, K_j(t)),$$

where $\mathcal{N}(\mathbf{m}, K)$ denotes the Gaussian density with mean $\mathbf{m}$ and covariance K . Assume that every $K_j(t)$ is differentiable and

$$K_j(t) \geq \kappa I, \quad \|\dot{K}_j(t)\| \leq \varepsilon.$$

Then

$$|h(p_1) - h(p_0)| \leq \frac{m\varepsilon}{2\kappa}. \quad (2.6)$$

The estimate is independent of the component means, weights, and number of components.

Proof. Components of weight zero may be discarded. All derivatives below are with respect to the sample variable $\mathbf{x}$ unless a t -derivative is shown explicitly. In each integrand K_j and $\dot{K}_j$ are evaluated at the same time t . The integrals in this proof are over $\mathbb{R}^m$ with respect to Euclidean volume.

Differentiate a Gaussian with respect to its covariance. For one component, direct differentiation gives

$$\begin{aligned} \partial_t p_{t,j}(\mathbf{x}) &= \frac{1}{2} p_{t,j}(\mathbf{x}) \left[(\mathbf{x} - \mathbf{m}_j)^T K_j^{-1} \dot{K}_j K_j^{-1} (\mathbf{x} - \mathbf{m}_j) - \text{Tr}(K_j^{-1} \dot{K}_j) \right] \\ &= \frac{1}{2} \text{Tr}(\dot{K}_j \nabla^2 p_{t,j}(\mathbf{x})). \end{aligned}$$

The second identity may also be checked from the Fourier transform of a Gaussian. Since $\int \partial_t p_t = 0$, differentiating the entropy and integrating by parts once gives

$$\frac{d}{dt} h(p_t) = \frac{1}{2} \sum_{j=1}^r \int (\nabla p_{t,j})^T \dot{K}_j \nabla \log p_t \, d\mathbf{x}. \quad (2.7)$$

The boundary terms vanish because all component densities and their first derivatives have Gaussian decay. Equivalently, one may first insert a smooth cutoff and then let its radius tend to infinity.

Compare the component scores with the mixture score. Define

$$\mathbf{s}_{t,j} = \nabla \log p_{t,j} = -K_j^{-1}(\mathbf{x} - \mathbf{m}_j), \quad \mathbf{s}_t = \nabla \log p_t.$$

At every $\mathbf{x}$,

$$\mathbf{s}_t(\mathbf{x}) = \sum_j \frac{p_{t,j}(\mathbf{x})}{p_t(\mathbf{x})} \mathbf{s}_{t,j}(\mathbf{x}).$$

The coefficients $p_{t,j}(\mathbf{x})/p_t(\mathbf{x})$ are the posterior component probabilities given the sample value $\mathbf{x}$; they are nonnegative and sum to one. Pointwise Jensen followed by integration gives

$$\int p_t |\mathbf{s}_t|^2 \leq \sum_j \int p_{t,j} |\mathbf{s}_{t,j}|^2 = \sum_j w_j \operatorname{Tr} K_j^{-1} \leq \frac{m}{\kappa}. \quad (2.8)$$

The middle equality is the elementary Gaussian identity $\mathbb{E}|K^{-1}(\mathbf{Z} - \mathbf{m})|^2 = \operatorname{Tr} K^{-1}$ for $\mathbf{Z} \sim N(\mathbf{m}, K)$.

Bound and integrate the entropy derivative. Using $\nabla p_{t,j} = p_{t,j} \mathbf{s}_{t,j}$, the operator-norm bound on $\dot{K}_j$, and Cauchy–Schwarz jointly in j and $\mathbf{x}$, we obtain

$$\begin{aligned} \left| \frac{d}{dt} h(p_t) \right| &\leq \frac{\varepsilon}{2} \sum_j \int p_{t,j} |\mathbf{s}_{t,j}| |\mathbf{s}_t| \\ &\leq \frac{\varepsilon}{2} \left(\sum_j \int p_{t,j} |\mathbf{s}_{t,j}|^2 \right)^{1/2} \left(\int p_t |\mathbf{s}_t|^2 \right)^{1/2} \\ &\leq \frac{\varepsilon}{2} \sum_j w_j \operatorname{Tr} K_j^{-1} \leq \frac{m\varepsilon}{2\kappa}. \end{aligned}$$

Integrating this uniform estimate over $0 \leq t \leq 1$ proves (2.6). Notice that the means never enter the bound; this is important later because the mixture components are centered at graph-dependent eigenvector coordinates. $\square$

Lemma 2.2 (Discretization of Gaussian mixtures). *Fix $m \geq 1$ and $\vartheta, \kappa > 0$, and put*

$$R_0 = N^\vartheta, \quad \eta_0 = N^{-\vartheta}, \quad a_0 = N^{-4\vartheta}, \quad K_0 = \lfloor R_0/a_0 \rfloor.$$

For $z \in \mathbb{R}$, let $J(z) = \lfloor z/a_0 \rfloor$. For an integer n , set $T(n) = \min\{K_0 - 1, \max\{-K_0, n\}\}$. Both maps act coordinatewise, so $TJ(\mathbf{Z})$ is the clipped random grid label, not the quantizer map itself. Suppose that $\mathbf{Z}$ is a finite Gaussian mixture in $\mathbb{R}^m$, every component mean has coordinate size at most $R_0/2$, and every component covariance K satisfies

$$\kappa \eta_0 I \leq K \leq \eta_0^{-1} I.$$

Then, for all sufficiently large N ,

$$|H(TJ(\mathbf{Z})) + m \log a_0 - h(\mathbf{Z})| \leq C_{m,\kappa} N^{-7\vartheta} + C_m N^{5\vartheta} e^{-c_m N^\vartheta}. \quad (2.9)$$

The constants are uniform in the number, weights, and means of the mixture components.

Proof. Write f for the density of $\mathbf{Z}$. The proof has two independent parts. First we compare differential entropy with the entropy of the unbounded grid label $J(\mathbf{Z})$. We then show that clipping this label changes entropy only by an exponentially small amount.

The exact cellwise entropy identity. Let

$$C_j = a_0(\mathbf{j} + [0, 1]^m), \quad p_j = \int_{C_j} f, \quad \bar{f}_j = p_j/a_0^m.$$

A direct expansion of the definitions gives

$$H(J(\mathbf{Z})) + m \log a_0 - h(\mathbf{Z}) = \sum_j \int_{C_j} f(\mathbf{x}) \log \frac{f(\mathbf{x})}{\bar{f}_j} d\mathbf{x}. \quad (2.10)$$

Every summand is nonnegative by Jensen's inequality. This also explains the correction $m \log a_0$: a small cell has probability approximately $a_0^m f$.

We next obtain a quantitative upper bound without requiring derivative bounds on individual component means. Concavity of the logarithm and symmetrization imply

$$\begin{aligned} 0 &\leq H(J(\mathbf{Z})) + m \log a_0 - h(\mathbf{Z}) \\ &\leq \frac{a_0^{-m}}{2} \int_{[-a_0, a_0]^m} \int_{\mathbb{R}^m} [f(\mathbf{x}) - f(\mathbf{x} + \mathbf{y})] \log \frac{f(\mathbf{x})}{f(\mathbf{x} + \mathbf{y})} d\mathbf{x} d\mathbf{y}. \end{aligned} \quad (2.11)$$

For clarity, this follows by applying $\log(a_0^{-m} \int_C f) \geq a_0^{-m} \int_C \log f$ in each cell, then averaging the resulting expression with the same expression in which the two points in the cell are interchanged. The integrand in (2.11) is nonnegative, so enlarging the set of pairs from points in the same cell to all pairs separated by a vector in $[-a_0, a_0]^m$ can only increase the integral.

The inner integral is the sum of the two relative entropies between f and its translate $f(\cdot + \mathbf{y})$. Relative entropy is jointly convex: the log-sum inequality bounds the relative entropy of two mixtures with the same weights by the weighted component relative entropies. Pair each Gaussian component with its translate. The two Gaussian relative entropies are both $\frac{1}{2} \mathbf{y}^T K^{-1} \mathbf{y}$, so their sum is $\mathbf{y}^T K^{-1} \mathbf{y}$. Consequently,

$$\int_{\mathbb{R}^m} [f(\mathbf{x}) - f(\mathbf{x} + \mathbf{y})] \log \frac{f(\mathbf{x})}{f(\mathbf{x} + \mathbf{y})} d\mathbf{x} \leq \frac{|\mathbf{y}|^2}{\kappa \eta_0}.$$

Substitution in (2.11) yields

$$0 \leq H(J(\mathbf{Z})) + m \log a_0 - h(\mathbf{Z}) \leq C_{m, \kappa} \frac{a_0^2}{\eta_0} = C_{m, \kappa} N^{-7\vartheta}. \quad (2.12)$$

The clipping probability. Let $B = \{TJ(\mathbf{Z}) \neq J(\mathbf{Z})\}$ and $p = \mathbb{P}(B)$. If B occurs, some coordinate of $\mathbf{Z}$ has absolute value at least $3R_0/4$ for all sufficiently large N . Its component mean has absolute value at most $R_0/2$ and its component variance is at most $\eta_0^{-1} = R_0$. The one-dimensional Gaussian tail bound, followed by averaging over components and a union bound over coordinates, gives

$$p \leq 2me^{-c_m R_0} = 2me^{-c_m N^\vartheta}. \quad (2.13)$$

This estimate is uniform in the mixture weights and means.

Entropy carried by the clipped tail. Because $TJ(\mathbf{Z})$ is a deterministic function of $J(\mathbf{Z})$ and the two labels agree on B^c , the chain rule applied also to the indicator $\mathbf{1}_B$ gives

$$H(J(\mathbf{Z})) - H(TJ(\mathbf{Z})) = H(J(\mathbf{Z}) | TJ(\mathbf{Z})) \leq H_2(p) + pH(J(\mathbf{Z}) | B).$$

Here $H(J(\mathbf{Z}) \mid B)$ denotes conditioning on the event B , not on its indicator; the product is taken to be zero when $p = 0$. For any $\mathbb{Z}^m$ -valued random vector $\mathbf{V}$, comparison with the probability mass function proportional to $e^{-|\mathbf{J}|_1}$ gives

$$H(\mathbf{V}) \leq C_m + \mathbb{E}|\mathbf{V}|_1. \quad (2.14)$$

Indeed, nonnegativity of relative entropy with respect to that reference law is exactly (2.14). Applying it to the conditional law of $J(\mathbf{Z})$ given the event B gives

$$H(J(\mathbf{Z})) - H(TJ(\mathbf{Z})) \leq H_2(p) + C_m p + \mathbb{E}[|J(\mathbf{Z})|_1 \mathbf{1}_B].$$

Since $|J(Z_r)| \leq a_0^{-1}|Z_r| + 1$ and the component means and variances satisfy the assumed bounds,

$$\mathbb{E}|J(\mathbf{Z})|_1^2 \leq C_m a_0^{-2}(R_0^2 + \eta_0^{-1}) + C_m \leq C_m N^{10\vartheta}.$$

Cauchy–Schwarz and (2.13) therefore imply

$$0 \leq H(J(\mathbf{Z})) - H(TJ(\mathbf{Z})) \leq C_m N^{5\vartheta} e^{-c_m N^\vartheta}. \quad (2.15)$$

Combining (2.12) and (2.15) proves (2.9). No independence between the coordinates of $\mathbf{Z}$ has been used. $\square$

2.1.3. Gaussian smoothing at high noise.

Lemma 2.3 (High-noise smoothing). *Let $\mathbf{Y}$ be an m -dimensional random vector satisfying*

$$\mathbb{E}\mathbf{Y} = 0, \quad \text{Cov}(\mathbf{Y}) = I_m, \quad |\mathbf{Y}| \leq B \quad \text{almost surely},$$

where $B \geq 2$. Let $\mathbf{G}$ be an independent standard Gaussian and put

$$f_s = \frac{\text{dLaw}(\sqrt{s}\mathbf{Y} + \sqrt{1-s}\mathbf{G})}{\text{d}\gamma_m}, \quad 0 < s < 1.$$

There are constants $c, C > 0$, depending only on m , such that

$$\|f_s - 1\|_{L^3(\gamma_m)} \leq C B s^{3/2}, \quad 0 < s \leq c B^{-2}. \quad (2.16)$$

Proof. The first two moment assumptions are used only to cancel the terms of orders $\sqrt{s}$ and s . The remaining third-order Taylor term is of order $s^{3/2}$. Throughout this proof, $\|\cdot\|_3$ is the $L^3(\gamma_m)$ norm in $\mathbf{z}$. After the Gaussian convolution is written as a density, every remaining expectation is over $\mathbf{Y}$.

Put $a = \sqrt{s}$. For fixed $\mathbf{y}$, the density of $a\mathbf{y} + \sqrt{1-a^2}\mathbf{G}$ relative to γ_m is

$$K(a; \mathbf{y}, \mathbf{z}) = (1 - a^2)^{-m/2} \exp\left(\frac{2a\mathbf{z} \cdot \mathbf{y} - a^2(|\mathbf{z}|^2 + |\mathbf{y}|^2)}{2(1 - a^2)}\right). \quad (2.17)$$

Consequently $f_s(\mathbf{z}) = \mathbb{E}K(a; \mathbf{Y}, \mathbf{z})$.

Cancellation of the first two Taylor coefficients. Direct differentiation at $a = 0$ gives

$$\begin{aligned} K(0; \mathbf{y}, \mathbf{z}) &= 1, \\ \partial_a K(0; \mathbf{y}, \mathbf{z}) &= \mathbf{z} \cdot \mathbf{y}, \\ \partial_a^2 K(0; \mathbf{y}, \mathbf{z}) &= (\mathbf{z} \cdot \mathbf{y})^2 - |\mathbf{y}|^2 - |\mathbf{z}|^2 + m. \end{aligned}$$

Now $\mathbb{E}(\mathbf{z} \cdot \mathbf{Y}) = 0$, $\mathbb{E}(\mathbf{z} \cdot \mathbf{Y})^2 = |\mathbf{z}|^2$, and $\mathbb{E}|\mathbf{Y}|^2 = m$. Hence the expectations of the first and second derivatives vanish. This is the precise reason that matching both the mean and covariance is necessary.

A uniform third-derivative bound. Write $\ell = \log K$. When $0 \leq t \leq a$, $aB \leq 1/4$, and $|\mathbf{y}| \leq B$, we have $t \leq 1/4$ and $t|\mathbf{y}| \leq 1/4$. From (2.17),

$$\partial_t \ell = \frac{mt}{1-t^2} + \frac{(1+t^2)\mathbf{z} \cdot \mathbf{y} - t(|\mathbf{z}|^2 + |\mathbf{y}|^2)}{(1-t^2)^2}.$$

Differentiating twice more shows, for $j = 1, 2, 3$,

$$|\partial_t^j \ell| \leq C_m(1 + |\mathbf{y}|)^j(1 + |\mathbf{z}|^2).$$

Also, the negative quadratic terms in the exponent and $t|\mathbf{y}| \leq 1/4$ give $K(t; \mathbf{y}, \mathbf{z}) \leq C_m e^{C_m |\mathbf{z}|}$. Since

$$\partial_t^3 K = K[(\partial_t \ell)^3 + 3(\partial_t \ell)(\partial_t^2 \ell) + \partial_t^3 \ell],$$

Gaussian integrability of every polynomial times $e^{C|\mathbf{z}|}$ yields

$$\|\partial_t^3 K(t; \mathbf{y}, \cdot)\|_{L^3(\gamma_m)} \leq C_m(1 + |\mathbf{y}|^3). \quad (2.18)$$

All constants in this estimate are independent of B once $t|\mathbf{y}| \leq 1/4$.

Taylor's integral remainder. Taylor's formula and the two cancellations give

$$f_s(\mathbf{z}) - 1 = \frac{1}{2} \mathbb{E} \int_0^a (a-t)^2 \partial_t^3 K(t; \mathbf{Y}, \mathbf{z}) dt.$$

Minkowski's inequality and (2.18) imply

$$\|f_s - 1\|_3 \leq Ca^3(1 + \mathbb{E}|\mathbf{Y}|^3).$$

Finally, $|\mathbf{Y}| \leq B$ and covariance normalization give $\mathbb{E}|\mathbf{Y}|^3 \leq B\mathbb{E}|\mathbf{Y}|^2 = mB$. Since $a^3 = s^{3/2}$ and $B \geq 2$, this proves (2.16) whenever $aB \leq 1/4$, which is ensured by $s \leq cB^{-2}$. $\square$

2.1.4. Gaussian star laws. Fix $d \geq 3$, $k \geq 1$ and $K \in (-2\sqrt{d-1}, 2\sqrt{d-1})$. Let $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_k) \in K^k$ be fixed parameters. For $\lambda \in K$, define the $(d+1) \times (d+1)$ covariance matrix

$$\begin{aligned} (\Sigma_\lambda)_{ii} &= 1 \quad (0 \leq i \leq d), & (\Sigma_\lambda)_{0i} &= (\Sigma_\lambda)_{i0} = \lambda/d \quad (1 \leq i \leq d), \\ (\Sigma_\lambda)_{ij} &= (\lambda^2 - d)/(d(d-1)) \quad (1 \leq i \neq j \leq d). \end{aligned} \quad (2.19)$$

Set

$$H_\lambda = \{\mathbf{y} \in \mathbb{R}^{d+1} : \sum_{i=1}^d y_i = \lambda y_0\}, \quad H_\lambda = \bigoplus_{a=1}^k H_{\lambda_a}.$$

The kernel of Σ_λ is spanned by $(-\lambda, 1, \dots, 1)^T$. Its nonzero eigenvalues are $(d + \lambda^2)/d$ once and $(d^2 - \lambda^2)/(d(d-1))$ with multiplicity $d-1$.

Here is the underlying decomposition. Vectors with zero root coordinate and leaf coordinates summing to zero form a $(d-1)$ -dimensional subspace on which the eigenvalue is $1 - (\lambda^2 - d)/(d(d-1))$. Its orthogonal complement is the root-constant-leaf plane. On that plane one eigenvalue is zero, because the vector $(-\lambda, 1, \dots, 1)$ is normal to H_λ ; the other follows either by a 2×2 calculation or by subtracting the transverse contribution from $\text{Tr } \Sigma_\lambda = d+1$. Thus the Gaussian is nondegenerate on H_λ , but singular in $\mathbb{R}^{d+1}$. Entropy must consequently be computed in H_λ , not in the ambient space. Let $\mathbf{Z}_\lambda$ have independent centered Gaussian components with covariances Σ_{λ_a} . For a law P on H_λ with finite entropies define

$$\mathcal{F}_\lambda(P) = h_{H_\lambda}(P) - \frac{d}{2} h(P_e),$$

where P_e is the marginal on the k root coordinates and their first neighbor coordinates, of total dimension $2k$, and subspace entropy uses induced Euclidean volume, as in [1, Section 7].

For one channel this is the functional of [1, Definition 11.1]; we use its joint version on the direct sum of the wave subspaces. As with entropy, $\mathcal{F}_\lambda(\mathbf{X})$ means $\mathcal{F}_\lambda(\text{Law}(\mathbf{X}))$ when the required star and edge densities exist. This functional is unchanged by a common scalar dilation.

Indeed, dilation by $r > 0$ adds $dk \log r$ to the star entropy and $2k \log r$ to the edge entropy, and these terms cancel. The product structure of $\mathbf{Z}_\lambda$ makes the functional additive in its k channels.

Lemma 2.4 (Entropy of the Gaussian star). *Let $L_d(\lambda) = \int \rho_d(t) \log |t - \lambda| dt$. Then, for every λ in the open Kesten–McKay bulk,*

$$L_d(\lambda) = \frac{d-1}{2} \log(d-1) - \frac{d-2}{4} \log(d^2 - \lambda^2), \quad (2.20)$$

and

$$\mathcal{F}_\lambda(\mathbf{Z}_\lambda) = \sum_{a=1}^k \left(\frac{1}{2} \log(d + \lambda_a^2) - L_d(\lambda_a) \right). \quad (2.21)$$

Proof. The logarithmic potential of the Kesten–McKay law. Put

$$g(z) = \frac{-z + \sqrt{z^2 - 4(d-1)}}{2(d-1)},$$

where $z \in \mathbb{C}^+$ and the square root is chosen to be asymptotic to z at infinity. The logarithms below are analytic branches fixed by their asymptotics as $z = iy$, $y \rightarrow \infty$. The tree resolvent recursion gives

$$(d-1)g^2 + zg + 1 = 0, \quad m_d(z) = \frac{g(z)}{1 - g(z)^2}.$$

Differentiating the recursion, or substituting it directly, gives

$$\frac{d}{dz} \left[-\log(-g(z)) - \frac{d-2}{2} \log(1 - g(z)^2) \right] = -m_d(z).$$

On the other hand, $\frac{d}{dz} \int \rho_d(t) \log(z - t) dt = -m_d(z)$ with our Stieltjes-transform convention. Comparison as $z \rightarrow \infty$ fixes the integration constant. Taking the real part of the boundary value at a bulk point λ , and using

$$|g(\lambda + i0)| = (d-1)^{-1/2}, \quad |1 - g(\lambda + i0)^2| = \frac{\sqrt{d^2 - \lambda^2}}{d-1},$$

proves (2.20). Taking this boundary value under the integral is justified by local integrability of the logarithmic singularity and boundedness of ρ_d near the fixed bulk point.

The star–edge Gaussian entropy difference. For one channel, the nonzero eigenvalues of Σ_λ are

$$\frac{d + \lambda^2}{d} \quad \text{and} \quad \frac{d^2 - \lambda^2}{d(d-1)} \quad \text{with multiplicity } d-1.$$

Therefore the determinant of the covariance restricted to the d -dimensional space H_λ is

$$\det(\Sigma_\lambda|_{H_\lambda}) = \frac{d + \lambda^2}{d} \left(\frac{d^2 - \lambda^2}{d(d-1)} \right)^{d-1}.$$

The root–first-neighbor covariance is $\begin{pmatrix} 1 & \lambda/d \\ \lambda/d & 1 \end{pmatrix}$, whose determinant is $(d^2 - \lambda^2)/d^2$. Using the Gaussian entropy formula on H_λ and in the 2-dimensional edge space, the factors $2\pi e$ cancel because $\dim H_\lambda = d = (d/2) \cdot 2$. We obtain

$$\mathcal{F}_\lambda(\mathbf{Z}_\lambda) = \frac{1}{2} \log \det(\Sigma_\lambda|_{H_\lambda}) - \frac{d}{4} \log \frac{d^2 - \lambda^2}{d^2}.$$

Substituting the two determinants and simplifying with (2.20) gives $\mathcal{F}_\lambda(\mathbf{Z}_\lambda) = \frac{1}{2} \log(d + \lambda^2) - L_d(\lambda)$. The k channels of $\mathbf{Z}_\lambda$ are independent, so both star and edge entropy are additive. Summing the one-channel identity proves (2.21). $\square$

2.1.5. Entropy of Gaussian resolvent noise.

Lemma 2.5 (Entropy per vertex of resolvent noise). *Let A be a fixed real symmetric $N \times N$ matrix with eigenvalues $d = \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_N \geq -d$ satisfying $\max_{2 \leq j \leq N} |\lambda_j - \gamma_j| \leq N^{-a}$ for some $a > 0$. Put $Q_\lambda = \text{Im}(A - \lambda - i\eta)^{-1}$, with $\eta = N^{-c_0}$ and $0 < c_0 < a$. For $\mathbf{Y} \sim N(0, Q_\lambda)$,*

$$\frac{1}{N} h(\mathbf{Y}) = \frac{1}{2} \log(2\pi e \eta) - L_d(\lambda) + O(N^{-a}/\eta + \eta + N^{-1} \log N)$$

uniformly for $\lambda \in K$.

Proof. Equation (2.3) gives exactly

$$\frac{1}{N} h(\mathbf{Y}) = \frac{1}{2} \log(2\pi e \eta) - \frac{1}{N} \sum_j \log |\lambda_j - \lambda - i\eta|.$$

The summand has Lipschitz constant at most η^{-1} , so replacing the nontrivial eigenvalues by their quantiles incurs an error of at most N^{-a}/η . Its total variation on the spectral support is $O(1 + |\log \eta|)$, which bounds the quantile Riemann-sum error by $O(N^{-1} \log N)$. The trivial eigenvalue contributes $O(N^{-1})$. It remains to bound the smoothing error:

$$\frac{1}{2} \int \rho_d(t) \log \left(1 + \frac{\eta^2}{(t - \lambda)^2} \right) dt \leq C\eta \int_{\mathbb{R}} \log(1 + r^{-2}) dr = O(\eta).$$

The deterministic approximation just used can also be viewed as a transport estimate. The quantiles give a Riemann sum for the probability measure $\rho_d(t) dt$. For a function of bounded variation on the spectral interval, its quantile Riemann error is bounded by its total variation divided by N . For $\log |t - \lambda - i\eta|$ that variation is $O(1 + |\log \eta|)$. The single trivial eigenvalue changes the normalized sum by at most $O(N^{-1} \log N)$ as well. Thus no microscopic rigidity estimate is being used in this entropy computation; the coarse displacement N^{-a} is sufficient because the noise height is larger. $\square$

2.1.6. Gaussian star geometry and fourth-chaos coercivity. We retain the fixed parameters and Gaussian star laws from Subsubsection 2.1.4. The weighted frame and Fisher-information argument adapt [1, Lemma 11.3 and Proposition 11.1] componentwise, allowing different parameters λ_a . The mixed-derivative calculation and fourth-chaos coercivity below are finite-dimensional statements, with constants uniform over $\boldsymbol{\lambda} \in K^k$.

The wave constraint depends on λ , so different star laws initially live on different hyperplanes. We want to move all of them to one fixed Euclidean space, with the standard Gaussian as reference. Let $U = \mathbb{R}^{d-1}$ contain the vertices $\mathbf{e}_1, \dots, \mathbf{e}_d$ of a regular unit simplex. Thus

$$\mathbf{e}_i \cdot \mathbf{e}_j = \begin{cases} 1, & i = j, \\ -1/(d-1), & i \neq j, \end{cases} \quad \sum_{i=1}^d \mathbf{e}_i = 0, \quad \sum_{i=1}^d \mathbf{e}_i \mathbf{e}_i^T = \frac{d}{d-1} I_U.$$

A concrete realization helps interpret these formulas. Identify U isometrically with the zero-sum hyperplane

$$\left\{ (u_1, \dots, u_d) \in \mathbb{R}^d : \sum_{i=1}^d u_i = 0 \right\}$$

and take

$$\mathbf{e}_i = \sqrt{\frac{d}{d-1}} \left(\boldsymbol{\varepsilon}_i - \frac{1}{d} \mathbf{1} \right),$$

where $\boldsymbol{\varepsilon}_i$ is the i th coordinate vector. The displayed simplex identities then follow by direct calculation. We use the abstract space $U \simeq \mathbb{R}^{d-1}$ only to avoid carrying the redundant zero-sum coordinate.

A *standard Gaussian on U* means a centered Gaussian whose covariance is the identity operator on the Euclidean space U . In the concrete model above, start with $\mathbf{G} \sim N(0, I_d)$ and orthogonally project it onto this realization of U :

$$\mathbf{G}_U = \mathbf{G} - \frac{1}{d} (\mathbf{1} \cdot \mathbf{G}) \mathbf{1}.$$

For every $\mathbf{u}, \mathbf{v} \in U$,

$$\mathbb{E}[(\mathbf{u} \cdot \mathbf{G}_U)(\mathbf{v} \cdot \mathbf{G}_U)] = \mathbf{u} \cdot \mathbf{v}.$$

Thus $\mathbf{G}_U$ is standard intrinsically on U , even though its d ambient coordinates are dependent. In particular, $\text{Cov}(\mathbf{e}_i \cdot \mathbf{G}_U, \mathbf{e}_j \cdot \mathbf{G}_U) = \mathbf{e}_i \cdot \mathbf{e}_j$.

For a single channel, the space is $\mathbb{R} + U$, where $+$ denotes the orthogonal direct sum. A point is a pair $(y_0, \mathbf{y})$ with $y_0 \in \mathbb{R}$ and $\mathbf{y} \in U$. The root coordinate is y_0 , and the i th leaf coordinate is

$$cy_0 + \rho \mathbf{e}_i \cdot \mathbf{y}, \quad c = \lambda/d, \quad \rho = \sqrt{1 - c^2}.$$

The leaf coordinates sum to $dcy_0 = \lambda y_0$, because the simplex vertices sum to zero. For a standard Gaussian pair $(y_0, \mathbf{y})$, the root-leaf covariance is c , each leaf has variance $c^2 + \rho^2 = 1$, and two distinct leaves have covariance

$$c^2 - \frac{\rho^2}{d-1} = \frac{\lambda^2 - d}{d(d-1)}.$$

Thus this map gives exactly the covariance Σ_λ .

For k channels, take k copies of the same space and set

$$E = (\mathbb{R} + U)^k, \quad V_i = (\mathbb{R} + \mathbb{R}\mathbf{e}_i)^k.$$

We write a point of E as $\mathbf{y} = ((y_{a,0}, \mathbf{y}_a))_{a=1}^k$, where $y_{a,0} \in \mathbb{R}$ and $\mathbf{y}_a \in U$. The inner product is the sum of the inner products in the k blocks. We identify $\mathbb{R}^k$ with the subspace in which every $\mathbf{y}_a = 0$, and U^k with the subspace in which every $y_{a,0} = 0$. These are the scalar and transverse coordinates, respectively. The space E and its edge subspaces V_i do not depend on λ .

Let P_i be the orthogonal projection onto V_i . It acts separately in each block:

$$P_i \mathbf{y} = ((y_{a,0}, (\mathbf{e}_i \cdot \mathbf{y}_a) \mathbf{e}_i))_{a=1}^k.$$

In this subsection E denotes the Euclidean space, not a spectral energy. Its dimension is dk , while $\dim V_i = 2k$. Write γ_E for the standard Gaussian law on E , with the analogous convention on its subspaces. A standard Gaussian on E has independent standard Gaussian scalar and transverse coordinates across all k blocks. For $1 \leq a \leq k$, put

$$c_a = \lambda_a/d, \quad \rho_a = (1 - c_a^2)^{1/2}.$$

Let $\mathbf{w}_a \in E$ have a th block $(1, \mathbf{0})$ and all other blocks zero. Let $\mathbf{v}_{i,a} \in E$ have a th block $(c_a, \rho_a \mathbf{e}_i)$ and all other blocks zero. Thus $\mathbf{w}_a$ and $\mathbf{v}_{i,a}$ represent the root and the i th leaf in channel a , and V_i is their span over all a .

The physical coordinate map $\mathcal{L}_\lambda$ is consequently

$$(\mathcal{L}_\lambda \mathbf{y})_{a,0} = y_{a,0}, \quad (\mathcal{L}_\lambda \mathbf{y})_{a,i} = c_a y_{a,0} + \rho_a \mathbf{e}_i \cdot \mathbf{y}_a.$$

Equivalently, its root coordinates are $\mathbf{w}_a \cdot \mathbf{y}$ and its neighbor coordinates are $\mathbf{v}_{i,a} \cdot \mathbf{y}$. For a fixed i , knowing these two physical coordinates in every channel is equivalent to knowing $P_i \mathbf{y}$. Indeed, the root coordinate gives $y_{a,0}$, and then

$$\mathbf{e}_i \cdot \mathbf{y}_a = \frac{(\mathcal{L}_\lambda \mathbf{y})_{a,i} - c_a (\mathcal{L}_\lambda \mathbf{y})_{a,0}}{\rho_a}.$$

This is why the physical oriented-edge marginal becomes the ordinary orthogonal projection onto the fixed subspace V_i after whitening. The one-channel calculation above applies independently in every block, so

$$\mathcal{L}_\lambda \mathcal{L}_\lambda^T = \text{diag}(\Sigma_{\lambda_1}, \dots, \Sigma_{\lambda_k}).$$

In each block, $\mathcal{L}_\lambda^T \mathcal{L}_\lambda$ has scalar and transverse eigenvalues $1 + dc_a^2$ and $d\rho_a^2/(d-1)$. Thus $\mathcal{L}_\lambda$ is invertible onto H_λ , the map and its inverse are uniformly bounded on K^k , and it sends a standard Gaussian on E to $\mathbf{Z}_\lambda$.

An explicit inverse clarifies this change of coordinates. For $\mathbf{x} \in H_\lambda$, the a th block of the inverse is

$$(\mathcal{L}_\lambda^{-1} \mathbf{x})_a = \left(x_{a,0}, \frac{d-1}{d\rho_a} \sum_{i=1}^d x_{a,i} \mathbf{e}_i \right).$$

Indeed, the simplex identities and $\sum_i x_{a,i} = dc_a x_{a,0}$ give

$$\mathbf{e}_j \cdot \left(\frac{d-1}{d\rho_a} \sum_{i=1}^d x_{a,i} \mathbf{e}_i \right) = \frac{x_{a,j} - c_a x_{a,0}}{\rho_a}.$$

Substitution in the physical coordinate map recovers every leaf coordinate. The change of variables adds a deterministic logarithmic Jacobian to star and edge entropies; these constants disappear when the functional of a law is subtracted from that of its Gaussian reference.

Let R_i be the orthogonal involution exchanging $\mathbf{w}_a, \mathbf{v}_{i,a}$ for all a and fixing $V_i^\perp$. It represents simultaneous endpoint reversal on V_i . Permuting the simplex vertices gives the orthogonal action of neighbor permutations on E . The statements below concern laws invariant under this action whose V_i marginals are invariant under R_i .

Define directions in V_i by

$$\begin{aligned} \mathbf{a}_{i,a} &= \frac{\mathbf{w}_a + \mathbf{v}_{i,a}}{2\sqrt{1+c_a}} + \frac{\mathbf{w}_a - \mathbf{v}_{i,a}}{2\sqrt{1-c_a}}, \\ \mathbf{b}_{i,a} &= \frac{\mathbf{w}_a + \mathbf{v}_{i,a}}{2\sqrt{1+c_a}} - \frac{\mathbf{w}_a - \mathbf{v}_{i,a}}{2\sqrt{1-c_a}}. \end{aligned}$$

The sum $\mathbf{w}_a + \mathbf{v}_{i,a}$ and difference $\mathbf{w}_a - \mathbf{v}_{i,a}$ are orthogonal, with squared lengths $2(1+c_a)$ and $2(1-c_a)$. Each summand in the definition of $\mathbf{a}_{i,a}$ therefore has squared length $1/2$. Their sum and difference, $\mathbf{a}_{i,a}$ and $\mathbf{b}_{i,a}$, are orthonormal. Different channels are orthogonal, so $\{\mathbf{a}_{i,a}, \mathbf{b}_{i,a} : 1 \leq a \leq k\}$ is an orthonormal basis of V_i . Endpoint reversal fixes the sum and negates the difference; consequently

$$R_i \mathbf{a}_{i,a} = \mathbf{b}_{i,a}, \quad R_i \mathbf{b}_{i,a} = \mathbf{a}_{i,a}.$$

Put

$$t_{1,a} = \frac{\rho_a - 1 + 2/d}{2\rho_a}, \quad t_{2,a} = \frac{\rho_a + 1 - 2/d}{2\rho_a}.$$

The open bulk condition gives $\rho_a > \sqrt{1 - 4(d-1)/d^2} = 1 - 2/d$. Since K is compactly contained in the bulk, both weights are bounded below by a positive constant depending only on d, K , and $t_{1,a} + t_{2,a} = 1$. The weights are chosen so that the following positive frame is exactly the identity:

$$\sum_{i=1}^d \sum_{a=1}^k \left(t_{1,a} \mathbf{a}_{i,a} \mathbf{a}_{i,a}^T + t_{2,a} \mathbf{b}_{i,a} \mathbf{b}_{i,a}^T \right) = I_E. \quad (2.22)$$

We verify this identity explicitly. Work in one channel and decompose vectors into their scalar part in $\mathbb{R}$ and their transverse part in U . A direct calculation from the definitions gives

$$|\mathbf{w}_a \cdot \mathbf{a}_{i,a}|^2 = \frac{1 + \rho_a}{2}, \quad |\mathbf{w}_a \cdot \mathbf{b}_{i,a}|^2 = \frac{1 - \rho_a}{2}.$$

Because both vectors have norm one, their transverse squared lengths are the two quantities in the opposite order. Therefore the weighted scalar coefficient contributed by a fixed neighbor i is

$$t_{1,a} \frac{1 + \rho_a}{2} + t_{2,a} \frac{1 - \rho_a}{2} = \frac{1}{d},$$

whereas the weighted transverse coefficient is

$$t_{1,a} \frac{1 - \rho_a}{2} + t_{2,a} \frac{1 + \rho_a}{2} = \frac{d - 1}{d}.$$

The scalar-transverse cross term is a scalar multiple of $\mathbf{e}_i$. It vanishes after summing over i because $\sum_i \mathbf{e}_i = 0$. Finally,

$$\sum_{i=1}^d \frac{1}{d} I_{\mathbb{R}} = I_{\mathbb{R}}, \quad \frac{d-1}{d} \sum_{i=1}^d \mathbf{e}_i \mathbf{e}_i^T = I_U.$$

This proves the frame identity in one channel, and the sum over the orthogonal channel blocks proves (2.22) on E .

The special case $\lambda_a = 0$ is a useful check. Then $c_a = 0$, $\rho_a = 1$, and, in the a th block,

$$\mathbf{a}_{i,a} = \mathbf{w}_a, \quad \mathbf{b}_{i,a} = (0, \mathbf{e}_i), \quad t_{1,a} = \frac{1}{d}, \quad t_{2,a} = \frac{d-1}{d}.$$

Thus (2.22) reduces to one copy of the scalar direction with total weight one and the standard simplex tight frame on U .

Let p be a smooth positive Lebesgue density on E with finite Fisher information, and let p_i be its marginal density on V_i . Assume that $p_i \circ R_i = p_i$. In this paragraph $\mathbb{E}_p$ means expectation under the density p . We regard $\log p_i(P_i \mathbf{x})$ as a function on E . If $\mathbf{Y}$ has density p , differentiating the marginal under the integral gives the score-projection identity

$$\mathbb{E}_p[P_i \nabla \log p(\mathbf{Y}) \mid P_i \mathbf{Y}] = \nabla \log p_i(P_i \mathbf{Y}). \quad (2.23)$$

For completeness, for any $\mathbf{u} \in V_i$ and smooth compactly supported ϕ on V_i , integration by parts gives

$$\mathbb{E}_p[\phi(P_i \mathbf{Y}) \partial_{\mathbf{u}} \log p(\mathbf{Y})] = -\mathbb{E}_p[\partial_{\mathbf{u}} \phi(P_i \mathbf{Y})] = \mathbb{E}_p[\phi(P_i \mathbf{Y}) \partial_{\mathbf{u}} \log p_i(P_i \mathbf{Y})],$$

which is exactly (2.23).

Write

$$I(p) = \int_E p |\nabla \log p|^2 \, d\mathbf{x}, \quad I(p_i) = \int_{V_i} p_i |\nabla \log p_i|^2.$$

For every unit vector $\mathbf{u} \in V_i$, conditional expectation is an orthogonal projection in $L^2(p)$, so

$$\begin{aligned} & \int_E p |\partial_{\mathbf{u}}(\log p - \log p_i \circ P_i)|^2 \, d\mathbf{x} \\ &= \int_E p |\partial_{\mathbf{u}} \log p|^2 \, d\mathbf{x} - \int_{V_i} p_i |\partial_{\mathbf{u}} \log p_i|^2. \end{aligned}$$

Sum this identity in the directions of the weighted frame. The full terms sum to $I(p)$ by (2.22). For the marginal terms, R_i preserves p_i and exchanges $\mathbf{a}_{i,a}$ with $\mathbf{b}_{i,a}$, hence

$$\int p_i |\partial_{\mathbf{a}_{i,a}} \log p_i|^2 = \int p_i |\partial_{\mathbf{b}_{i,a}} \log p_i|^2.$$

Since $t_{1,a} + t_{2,a} = 1$ and the paired directions form an orthonormal basis of V_i , their weighted marginal contribution is $I(p_i)/2$. We have proved the exact sum-of-squares identity

$$\begin{aligned} I(p) - \frac{1}{2} \sum_{i=1}^d I(p_i) &= \sum_{i=1}^d \sum_{a=1}^k \int_E p (t_{1,a} |\partial_{\mathbf{a}_{i,a}}(\log p - \log p_i \circ P_i)|^2 \\ &\quad + t_{2,a} |\partial_{\mathbf{b}_{i,a}}(\log p - \log p_i \circ P_i)|^2) \, d\mathbf{x} \geq 0. \end{aligned} \tag{2.24}$$

No independence among the k channels is used here.

The same proof applies to scores relative to the standard Gaussian: replace $\log p$ by the logarithm of the density of the same law relative to γ_E , and use the analogous relative marginal density on V_i . The Gaussian score $-\mathbf{x}$ projects compatibly under P_i , so the conditional-score calculation and the frame identity are unchanged. This relative version is the one used for Ornstein–Uhlenbeck interpolation below.

Lemma 2.6 (Kernel of mixed third derivatives). *Let T be a symmetric trilinear form on $E = (\mathbb{R} + U)^k$. Suppose that*

$$T((I - P_i)\mathbf{x}, P_i\mathbf{y}, \mathbf{z}) = 0 \quad \text{for every } i \text{ and every } \mathbf{x}, \mathbf{y}, \mathbf{z} \in E.$$

Then T depends only on the scalar coordinates of its arguments. Equivalently, $T(\mathbf{x}, \mathbf{y}, \mathbf{z}) = 0$ whenever at least one of its arguments belongs to U^k .

Proof. Symmetry means that the value of T is unchanged by any permutation of its three arguments. Fixing the last argument $\mathbf{z}$ gives a self-adjoint operator $B_{\mathbf{z}}$ on E , defined by

$$\langle B_{\mathbf{z}}\mathbf{x}, \mathbf{y} \rangle = T(\mathbf{x}, \mathbf{y}, \mathbf{z}).$$

The hypothesis says that this operator has no block between V_i and $V_i^\perp$. More explicitly, $(I - P_i)B_{\mathbf{z}}P_i = 0$; taking adjoints gives $P_iB_{\mathbf{z}}(I - P_i) = 0$. Hence $B_{\mathbf{z}}$ commutes with every P_i . We first determine the operators with this property, and then use symmetry in the third argument.

Separate scalar and transverse coordinates. The simplex identities give, for $\mathbf{y} = ((y_{a,0}, \mathbf{y}_a))_{a=1}^k$,

$$\sum_{i=1}^d P_i \mathbf{y} = \left(\left(dy_{a,0}, \frac{d}{d-1} \mathbf{y}_a \right) \right)_{a=1}^k.$$

Thus $\mathbb{R}^k$ and U^k are the eigenspaces of $\sum_i P_i$ with eigenvalues d and $d/(d-1)$, respectively. These eigenvalues are distinct because $d \geq 3$. Since B_z commutes with their sum, it preserves both subspaces. Consequently,

$$T(\mathbf{x}, \mathbf{y}, \mathbf{z}) = 0 \quad (\mathbf{x} \in \mathbb{R}^k, \mathbf{y} \in U^k, \mathbf{z} \in E).$$

By symmetry, T vanishes whenever one argument is scalar and another is transverse. Only the case in which all three arguments are transverse remains to be considered.

Determine the action on transverse coordinates. The intersection $V_i \cap U^k$ consists of the vectors $((0, a_a \mathbf{e}_i))_{a=1}^k$ with $\mathbf{a} \in \mathbb{R}^k$. It is invariant under B_z , so there is a self-adjoint $k \times k$ matrix $M_i(\mathbf{z})$ such that

$$B_z((0, a_a \mathbf{e}_i))_{a=1}^k = ((0, (M_i(\mathbf{z})\mathbf{a})_a \mathbf{e}_i))_{a=1}^k.$$

Self-adjointness follows because $\mathbf{a} \mapsto ((0, a_a \mathbf{e}_i))_{a=1}^k$ preserves inner products. For $i \neq j$, self-adjointness of B_z further gives

$$(\mathbf{e}_i \cdot \mathbf{e}_j) \langle M_i(\mathbf{z})\mathbf{a}, \mathbf{b} \rangle = (\mathbf{e}_i \cdot \mathbf{e}_j) \langle \mathbf{a}, M_j(\mathbf{z})\mathbf{b} \rangle.$$

Since $\mathbf{e}_i \cdot \mathbf{e}_j = -1/(d-1) \neq 0$, all the matrices $M_i(\mathbf{z})$ agree; write their common value as M_z . The simplex vertices span U , so for arbitrary transverse vectors $\mathbf{x} = ((0, \mathbf{x}_a))_{a=1}^k$ and $\mathbf{y} = ((0, \mathbf{y}_a))_{a=1}^k$, linearity yields

$$T(\mathbf{x}, \mathbf{y}, \mathbf{z}) = \sum_{a,b=1}^k (M_z)_{ab} \mathbf{x}_b \cdot \mathbf{y}_a.$$

In words, the same matrix mixes the k channels in every direction of U .

Use symmetry to eliminate the transverse part. Choose a unit vector $\mathbf{u} \in U$ and put $\mathbf{z} = ((0, z_a \mathbf{u}))_{a=1}^k$, with arbitrary scalar coefficients. Since $\dim U \geq 2$, choose a unit vector $\mathbf{v} \perp \mathbf{u}$. For arbitrary $\mathbf{a}, \mathbf{b} \in \mathbb{R}^k$, take

$$\mathbf{x} = ((0, a_a \mathbf{v}))_{a=1}^k, \quad \mathbf{y} = ((0, b_a \mathbf{v}))_{a=1}^k.$$

The preceding formula and symmetry give

$$\langle M_z \mathbf{a}, \mathbf{b} \rangle = T(\mathbf{x}, \mathbf{y}, \mathbf{z}) = T(\mathbf{x}, \mathbf{z}, \mathbf{y}) = 0.$$

For the last equality, every inner product between a transverse block of $\mathbf{x}$ and a transverse block of $\mathbf{z}$ is zero, since $\mathbf{v} \perp \mathbf{u}$. Thus $M_z = 0$ for all such $\mathbf{z}$. These vectors span U^k , so linearity in the last argument proves that T vanishes when all three arguments are transverse as well. Together with the first step, this is the claimed conclusion.

The assumption $d \geq 3$ has been used twice: to separate the scalar and transverse eigenspaces of $\sum_i P_i$, and to choose two orthogonal directions in U . Preservation of every edge subspace alone does not eliminate the transverse part; symmetry in all three arguments is essential. $\square$

We recall the finite-dimensional Gaussian terminology used in the next lemma. The n th Gaussian chaos is the span of multivariate Hermite polynomials of total degree n . These are ordinary polynomials with lower-degree corrections chosen so that different chaoses are orthogonal in $L^2(\gamma_E)$. The Ornstein–Uhlenbeck operator $\Delta - \mathbf{x} \cdot \nabla$ acts as multiplication by $-n$ on the n th chaos. Gaussian integration by parts therefore gives $\int |\nabla h|^2 d\gamma_E = n \int h^2 d\gamma_E$. Conditional expectation onto a linear subspace is an orthogonal projection in L^2 and preserves

each chaos: in coordinates adapted to the subspace, all Hermite factors of positive degree in a discarded coordinate have expectation zero.

Let $\mathcal{C}_i$ denote Gaussian conditional expectation onto $V_i = (\mathbb{R} + \mathbb{R}\mathbf{e}_i)^k$. Thus $\mathcal{C}_i h$ keeps the scalar coordinates and the components of each transverse block along $\mathbf{e}_i$, and averages over the remaining Gaussian coordinates. More explicitly, if $\mathbf{G}$ is standard Gaussian on E , then

$$(\mathcal{C}_i h)(\mathbf{y}) = \mathbb{E}[h(\mathbf{G}) \mid P_i \mathbf{G} = P_i \mathbf{y}].$$

The right side denotes the Gaussian conditional integral on the orthogonal complement of V_i , and is lifted back to a function on E . In particular, if f is a density relative to γ_E , then $\mathcal{C}_i f$ is its marginal density relative to γ_{V_i} . Integrating a lifted marginal over γ_E is the same as integrating it over γ_{V_i} . For a function h on E , put

$$Q(h) = \|h\|_2^2 - \frac{d}{2} \|\mathcal{C}_1 h\|_2^2.$$

Until the end of Subsubsection 2.1.7, these function norms are with respect to γ_E , and $Q(h)$ denotes this quadratic functional, not a covariance matrix.

Lemma 2.7 (Fourth-chaos coercivity). *Suppose h belongs to Gaussian chaos $n \geq 1$, is invariant under neighbor permutations, and satisfies $(\mathcal{C}_i h) \circ R_i = \mathcal{C}_i h$ for every i . Then $Q(h) \geq 0$. If $n = 4$, there is $\kappa > 0$, uniform for $\boldsymbol{\lambda} \in K^k$, such that*

$$Q(h) \geq \kappa \|h\|_2^2. \quad (2.25)$$

Proof. The frame identity, Gaussian conditional expectation, and the reflection symmetry of the marginals give

$$nQ(h) = \sum_{i,a} (t_{1,a} \|(I - \mathcal{C}_i) \partial_{\mathbf{a}_{i,a}} h\|_2^2 + t_{2,a} \|(I - \mathcal{C}_i) \partial_{\mathbf{b}_{i,a}} h\|_2^2).$$

Indeed, the full derivative energy is $n\|h\|_2^2$. Reflection equates the paired projected derivative energies, so their weighted sum is $(n/2)\|\mathcal{C}_i h\|_2^2$ for each i ; neighbor symmetry then gives the displayed identity. This proves nonnegativity.

For degree four it suffices to show that $Q(h) = 0$ forces $h = 0$. Since the weights are positive, equality implies $P_i \nabla(h - \mathcal{C}_i h) = 0$ for every i . Thus every derivative of h in a direction of V_i depends only on $P_i \mathbf{x}$. Differentiating in a direction perpendicular to V_i and then in an arbitrary direction gives

$$D^3 h(\mathbf{x})[(I - P_i)\mathbf{u}, P_i \mathbf{v}, \mathbf{z}] = 0 \quad (\mathbf{x}, \mathbf{u}, \mathbf{v}, \mathbf{z} \in E).$$

Here $D^3 h(\mathbf{x})$ is the symmetric trilinear form obtained by taking three directional derivatives of h at $\mathbf{x}$. Apply Lemma 2.6 at each point $\mathbf{x}$: this third derivative vanishes whenever one of its arguments belongs to U^k . Differentiating once more, and using symmetry of fourth derivatives, shows that $D^4 h$ vanishes whenever any of its four arguments is transverse.

A fourth-chaos polynomial is determined by its fourth derivative: in the product Hermite basis, each basis element has a distinct leading monomial of total degree four. The preceding vanishing therefore means that h depends only on the k scalar coordinates, with no transverse coordinate present. These scalar coordinates are retained by every P_i , so $\mathcal{C}_i h = h$ for every i . It follows that

$$0 = Q(h) = \left(1 - \frac{d}{2}\right) \|h\|_2^2,$$

and hence $h = 0$, since $d \geq 3$. The unit sphere of fourth chaos is finite dimensional, and the symmetry constraints depend continuously on $\boldsymbol{\lambda}$. Compactness over $\boldsymbol{\lambda} \in K^k$ now

proves (2.25). To spell out the uniform compactness argument, suppose no positive uniform κ existed. There would be $\boldsymbol{\lambda}^{(r)} \in K^k$ and admissible fourth-chaos polynomials h_r with $\|h_r\|_2 = 1$ and $Q(h_r) \rightarrow 0$. Passing to subsequences in the compact parameter set and the finite-dimensional unit sphere gives $\boldsymbol{\lambda}^{(r)} \rightarrow \boldsymbol{\lambda}$ and $h_r \rightarrow h$ in L^2 . The reflection and permutation equations pass to the limit, as do conditional expectation and Q . Thus h is admissible, $\|h\|_2 = 1$, and $Q(h) = 0$, contradicting the kernel computation. This is a finite-dimensional spectral gap, not a quantitative stability theorem for arbitrary probability laws. $\square$

2.1.7. An abstract fourth-moment estimate. The following proposition gives a quantitative fourth-moment refinement of the Gaussian maximality argument of Backhausz–Szegedy [1, Theorem 11.2]. In the scalar setting, their theorem identifies the Gaussian law as the unique maximizer of the star-edge entropy functional under the prescribed covariance and symmetry constraints. The heat-flow comparison below follows [1, Proposition 11.1], with the weighted-frame calculation carried out in the present block coordinates $E = (\mathbb{R} + U)^k$.

To extract an explicit moment bound, we combine the fourth-chaos coercivity in Lemma 2.7 with the high-noise remainder estimate in Lemma 2.3. Uniqueness of the Gaussian maximizer alone does not provide this coercivity or a power-law estimate on moments. The argument applies to the joint law of k channels with possibly different eigenvalue parameters, uniformly for $\boldsymbol{\lambda} \in K^k$.

Let $\mathbf{Y}$ be an E -valued vector with the neighbor and edge-reflection symmetries above, and suppose

$$\mathbb{E}\mathbf{Y} = 0, \quad \text{Cov}(\mathbf{Y}) = I_E, \quad |\mathbf{Y}| \leq B, \quad B \geq 2.$$

For an independent standard Gaussian $\mathbf{G}$ on E , define

$$f_s = \frac{d\text{Law}(\sqrt{s}\mathbf{Y} + \sqrt{1-s}\mathbf{G})}{d\gamma_E}, \quad \Delta_s = \int f_s \log f_s \, d\gamma_E - \frac{d}{2} \int (\mathcal{C}_1 f_s) \log(\mathcal{C}_1 f_s) \, d\gamma_E.$$

Proposition 2.8 (Fourth moments from an entropy deficit). *If $0 \leq \Delta_{1/2} \leq \varepsilon \leq cB^{-6}$, then every homogeneous polynomial P of degree four whose coefficients in the fixed orthonormal coordinates of E are bounded by one satisfies*

$$|\mathbb{E}P(\mathbf{Y}) - \mathbb{E}P(\mathbf{G})| \leq CB^{4/3}\varepsilon^{1/18}. \quad (2.26)$$

Moreover, $0 \leq \Delta_s \leq \varepsilon$ for $0 < s \leq 1/2$. The constants depend only on d, k, K .

The conclusion controls fourth moments, rather than a distance between the full law and its Gaussian reference. We do not optimize the exponent $1/18$; it results from balancing the entropy deficit against the high-noise remainder in the proof.

Proof. 1. Heat-flow monotonicity. We follow the Fisher-information argument of [1, Proposition 11.1] in the block coordinates introduced above. For a positive density f relative to γ_E , write $I(f) = \int |\nabla f|^2 / f \, d\gamma_E$ for its relative Gaussian Fisher information. The reference measure here is Gaussian, whereas it was Lebesgue measure in (2.24). The same projected-score calculation gives $I(f) \geq (d/2)I(\mathcal{C}_1 f)$ under the stated symmetries. Under Ornstein–Uhlenbeck evolution,

$$\frac{d}{dt}\Delta_{se^{-2t}} = -I(f_{se^{-2t}}) + \frac{d}{2}I(\mathcal{C}_1 f_{se^{-2t}}) \leq 0.$$

Boundedness of $\mathbf{Y}$ justifies differentiation and integration by parts. Lemma 2.3 gives $f_s \rightarrow 1$ in L^2 as $s \downarrow 0$; the relative entropy is at most $\|f_s - 1\|_2^2$, giving $\Delta_s \rightarrow 0$ and $0 \leq \Delta_s \leq \varepsilon$.

2. *Isolating the fourth chaos.* Lemma 2.7 gives $Q \geq 0$ on each positive Hermite degree with the stated symmetries and $Q(h) \geq \kappa \|h\|_2^2$ on degree four.

Use probabilists' multivariate Hermite polynomials normalized by $\int H_\alpha H_\beta d\gamma_E = \alpha! \mathbf{1}_{\alpha=\beta}$. Set

$$g_4 = \sum_{|\alpha|=4} \frac{\mathbb{E}H_\alpha(\mathbf{Y})}{\alpha!} H_\alpha, \quad A_4 = \|g_4\|_2^2.$$

The Hermite generating function gives, for every multi-index,

$$\mathbb{E}H_\alpha(\sqrt{s}\mathbf{Y} + \sqrt{1-s}\mathbf{G}) = s^{|\alpha|/2} \mathbb{E}H_\alpha(\mathbf{Y}).$$

Consequently the fourth chaos of $f_s - 1$ is $s^2 g_4$. The zeroth chaos vanishes because $\int f_s = 1$; the first and second vanish because the smoothed vector is centered and isotropic. If g_n denotes the corresponding coefficient of degree n , then orthogonality and preservation of degree by $\mathcal{C}_1$ give

$$Q(f_s - 1) = \sum_{n \geq 3} s^n Q(g_n).$$

Each summand is nonnegative by Lemma 2.7, including the possible third-degree term. We may therefore retain only degree four. Hence

$$Q(f_s - 1) \geq \kappa s^4 A_4. \quad (2.27)$$

3. *Entropy expansion and choice of the noise level.* Lemma 2.3 gives the bound (2.16) for $0 < s \leq cB^{-2}$. This compares the quadratic deficit with entropy. The global remainder estimate

$$|(1+u)\log(1+u) - u - u^2/2| \leq C|u|^3, \quad u \geq -1,$$

applies to $f_s - 1$ and its marginal. Since conditional expectation contracts L^3 , equations (2.27) and (2.16) yield

$$\varepsilon \geq \Delta_s \geq \frac{1}{2}Q(f_s - 1) - C\|f_s - 1\|_3^3 \geq \frac{\kappa}{2}s^4 A_4 - CB^3 s^{9/2}.$$

Dividing the preceding inequality by s^4 gives $A_4 \leq C(\varepsilon s^{-4} + B^3 s^{1/2})$. For $\varepsilon > 0$, balance the two terms by taking $s = (\varepsilon/B^3)^{2/9}$. After decreasing the constant in $\varepsilon \leq cB^{-6}$, this choice satisfies $s \leq \min(1/2, cB^{-2})$. Both terms then equal a constant times $B^{8/3}\varepsilon^{1/9}$, so $A_4 \leq CB^{8/3}\varepsilon^{1/9}$; for $\varepsilon = 0$, let $s \downarrow 0$. A homogeneous quartic has Hermite degrees four, two and zero. Covariance normalization fixes the last two, so its moment error is at most $C\sqrt{A_4}$, proving (2.26). $\square$

2.2. Empirical stars and spectral inputs

2.2.1. Sampling a fixed graph. Fix a simple d -regular graph $\mathcal{G}$ and an orthonormal selection $\mathbf{u}_1, \dots, \mathbf{u}_k$ of real eigenvectors with eigenvalues $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_k) \in K^k$. The graph, eigenvectors, and eigenvalues are held fixed throughout this subsection. Here $a = 1, \dots, k$ labels the selected eigenpairs; it does not refer to the full ordered spectrum, whose first eigenvalue is d . Choose v_0 uniformly from $\llbracket N \rrbracket$, then order its neighbors $v_1, \dots, v_d$ uniformly among their $d!$ permutations, and put $X_{a,j} = \sqrt{N}\mathbf{u}_a(v_j)$. The empirical star law $\nu_{\mathbf{X}}$ is the distribution under this sampling alone.

Write $\mathbb{E}_\star$ and $\text{Cov}_\star$ for expectation and covariance over the sampled root and neighbor ordering. In particular,

$$\mathbb{E}_\star f(X_{a,0}) = \frac{1}{N} \sum_{v=1}^N f(\sqrt{N}\mathbf{u}_a(v)).$$

For the full star, $\mathbb{E}_\star$ additionally averages the $d!$ neighbor orderings at each vertex. Thus all empirical moments below are finite averages on a fixed graph. The pair (v_0, v_1) is a uniform oriented edge. Probabilities of good graph events will be specified separately.

Lemma 2.9 (Exact empirical covariance).

$$\mathbb{E}_\star X_{a,i} = 0, \quad \mathbb{E}_\star [X_{a,i} X_{b,j}] = \mathbf{1}_{a=b} (\Sigma_{\lambda_a})_{ij}. \quad (2.28)$$

Proof. Every vertex appears as the root with probability $1/N$. It also appears at any fixed neighbor position with probability $1/N$: it has d possible roots, and each such root places it in that position with probability $1/d$. Therefore, for every $0 \leq i \leq d$,

$$\mathbb{E}_\star X_{a,i} X_{b,i} = \sum_{v=1}^N \mathbf{u}_a(v) \mathbf{u}_b(v) = \mathbf{1}_{a=b}.$$

Since each $\lambda_a \in K$ differs from d , its eigenvector is orthogonal to the constant vector. The same uniformity proves $\mathbb{E}_\star X_{a,i} = N^{-1/2} \sum_v \mathbf{u}_a(v) = 0$.

For a root and a neighbor, use the eigenrelation at the root:

$$\mathbb{E}_\star X_{a,0} X_{b,1} = \frac{1}{d} \sum_{v=1}^N \mathbf{u}_a(v) \sum_{w \sim v} \mathbf{u}_b(w) = \frac{\lambda_b}{d} \mathbf{1}_{a=b}.$$

Interchanging the two endpoints gives the same result with the positions reversed. For two distinct neighbor positions, multiply the eigenrelations and subtract terms with the same neighbor:

$$\begin{aligned} d(d-1) \mathbb{E}_\star X_{a,1} X_{b,2} &= \mathbb{E}_\star \left[\left(\sum_{r=1}^d X_{a,r} \right) \left(\sum_{s=1}^d X_{b,s} \right) - \sum_{r=1}^d X_{a,r} X_{b,r} \right] \\ &= (\lambda_a \lambda_b - d) \mathbf{1}_{a=b}. \end{aligned}$$

These three cases give exactly (2.28). Uniform neighbor ordering makes the law invariant under simultaneous permutations of all channels' neighbors. Finally, (v_0, v_1) is a uniform oriented edge, and its reversal is uniform as well. This proves the endpoint symmetry of the joint edge marginal. All these identities are deterministic finite averages on the given graph; no random-graph estimate has been used. $\square$

The eigenrelations imply $\mathbf{X} \in H_\lambda$. Define $\mathbf{Y} = \mathcal{L}_\lambda^{-1} \mathbf{X}$ using the deterministic map of Subsubsection 2.1.6. Equation (2.28) gives

$$\mathbb{E}_\star \mathbf{Y} = 0, \quad \text{Cov}_\star(\mathbf{Y}) = I_E.$$

Uniform neighbor ordering gives neighbor-permutation symmetry, and reversal of a uniform oriented edge gives the R_i -invariance of each V_i marginal after whitening. These are precisely the hypotheses on symmetry and covariance in Proposition 2.8; they hold for every fixed graph and orthonormal selection as above.

2.2.2. Spectral inputs for the random graph. We now let $\mathcal{G}$ be uniformly random. Probabilities in Proposition 2.10 refer to this graph randomness. Once a graph in the stated event is chosen, all its spectral estimates hold deterministically, and the empirical-star expectations remain $\mathbb{E}_\star$. In this subsection, $\lambda_1 \geq \dots \geq \lambda_N$ again denotes the full ordered spectrum of A .

For a fixed energy $E \in (-2\sqrt{d-1}, 2\sqrt{d-1})$, define

$$\sigma = \pi \rho_d(E), \quad a_E = -\frac{(d-2)E}{2(d^2 - E^2)}, \quad x_j = N\sigma(\lambda_j - E).$$

The resolvents and microscopic variables are

$$G(z) = (A - z)^{-1}, \quad m_N(z) = \frac{1}{N} \operatorname{Tr} G(z), \quad \mu_N = \sum_j \delta_{x_j},$$

$$s_N(w) = \frac{m_N(E + w/(N\sigma)) - a_E}{\sigma}.$$

Thus $s_N(w) = \sum_j (x_j - w)^{-1} - a_E/\sigma$. All complex spectral parameters lie off the real axis, with reflection to the lower half-plane. The measure μ_N is exactly the point measure in the main point-process theorem. Its coordinates agree with the intensity- $1/\pi$ convention of [9], and its claimed limit is Θ_1 . The sum here includes all N eigenvalues.

Proposition 2.10 (Spectral and geometric input events). *There are $a, b_0 > 0$ and $C < \infty$ such that, on a graph event of probability $1 - O(N^{-b_0})$,*

$$\max_{2 \leq j \leq N} |\lambda_j - \gamma_j| \leq N^{-a}, \quad \sup_{\|u\|_2=1, Au=\lambda u} \sqrt{N} \|u\|_\infty \leq (\log N)^C. \quad (2.29)$$

For each fixed $0 < c_0 < 1/2$, on such an event all but at most $N^{1/2}$ roots satisfy, uniformly for $\lambda \in K$,

$$\max_{v, w \in B_1(o)} |G_{vw}(\lambda + iN^{-c_0}) - G_{\mathbb{T}_d, vw}(\lambda + iN^{-c_0})| \leq N^{-1/128}. \quad (2.30)$$

The comparison entries use distances zero, one, and two on a tree star. For every fixed $\alpha > 0$, another such event satisfies

$$\operatorname{Im} m_N(x + i\eta) \leq C_K, \quad x \in K, \quad N^{-1+\alpha} \leq \eta \leq 1. \quad (2.31)$$

For every fixed radius, the fraction of roots whose neighborhood is not a tree tends to zero on these events. All spectral-parameter bounds are uniform on their stated domains.

Proof. We state explicitly the normalization change needed to use [23]. If $H = A/\sqrt{d-1}$ and G_H is its resolvent, then

$$G_A(z) = \frac{1}{\sqrt{d-1}} G_H\left(\frac{z}{\sqrt{d-1}}\right).$$

Since d is fixed and K stays a positive distance from the spectral edges, this change only modifies constants. The input theorem is uniform over its spectral domain; evaluating it later at a random eigenvalue therefore requires no additional union bound.

The estimates in [23, Theorems 1.2 and 1.4], with eigenvalues multiplied by $\sqrt{d-1}$, give (2.29) after decreasing a if necessary. For (2.30), take geometric parameter $1/2$ and approximation radius $(\log_{d-1} N)/64$. Except at $N^{1/2}$ roots, the radius $(\log_{d-1} N)/8$ neighborhood is a tree, so the tree extension of the approximation neighborhood of a star is $\mathbb{T}_d$. The error from [23, Theorem 4.2 and (4.2)–(4.6)] is at most

$$C(\log N)^C (N^{-1/64} + N^{-(1-c_0)/2} + N^{-2(1-c_0)/3}) \leq N^{-1/128}.$$

The stability denominator is bounded below in K , and the slack in the exponents absorbs the fixed factors from the adjacency normalization. The same theorem and its averaged estimate (4.7) give (2.31). The geometric event has polynomial failure probability by [23, Proposition 2.1], while the local-law failures conditional on it have arbitrarily large polynomial exponents. In each application we take the intersection of the finitely many required events. Since the estimates are uniform, they apply at graph-dependent eigenvalues and to all orthonormal selections. $\square$

For each graph satisfying (2.29), Lemma 2.5 applies with its adjacency matrix held fixed. Its Gaussian vector is auxiliary noise; the entropy in that lemma is taken over this Gaussian law with the graph fixed.

2.3. Counting and a uniform Gaussian channel

This subsection proves a uniform bound on the number of colorings with a prescribed local profile and uses it to derive an entropy inequality for correlated Gaussian noise. The counting estimate produces a single graph event on which the Gaussian inequality holds for every admissible signal and covariance.

Throughout this subsection, $\mathcal{G}$ is a uniformly random simple d -regular graph on $\llbracket N \rrbracket$, where $d \geq 3$ is fixed and dN is even. All logarithms are natural. A coloring with q colors means an arbitrary map from $\llbracket N \rrbracket$ to $\llbracket q \rrbracket$; it need not be a proper coloring.

For a fixed graph and a fixed coloring, choose a uniform vertex and uniformly order its d neighbors. Let $\mathbf{p}$ be the law of the resulting ordered list of $d+1$ colors, with the root color first. Let $\mathbf{p}_e$ be its marginal on the first two coordinates. Equivalently, $\mathbf{p}_e$ is the law of the colors at the endpoints of a uniform oriented edge. Here the graph and the coloring are fixed: the only randomness is the choice of the root and the ordering of its neighbors.

Write $Z_{\mathbf{p}}$ for the number of colorings of the fixed graph with star law $\mathbf{p}$. The next estimate is a finite- N version of the counting argument in [1, Lemma 6.3], with explicit dependence on the palette size.

Lemma 2.11 (Uniform counting of empirical color profiles). *For every integer $q \geq 1$ and every $t > 0$, except on a graph event of probability at most $C_d e^{-t}$,*

$$Z_{\mathbf{p}} \leq \exp \left(N \left[H(\mathbf{p}) - \frac{d}{2} H(\mathbf{p}_e) \right] + C_d q^{d+1} \log(N+1) + t \right) \quad \text{for every } \mathbf{p}. \quad (2.32)$$

There are at most $(N+1)^{q^{d+1}}$ possible star profiles. The constants depend only on d , and q may depend on N .

Proof. We first work in the configuration model. Each vertex has d labelled half-edges, and we choose a uniform perfect matching of the Nd half-edges. Until we condition on simplicity, loops and multiple edges are allowed. The half-edge labels give a fixed ordering of the d neighbor slots at each vertex. In this proof, $\mathbb{P}_{\text{conf}}$ and $\mathbb{E}_{\text{conf}}$ refer to the uniform matching of these labelled half-edges, before conditioning on simplicity. The palette is fixed during the count, but may vary with N .

Counting ordered profiles. For an ordered star word $s = (s_0, \dots, s_d) \in \llbracket q \rrbracket^{d+1}$, let n_s be the number of vertices with this word in the fixed half-edge ordering. Thus

$$\sum_s n_s = N, \quad \mathbf{p}^{\text{ord}}(s) = \frac{n_s}{N}.$$

For $a, b \in \llbracket q \rrbracket$, set

$$m_{ab} = \sum_{s: s_0=a} n_s \# \{i \in \llbracket d \rrbracket : s_i = b\}.$$

This is the number of half-edges whose incident vertex has color a and whose prescribed neighbor has color b . A necessary consistency condition is

$$m_{ab} = m_{ba}, \quad m_{aa} \text{ is even}, \quad \sum_{a,b} m_{ab} = Nd.$$

Let $Z_{\mathbf{p}^{\text{ord}}}^{\text{ord}}$ be the number of colorings with this ordered profile. For a consistent profile,

$$\mathbb{E}_{\text{conf}} Z_{\mathbf{p}^{\text{ord}}}^{\text{ord}} = \frac{N!}{\prod_s n_s!} \frac{\prod_{a < b} m_{ab}! \prod_a (m_{aa} - 1)!!}{(Nd - 1)!!}. \quad (2.33)$$

Indeed, the first factor assigns the prescribed words to the labelled vertices. Once the words are assigned, they determine the vertex colors and the color demanded by each half-edge. For $a < b$, the m_{ab} half-edges demanding an a - b edge must be matched bijectively to the m_{ba} half-edges demanding a b - a edge, giving $m_{ab}!$ choices. The m_{aa} half-edges demanding an a - a edge can be paired in $(m_{aa} - 1)!!$ ways. Finally, $(Nd - 1)!!$ is the total number of perfect matchings. We use the conventions $0! = 1$ and $(-1)!! = 1$. An inconsistent profile has expected count zero.

The oriented-edge law associated with this ordered profile is

$$\mathbf{p}_e(a, b) = \frac{1}{d} \sum_{i=1}^d \sum_{\substack{s: s_0=a \\ s_i=b}} \mathbf{p}^{\text{ord}}(s) = \frac{m_{ab}}{Nd}.$$

Thus, before symmetrization, $\mathbf{p}_e$ is the average of the root-neighbor marginals over all d slots, rather than necessarily the marginal on the first two coordinates.

Stirling's estimate, uniformly for integers between 0 and Nd , gives

$$\log \frac{N!}{\prod_s n_s!} = NH(\mathbf{p}^{\text{ord}}) + O_d(q^{d+1} \log(N + 1)).$$

For the matching factor, symmetry of (m_{ab}) gives

$$\begin{aligned} & \log \frac{\prod_{a < b} m_{ab}! \prod_a (m_{aa} - 1)!!}{(Nd - 1)!!} \\ &= \frac{1}{2} \sum_{a, b} m_{ab} \log m_{ab} - \frac{Nd}{2} \log(Nd) + O_d(q^2 \log(N + 1)) \\ &= -\frac{Nd}{2} H(\mathbf{p}_e) + O_d(q^2 \log(N + 1)). \end{aligned}$$

Here and below, $0 \log 0 = 0$. Consequently,

$$\mathbb{E}_{\text{conf}} Z_{\mathbf{p}^{\text{ord}}}^{\text{ord}} \leq \exp \left(N \left[H(\mathbf{p}^{\text{ord}}) - \frac{d}{2} H(\mathbf{p}_e) \right] + C_d q^{d+1} \log(N + 1) \right). \quad (2.34)$$

Symmetrization and the simultaneous bound. Averaging the ordered profile over all permutations of the neighbor slots gives the star law used in the statement:

$$\mathbf{p}(s_0, \dots, s_d) = \frac{1}{d!} \sum_{\pi} \mathbf{p}^{\text{ord}}(s_0, s_{\pi(1)}, \dots, s_{\pi(d)}),$$

where the sum is over all permutations of $\llbracket d \rrbracket$. Concavity of entropy gives

$$H(\mathbf{p}) \geq H(\mathbf{p}^{\text{ord}}),$$

and symmetrization preserves the averaged oriented-edge law $\mathbf{p}_e$.

There are at most $(N + 1)^{q^{d+1}}$ ordered count vectors $(n_s)_s$, since each n_s is an integer between 0 and N . Every symmetrized profile is determined by such a vector, so the same bound holds for the number of possible symmetrized profiles. This bound is over all configurations, not just those arising from one fixed graph.

For each symmetrized profile $\mathbf{p}$, sum (2.34) over the ordered profiles whose symmetrization is $\mathbf{p}$. After increasing C_d , we obtain

$$\mathbb{E}_{\text{conf}} Z_{\mathbf{p}} \leq \exp \left(N \left[H(\mathbf{p}) - \frac{d}{2} H(\mathbf{p}_e) \right] + C_d q^{d+1} \log(N + 1) \right).$$

For a fixed $\mathbf{p}$, Markov's inequality therefore gives

$$\begin{aligned} \mathbb{P}_{\text{conf}} \left(Z_{\mathbf{p}} > \exp \left(N \left[H(\mathbf{p}) - \frac{d}{2} H(\mathbf{p}_e) \right] + (C_d + 1) q^{d+1} \log(N + 1) + t \right) \right) \\ \leq e^{-t} (N + 1)^{-q^{d+1}}. \end{aligned}$$

Taking a union bound over all possible profiles proves (2.32) in the configuration model, with failure probability at most e^{-t} .

Conditioning on simplicity. The standard configuration-model enumeration [7] gives

$$\mathbb{P}_{\text{conf}}(\text{simple}) \longrightarrow \exp \left(-\frac{d^2 - 1}{4} \right) > 0.$$

Conditioning on simplicity therefore increases the failure probability by at most a factor depending only on d . Moreover, every simple labelled d -regular graph has exactly $(d!)^N$ half-edge representations: at each vertex, assign its d incident edges bijectively to its d labelled half-edges. Thus the conditional graph law is uniform, which proves the lemma. $\square$

We next adapt the entropy argument of [1, Theorem 6.1] to correlated Gaussian noise on a finite graph. Both the signal and the covariance may depend on the graph. The proof first bounds the entropy of the entire discretized coloring by its empirical star and edge entropies. Concavity handles the star entropy; only the edge entropy requires a concentration estimate.

Lemma 2.12 (A uniform correlated Gaussian channel). *Fix an integer $k \geq 1$ and $\kappa > 0$, and put*

$$m = k(d + 1), \quad \vartheta = \frac{1}{100(m + 1)}.$$

For sufficiently large N , there is a graph event of probability at least

$$1 - C_d \exp(-N^{1-\vartheta})$$

on which the following holds simultaneously for every $\mathbf{x} \in \mathbb{R}^{kN}$ and every covariance matrix Q satisfying

$$\|\mathbf{x}\|_{\infty} \leq \frac{1}{2} N^{\vartheta}, \quad \kappa N^{-\vartheta} I \leq Q \leq N^{\vartheta} I.$$

Fix a graph in this event and an admissible pair $\mathbf{x}, Q$. Index the kN scalar coordinates by (a, v) , and write $\mathbf{x}_v = (x_{a,v})_{a=1}^k$ and $\mathbf{Y}_v = (Y_{a,v})_{a=1}^k$ for the vertex blocks. Introduce an auxiliary Gaussian vector $\mathbf{Y} \sim N(0, Q)$. Independently of $\mathbf{Y}$, choose a uniform vertex v and a uniform ordering $v_1, \dots, v_d$ of its neighbors. Let

$$\begin{aligned} P_S &= \text{Law}(\mathbf{x}_v + \mathbf{Y}_v, \mathbf{x}_{v_1} + \mathbf{Y}_{v_1}, \dots, \mathbf{x}_{v_d} + \mathbf{Y}_{v_d}), \\ P_e &= \text{Law}(\mathbf{x}_v + \mathbf{Y}_v, \mathbf{x}_{v_1} + \mathbf{Y}_{v_1}). \end{aligned}$$

These laws average over both the Gaussian noise and the local sampling, with the graph and $\mathbf{x}, Q$ fixed. Then

$$h(P_S) - \frac{d}{2} h(P_e) \geq \frac{1}{N} h(\mathbf{Y}) - C_{d,k,\kappa} N^{-\vartheta} \log N. \quad (2.35)$$

The same graph event works for all admissible $\mathbf{x}, Q$, including graph-dependent choices.

Proof. Set

$$\delta = N^{-\vartheta}, \quad a_0 = N^{-4\vartheta}, \quad K_0 = \lfloor N^{5\vartheta} \rfloor, \quad q_0 = 2K_0, \quad q = q_0^k.$$

We use the quantizer from Lemma 2.2: each scalar coordinate is divided by a_0 , rounded down, and then clipped to $\{-K_0, \dots, K_0 - 1\}$. A vertex carries k scalar labels, so the vertex palette has $q = q_0^k$ colors.

The choice of ϑ ensures that

$$(5m + 1)\vartheta < 1, \quad (10k + 8)\vartheta < \frac{1}{2}.$$

In particular, the counting error satisfies

$$q^{d+1} \log(N + 1) \leq C_{d,k} N^{5m\vartheta} \log(N + 1) = o(N\delta). \quad (2.36)$$

Apply Lemma 2.11 with this palette and $t = N\delta = N^{1-\vartheta}$. Fix a graph in the resulting simultaneous event, and then fix any admissible $\mathbf{x}, Q$. From this point onward, $\mathbb{E}_{\mathbf{Y}}$, $\mathbb{P}_{\mathbf{Y}}$ and $\text{Var}_{\mathbf{Y}}$ refer only to $\mathbf{Y} \sim N(0, Q)$, with $\mathcal{G}$, $\mathbf{x}$ and Q held fixed. The empirical profiles already include the finite averages over vertices and neighbor orderings.

1. *Entropy of the entire coloring.* Let

$$\mathbf{J} = \left\lfloor \frac{\mathbf{x} + \mathbf{Y}}{a_0} \right\rfloor$$

coordinatewise, and let $\mathbf{C}$ be obtained by clipping each coordinate to $\{-K_0, \dots, K_0 - 1\}$. Thus $\mathbf{J} = J(\mathbf{x} + \mathbf{Y})$ and $\mathbf{C} = T\mathbf{J}$. Write $\mathbf{C}(v)$ for the k -tuple of scalar labels at vertex v . For each realization of $\mathbf{Y}$, let $\hat{\mathbf{p}}_S, \hat{\mathbf{p}}_e$ be the star and oriented-edge laws of the resulting fixed coloring $\mathbf{C}$. Explicitly, for a star word $\mathbf{s}$ of $d + 1$ vertex colors and vertex colors $\mathbf{a}, \mathbf{b}$,

$$\begin{aligned} \hat{\mathbf{p}}_S(\mathbf{s}) &= \frac{1}{Nd!} \sum_{v_0=1}^N \sum_{\substack{(v_1, \dots, v_d) \text{ orders} \\ \text{the neighbors of } v_0}} \mathbf{1}_{\{(\mathbf{C}(v_0), \dots, \mathbf{C}(v_d)) = \mathbf{s}\}}, \\ \hat{\mathbf{p}}_e(\mathbf{a}, \mathbf{b}) &= \frac{1}{Nd} \sum_{v=1}^N \sum_{u \sim v} \mathbf{1}_{\{\mathbf{C}(v) = \mathbf{a}, \mathbf{C}(u) = \mathbf{b}\}}. \end{aligned}$$

These formulas already perform the root and neighbor-ordering averages. The profiles are random only because the coloring depends on $\mathbf{Y}$.

The star profile is a function of the coloring and takes at most $(N + 1)^{q^{d+1}}$ possible values, by Lemma 2.11. Therefore

$$H(\text{Law}_{\mathbf{Y}}(\hat{\mathbf{p}}_S)) \leq q^{d+1} \log(N + 1).$$

This is the entropy of the random profile as an outcome, not the random number $H(\hat{\mathbf{p}}_S)$ obtained by computing the entropy of its realized entries.

For each profile $\mathbf{p}$ with positive probability, the conditional coloring is supported on the $Z_{\mathbf{p}}$ colorings having that profile. A distribution on at most $Z_{\mathbf{p}}$ points has entropy at most $\log Z_{\mathbf{p}}$, whether or not it is uniform. Hence

$$\begin{aligned} H(\mathbf{C} \mid \hat{\mathbf{p}}_S = \mathbf{p}) &\leq \log Z_{\mathbf{p}}, \\ H(\mathbf{C} \mid \hat{\mathbf{p}}_S) &= \sum_{\mathbf{p}} \mathbb{P}_{\mathbf{Y}}(\hat{\mathbf{p}}_S = \mathbf{p}) H(\mathbf{C} \mid \hat{\mathbf{p}}_S = \mathbf{p}) \\ &\leq \mathbb{E}_{\mathbf{Y}} \log Z_{\hat{\mathbf{p}}_S}. \end{aligned}$$

The sum runs only over profiles of positive probability, for which $Z_{\mathbf{p}} \geq 1$. Since the profile is determined by the coloring, its conditional entropy given the coloring is zero. The chain rule

for the pair consisting of the coloring and its profile gives

$$\begin{aligned}
H(\mathbf{C}) &= H(\text{Law}_{\mathbf{Y}}(\widehat{\mathbf{p}}_S)) + H(\mathbf{C} \mid \widehat{\mathbf{p}}_S) \\
&\leq q^{d+1} \log(N+1) + \mathbb{E}_{\mathbf{Y}} \log Z_{\widehat{\mathbf{p}}_S} \\
&\leq N \mathbb{E}_{\mathbf{Y}} \left[H(\widehat{\mathbf{p}}_S) - \frac{d}{2} H(\widehat{\mathbf{p}}_e) \right] + C_d q^{d+1} \log(N+1) + N\delta.
\end{aligned} \tag{2.37}$$

The last line applies the simultaneous counting bound to every realized profile and absorbs the profile-entropy bound into C_d . Inside the expectation, for example, $H(\widehat{\mathbf{p}}_S) = -\sum_s \widehat{\mathbf{p}}_S(s) \log \widehat{\mathbf{p}}_S(s)$. We compute this number for the realized coloring before averaging over $\mathbf{Y}$.

For the unbounded grid coloring, the cellwise Jensen argument used in Lemma 2.2 gives

$$H(\mathbf{J}) + kN \log a_0 \geq h(\mathbf{x} + \mathbf{Y}) = h(\mathbf{Y}).$$

Indeed, each grid cell has volume a_0^{kN} , and the difference on the left is the relative entropy of the Gaussian density with respect to its average on each cell.

Clipping acts coordinatewise. Hence conditional subadditivity and the fact that conditioning reduces discrete entropy give

$$\begin{aligned}
H(\mathbf{J}) - H(\mathbf{C}) &= H(\mathbf{J} \mid \mathbf{C}) \\
&\leq \sum_{a=1}^k \sum_{v=1}^N H(\mathbf{J}_{a,v} \mid \mathbf{C}) \\
&\leq \sum_{a=1}^k \sum_{v=1}^N H(\mathbf{J}_{a,v} \mid \mathbf{C}_{a,v}) \\
&\leq CkN^{1+5\vartheta} e^{-cN^\vartheta}.
\end{aligned}$$

The last inequality is the scalar clipping estimate from the proof of Lemma 2.2. It applies uniformly to each coordinate, since its mean has absolute value at most $N^\vartheta/2$ and its variance is at most N^ϑ . No independence between coordinates is needed.

Combining these estimates with (2.37) and (2.36), we obtain

$$\mathbb{E}_{\mathbf{Y}} \left[H(\widehat{\mathbf{p}}_S) - \frac{d}{2} H(\widehat{\mathbf{p}}_e) \right] \geq \frac{h(\mathbf{Y})}{N} - k \log a_0 - C_{d,k} \delta. \tag{2.38}$$

All discrete and differential entropies used here are finite by the Gaussian density and moment bounds.

2. *Passing to the averaged profiles.* Put

$$\overline{\mathbf{p}}_S = \mathbb{E}_{\mathbf{Y}} \widehat{\mathbf{p}}_S, \quad \overline{\mathbf{p}}_e = \mathbb{E}_{\mathbf{Y}} \widehat{\mathbf{p}}_e.$$

These are exactly the laws obtained by applying the clipped quantizer to P_S and P_e , respectively.

Concavity gives $H(\overline{\mathbf{p}}_S) \geq \mathbb{E}_{\mathbf{Y}} H(\widehat{\mathbf{p}}_S)$. Consequently,

$$\begin{aligned}
H(\overline{\mathbf{p}}_S) - \frac{d}{2} H(\overline{\mathbf{p}}_e) &\geq \mathbb{E}_{\mathbf{Y}} \left[H(\widehat{\mathbf{p}}_S) - \frac{d}{2} H(\widehat{\mathbf{p}}_e) \right] \\
&\quad - \frac{d}{2} [H(\overline{\mathbf{p}}_e) - \mathbb{E}_{\mathbf{Y}} H(\widehat{\mathbf{p}}_e)].
\end{aligned} \tag{2.39}$$

Thus we only need to control the entropy gap for the edge profile. There is no need to prove concentration of the star profile.

3. *Smoothing and concentration of the edge profile.* The hard quantizer is discontinuous, so we first replace its label indicators by Lipschitz functions. Set

$$\ell = \frac{\delta^{3/2}}{10q_0}.$$

The scalar clipped quantizer has $q_0 - 1$ jump points, spaced a_0 apart. Since

$$\frac{\ell}{a_0} = O(N^{-5\vartheta/2}),$$

the intervals of radius ℓ around these jumps are disjoint for sufficiently large N .

For each scalar label $b \in \{-K_0, \dots, K_0 - 1\}$, let $\varphi_b : \mathbb{R} \rightarrow [0, 1]$ agree with the hard label indicator away from these intervals. At the jump point ba_0 between labels $b - 1$ and b , replace the two indicators on $[ba_0 - \ell, ba_0 + \ell]$ by

$$\varphi_{b-1}(z) = \frac{ba_0 + \ell - z}{2\ell}, \quad \varphi_b(z) = \frac{z - ba_0 + \ell}{2\ell},$$

and set all other labels equal to zero there. Thus

$$\sum_b \varphi_b(z) = 1, \quad 0 \leq \varphi_b(z) \leq 1, \quad \text{Lip}(\varphi_b) \leq \frac{1}{2\ell}.$$

An edge word consists of $2k$ scalar labels. For $w = (w_1, \dots, w_{2k})$, define

$$S_w(\mathbf{Y}) = \frac{1}{Nd} \sum_{\substack{u, v \in \llbracket N \rrbracket \\ u \sim v}} \prod_{r=1}^k \varphi_{w_r}(x_{r,u} + Y_{r,u}) \varphi_{w_{k+r}}(x_{r,v} + Y_{r,v}),$$

where the sum is over oriented edges, so each unoriented edge appears twice. Write $\mathbf{S} = (S_w)_w$. The partition-of-unity property implies that $\mathbf{S}$ is a probability vector for every realization of $\mathbf{Y}$.

For a fixed scalar coordinate $Y_{r,v}$, only the d oriented edges starting at v and the d oriented edges ending at v contribute to its derivative. Since all other factors are at most one, almost everywhere

$$\left| \frac{\partial S_w}{\partial Y_{r,v}} \right| \leq \frac{2d}{Nd} \frac{1}{2\ell} = \frac{1}{N\ell}.$$

Summing over the kN scalar coordinates gives

$$|\nabla S_w|^2 \leq \frac{k}{N\ell^2}.$$

By the Gaussian Poincaré inequality (2.5) and $Q \leq \delta^{-1}I$,

$$\text{Var}_{\mathbf{Y}}(S_w) \leq \mathbb{E}_{\mathbf{Y}}[(\nabla S_w)^T Q \nabla S_w] \leq \frac{k}{N\delta\ell^2}.$$

The inequality applies to these Lipschitz functions by smooth approximation.

There are q_0^{2k} edge words. Cauchy–Schwarz therefore gives

$$\begin{aligned} \mathbb{E}_{\mathbf{Y}} \|\mathbf{S} - \mathbb{E}_{\mathbf{Y}} \mathbf{S}\|_1 &\leq \sum_w \sqrt{\text{Var}_{\mathbf{Y}}(S_w)} \\ &\leq \frac{C_k q_0^{2k}}{\sqrt{N\delta}\ell} = \frac{C_k q_0^{2k+1}}{\sqrt{N}\delta^2} \\ &\leq C_k N^{-1/2+(10k+7)\vartheta} \leq C_k \delta. \end{aligned} \tag{2.40}$$

The last inequality uses $(10k + 8)\vartheta < 1/2$. This estimate uses the covariance bound, not independence of the coordinates of $\mathbf{Y}$.

We now compare the soft profile with the hard profile. Each scalar coordinate of $\mathbf{x} + \mathbf{Y}$ has variance at least $\kappa\delta$, and hence its density is bounded by $(2\pi\kappa\delta)^{-1/2}$. The total length of the transition intervals is at most $2q_0\ell$. Thus, uniformly in v, r ,

$$\mathbb{P}_{\mathbf{Y}}(x_{r,v} + Y_{r,v} \text{ belongs to a transition interval}) \leq \frac{2q_0\ell}{\sqrt{2\pi\kappa\delta}} \leq C_\kappa\delta.$$

For a fixed edge, the hard word and its soft distribution agree unless at least one of the $2k$ scalar coordinates belongs to a transition interval. Their ℓ^1 distance is always at most two, so it is bounded by twice the number of coordinates in transition intervals. Averaging over oriented edges and then over $\mathbf{Y}$ gives

$$\mathbb{E}_{\mathbf{Y}}\|\hat{\mathbf{p}}_e - \mathbf{S}\|_1 \leq C_{k,\kappa}\delta.$$

Jensen's inequality also gives

$$\|\bar{\mathbf{p}}_e - \mathbb{E}_{\mathbf{Y}}\mathbf{S}\|_1 \leq \mathbb{E}_{\mathbf{Y}}\|\hat{\mathbf{p}}_e - \mathbf{S}\|_1.$$

Combining these inequalities with (2.40), we obtain

$$\begin{aligned} \mathbb{E}_{\mathbf{Y}}\|\hat{\mathbf{p}}_e - \bar{\mathbf{p}}_e\|_1 &\leq \mathbb{E}_{\mathbf{Y}}\|\hat{\mathbf{p}}_e - \mathbf{S}\|_1 + \mathbb{E}_{\mathbf{Y}}\|\mathbf{S} - \mathbb{E}_{\mathbf{Y}}\mathbf{S}\|_1 + \|\mathbb{E}_{\mathbf{Y}}\mathbf{S} - \bar{\mathbf{p}}_e\|_1 \\ &\leq C_{k,\kappa}\delta. \end{aligned} \tag{2.41}$$

4. Controlling the edge entropy gap. The edge profiles are probability vectors on an alphabet of size q_0^{2k} . Apply the entropy continuity bound (2.2), with total variation distance $\frac{1}{2}\|\hat{\mathbf{p}}_e - \bar{\mathbf{p}}_e\|_1$. Concavity of entropy and then concavity of H_2 give

$$\begin{aligned} 0 &\leq H(\bar{\mathbf{p}}_e) - \mathbb{E}_{\mathbf{Y}}H(\hat{\mathbf{p}}_e) \\ &\leq \mathbb{E}_{\mathbf{Y}}|H(\bar{\mathbf{p}}_e) - H(\hat{\mathbf{p}}_e)| \\ &\leq H_2\left(\frac{1}{2}\mathbb{E}_{\mathbf{Y}}\|\hat{\mathbf{p}}_e - \bar{\mathbf{p}}_e\|_1\right) + \frac{1}{2}\log(q_0^{2k})\mathbb{E}_{\mathbf{Y}}\|\hat{\mathbf{p}}_e - \bar{\mathbf{p}}_e\|_1 \\ &\leq C_{k,\kappa}\delta \log N. \end{aligned}$$

In the last step we used (2.41), $\log q_0 = O(\log N)$, and $H_2(u) \leq u \log(e/u)$ for sufficiently small u .

Substituting this estimate and (2.38) into (2.39), we find

$$H(\bar{\mathbf{p}}_S) - \frac{d}{2}H(\bar{\mathbf{p}}_e) \geq \frac{h(\mathbf{Y})}{N} - k \log a_0 - C_{d,k,\kappa}\delta \log N. \tag{2.42}$$

5. Returning to differential entropy. With the graph fixed, P_S and P_e are finite Gaussian mixtures, indexed by the sampled stars and oriented edges. The graph is simple, so the vertices in a sampled star are distinct. Consequently, each component covariance is a principal submatrix of Q , up to a permutation of coordinates. It therefore lies between $\kappa\delta I$ and $\delta^{-1}I$. The component means also satisfy the coordinate bound $N^\vartheta/2$.

Lemma 2.2 thus applies in dimensions $m = k(d + 1)$ and $2k$. Since $\bar{\mathbf{p}}_S$ and $\bar{\mathbf{p}}_e$ are the corresponding clipped quantization laws, it gives

$$\begin{aligned} &|H(\bar{\mathbf{p}}_S) + m \log a_0 - h(P_S)| + |H(\bar{\mathbf{p}}_e) + 2k \log a_0 - h(P_e)| \\ &\leq C_{d,k,\kappa}N^{-7\vartheta} + C_{d,k}N^{5\vartheta}e^{-c_{d,k}N^\vartheta} \leq C_{d,k,\kappa}\delta. \end{aligned}$$

The logarithmic terms combine according to

$$m - \frac{d}{2}(2k) = k(d+1) - dk = k.$$

Hence

$$\begin{aligned} h(P_S) - \frac{d}{2}h(P_e) &\geq H(\bar{\mathbf{p}}_S) - \frac{d}{2}H(\bar{\mathbf{p}}_e) + k \log a_0 - C_{d,k,\kappa}\delta \\ &\geq \frac{h(\mathbf{Y})}{N} - C_{d,k,\kappa}\delta \log N, \end{aligned}$$

where the last inequality follows from (2.42). This is (2.35).

Finally, the only exceptional graph event used in the proof is the simultaneous counting event from Lemma 2.11. After fixing a graph in that event, every subsequent estimate holds for each admissible $\mathbf{x}, Q$. In particular, we have not introduced an exceptional event depending on $\mathbf{x}$ or Q , nor taken a union bound over these parameters. This proves the asserted uniformity, including for graph-dependent choices. $\square$

2.4. The graph entropy deficit at a fixed noise level

The lower entropy comparison for smooth typical eigenvector processes is given in [1, Theorem 10.3]. For our finite- N argument, the error must be polynomially small and uniform over exact eigenvectors. Lemma 2.12, applied with graph-dependent resolvent noise, gives the required comparison at the single noise level used in Proposition 2.8.

Fix the graph and the selected eigenvectors. In this subsection $\mathbf{Y} = \mathcal{L}_\lambda^{-1}\mathbf{X}$ is the whitened empirical star, not the global Gaussian noise from Lemma 2.12. The independent Gaussian star is $\mathbf{Z}_\lambda$. Whitening the star and edge and then dividing by $\sqrt{2}$ gives

$$\mathcal{F}_\lambda(\mathbf{Z}_\lambda) - \mathcal{F}_\lambda(\mathbf{X} + \mathbf{Z}_\lambda) = \Delta_{1/2}. \quad (2.43)$$

Here $\Delta_{1/2}$ is the relative-entropy deficit defined in Subsubsection 2.1.7. The whitening Jacobians cancel between the two laws, as in [1, proof of Theorem 11.2 and Lemma B.2]. The Gaussian quadratic terms cancel because the normalized law has covariance I_E , and $dk = (d/2)(2k)$ makes the star-edge functional invariant under common scalar dilation.

Proposition 2.13 (Polynomial entropy deficit for empirical stars). *For fixed d, k, K , there are $b, \gamma > 0$ and $C < \infty$ such that, on a graph event of probability at least $1 - CN^{-b}$, every orthonormal selection of k real eigenvectors with eigenvalues in K satisfies*

$$0 \leq \mathcal{F}_\lambda(\mathbf{Z}_\lambda) - \mathcal{F}_\lambda\left(\frac{\mathbf{X} + \mathbf{Z}_\lambda}{\sqrt{2}}\right) \leq N^{-\gamma}. \quad (2.44)$$

For each graph in this event, $\mathbf{X}$ has the empirical law under $\mathbb{E}_$, and $\mathbf{Z}_\lambda$ is an independent auxiliary Gaussian. The entropies average these two sources of randomness while holding the graph and eigenvectors fixed.*

Proof. The proof compares the entropy of a Gaussian perturbation in two ways. Globally, its entropy is a log determinant of the resolvent covariance. Locally, the same noise looks like a Gaussian wave in the tangent space H_λ , with a much smaller Gaussian component normal to that space. The normal entropy is singular as $\eta \rightarrow 0$, but it is explicitly computable and cancels the corresponding global term. We keep this normal component until the cancellation has been performed.

Let $a_* > 0$ be the coarse-rigidity exponent in Proposition 2.10, and let $\vartheta > 0$ be the exponent in Lemma 2.12. Fix

$$0 < c < \min(a_*/2, \vartheta/4, 1/2), \quad \eta = N^{-c}.$$

Work on the intersection of the counting and spectral events for this c ; its complement has polynomially small graph probability. Fix a graph in this event and a selected eigenvector tuple for the remainder of the proof. All local-law errors below use the explicit exponent $1/128$ from (2.30).

For $1 \leq a \leq k$ put

$$s_a = \pi \rho_d(\lambda_a), \quad G_a = (A - \lambda_a - i\eta)^{-1}, \quad Q_a = \frac{1}{s_a} \operatorname{Im} G_a = \frac{\eta}{s_a} ((A - \lambda_a)^2 + \eta^2)^{-1}.$$

Independently of the empirical root and neighbor ordering, let ζ^a be independent centered Gaussian vectors on $\mathbb{R}^N$ with covariances Q_a . Write $\zeta = (\zeta^1, \dots, \zeta^k)$ for the full noise vector. Since s_a is bounded above and below on K , $c_K \eta I \leq Q_a \leq C_K \eta^{-1} I$. Their direct sum satisfies Lemma 2.12 for all large N . The signal bound follows from delocalization for $x_{a,v} = \sqrt{N} \mathbf{u}_a(v)$. Averaging $\mathbf{x} + \zeta$ over the full star and edge samples and the auxiliary Gaussian noise, always with the graph fixed, gives

$$h((\mathbf{x} + \zeta)_S) - \frac{d}{2} h((\mathbf{x} + \zeta)_e) \geq \frac{1}{N} h(\zeta^1, \dots, \zeta^k) - CN^{-\vartheta} \log N. \quad (2.45)$$

Lemma 2.5, with the scalar covariance factor $1/s_a$, yields

$$\frac{1}{N} h(\zeta^1, \dots, \zeta^k) = \sum_a \left(\frac{1}{2} \log \frac{2\pi e \eta}{s_a} - L_d(\lambda_a) \right) + O(N^{-a^*}/\eta + \eta + N^{-1} \log N). \quad (2.46)$$

On a good-root star from (2.30), $\mathbf{n}_a = (-\lambda_a, 1, \dots, 1) = (A - \lambda_a) \mathbf{e}_o$ when embedded in the graph. The useful identities are exact, not consequences of the local law. Writing $T = A - \lambda_a$, spectral calculus gives

$$T \operatorname{Im}(T - i\eta)^{-1} = \eta \operatorname{Re}(T - i\eta)^{-1}, \quad T \operatorname{Im}(T - i\eta)^{-1} T = \eta I - \eta^2 \operatorname{Im}(T - i\eta)^{-1}.$$

Take the o, v or o, o entry, respectively, and divide by s_a . This gives

$$\begin{aligned} \mathbf{n}_a^T (Q_a)_S \mathbf{n}_a &= \frac{\eta}{s_a} (1 - \eta \operatorname{Im}(G_a)_{oo}), \\ \mathbf{n}_a^T (Q_a)_S \mathbf{e}_v &= \frac{\eta}{s_a} \operatorname{Re}(G_a)_{ov}, \quad v \in S. \end{aligned} \quad (2.47)$$

On the tree, the boundary imaginary part of the resolvent is $s_a \Sigma_{\lambda_a}$ on a star. Indeed, the diagonal is s_a , the resolvent equation at the root gives the root-neighbor value $s_a \lambda_a / d$, and applying it at a neighbor gives the distance-two value $s_a (\lambda_a^2 - d) / (d(d-1))$. The explicit tree recursion varies by $O_K(\eta)$ on this fixed star between height zero and height η . Thus

$$(Q_a)_S = \Sigma_{\lambda_a} + O_K(N^{-1/128} + \eta).$$

Let J be the uniform root conditioned to be good, together with its uniform neighbor ordering; this is conditioning the sampling on the fixed graph. Let $\mathbf{X}^g$ and $\mathbf{X}^g$ be the signal star and the noisy star sampled with this conditional root law. Write $(\mathbf{x}_\parallel, \mathbf{x}_\perp)$ for $\mathbf{X}^g$ in orthonormal tangent coordinates on H_λ and unit normal coordinates $\mathbf{n}_a / \sqrt{d + \lambda_a^2}$. Put

$$\Sigma_\parallel = \bigoplus_a \Sigma_{\lambda_a} |_{H_{\lambda_a}}, \quad Q_* = \operatorname{diag}_a \frac{\eta}{s_a(d + \lambda_a^2)}.$$

The label J specifies both the good root and its neighbor ordering. Given J , the signal is deterministic and only the auxiliary Gaussian noise remains random. Without conditioning on

J , local entropies average over this finite sampling law and the Gaussian noise. Its conditional covariance has blocks

$$\begin{pmatrix} Q_{\parallel,J} & Q_{\times,J} \\ Q_{\times,J}^T & Q_{\perp,J} \end{pmatrix}, \quad \begin{aligned} Q_{\parallel,J} &= \Sigma_{\parallel} + O(N^{-1/128} + \eta), \\ Q_{\times,J} &= O(\eta), \\ Q_{\perp,J} &= Q_*^{1/2}(I + O(\eta))Q_*^{1/2}. \end{aligned}$$

Equation (2.47) and bounded good-root resolvent entries give the last two estimates. The eigenrelation gives $\mathbb{E}[\mathbf{X}_{\perp} | J] = 0$, and $Q_{\parallel,J}^{-1} = O(1)$ gives the conditional normal covariance

$$Q_{\perp,J} - Q_{\times,J}^T Q_{\parallel,J}^{-1} Q_{\times,J} = Q_*^{1/2}(I + O(\eta))Q_*^{1/2}.$$

Equations (2.3)–(2.4) and the covariance averaged over J bound the two outer terms in

$$h(\mathbf{X}_{\perp} | \mathbf{X}_{\parallel}, J) \leq h(\mathbf{X}_{\perp} | \mathbf{X}_{\parallel}) \leq h(\mathbf{X}_{\perp}).$$

For the leftmost term, condition on J and on the tangent coordinate, and use the Gaussian Schur complement. Its entropy is $\frac{1}{2} \log \det(2\pi e Q_*) + O(\eta)$ uniformly in J . For the rightmost term, its covariance is $Q_*^{1/2}(I + O(\eta))Q_*^{1/2}$ because the normal mean vanishes for every J . Gaussian maximum entropy gives the same expression as an upper bound. The entropy sandwich therefore gives

$$h(\mathbf{X}_{\perp} | \mathbf{X}_{\parallel}) = \frac{1}{2} \log \det(2\pi e Q_*) + O(\eta).$$

In this argument it is important that the signal is exactly tangent. An uncontrolled normal component of the signal would contribute to the covariance and destroy this comparison. Apply Lemma 2.1 directly to the tangential law; its constant is independent of the signal means. The entropy chain rule gives

$$\begin{aligned} h(\mathbf{X}^g) &= h_{H_{\lambda}}(\mathbf{X}^g + \mathbf{Z}_{\lambda}) + \frac{1}{2} \sum_a \log \frac{2\pi e \eta}{s_a(d + \lambda_a^2)} + O(N^{-1/128} + \eta), \\ h((\mathbf{X}^g)_e) &= h((\mathbf{X}^g + \mathbf{Z}_{\lambda})_e) + O(N^{-1/128} + \eta). \end{aligned} \tag{2.48}$$

For the edge law, use the root and its first uniformly ordered neighbor in the same good-root sample. Its covariance is uniformly nondegenerate, so the same comparison applies directly.

We now restore the omitted fraction $p \leq N^{-1/2}$ of roots. The actual component covariances are principal submatrices of $\bigoplus_a Q_a$ because star vertices are distinct. Their eigenvalues lie between $c_K \eta$ and $C_K \eta^{-1}$; the ideal covariances on H_{λ} are uniformly bounded and nondegenerate. The means have at most polynomial size. Entropy concavity and the Gaussian covariance bound therefore give $|h| \leq C \log N$ for every conditional mixture. The mixture entropy bounds (2.1), applied to the good-root and bad-root components, now restore the full sampling law. Thus (2.48) holds with $\mathbf{X}^g$ and $\mathbf{X}^g$ replaced by the full-sample stars $(\mathbf{x} + \boldsymbol{\zeta})_S$ and $\mathbf{X}$, after adding an error of at most $Cp \log N + H_2(p) = O(N^{-1/2} \log N)$. The star entropies throughout use their respective ambient spaces: the actual law has full dimension $k(d+1)$ and the ideal tangential law has dimension dk .

The local comparison expresses the left side of (2.45) as $\mathcal{F}_{\lambda}(\mathbf{X} + \mathbf{Z}_{\lambda})$ plus the explicit normal entropy in (2.48), up to the displayed errors. Subtract that normal entropy from (2.46), channel by channel:

$$\frac{1}{2} \log \frac{2\pi e \eta}{s_a} - L_d(\lambda_a) - \frac{1}{2} \log \frac{2\pi e \eta}{s_a(d + \lambda_a^2)} = \frac{1}{2} \log(d + \lambda_a^2) - L_d(\lambda_a).$$

Thus all singular normal terms cancel exactly, leaving

$$\begin{aligned}\mathcal{F}_\lambda(\mathbf{X} + \mathbf{Z}_\lambda) &\geq \sum_a \left(\frac{1}{2} \log(d + \lambda_a^2) - L_d(\lambda_a) \right) \\ &\quad - C(N^{-\vartheta} \log N + N^{-a_*+c} + N^{-1/128} + N^{-c} + N^{-1/2} \log N).\end{aligned}$$

The sum is $\mathcal{F}_\lambda(\mathbf{Z}_\lambda)$ by Lemma 2.4, specifically (2.21). Any $0 < \gamma < \min(\vartheta, a_* - c, 1/128, c, 1/2)$ absorbs constants and logarithms for large N . The Ornstein–Uhlenbeck argument in the proof of Proposition 2.8 shows that Δ_s increases from zero as s increases. Thus (2.43) gives the lower bound. Scale invariance permits division by $\sqrt{2}$ in (2.44). All estimates hold on the same graph event, uniformly in the selected vectors. $\square$

2.5. Fourth moments and the qualitative Gaussian limit

We return to the physical star coordinates to prove Theorem 1.2 and Corollary 1.3.

Proof of Theorem 1.2. The abstract fourth-moment estimate applies to the whitened empirical law, while the theorem is stated in physical coordinates. We check both the common graph event and this change of coordinates explicitly. Intersect the events of Proposition 2.13 for $1 \leq k \leq 4$ with the spectral event, and use a common $\gamma > 0$. Fix any graph in this event. Under the empirical sampling law $\mathbb{E}_*$, $\mathbf{Y} = \mathcal{L}_\lambda^{-1} \mathbf{X}$ has the required symmetries, mean zero and covariance I . Delocalization and the uniform bound on the whitening map give $|\mathbf{Y}| \leq B = (\log N)^C$ for every selection.

Equations (2.43) and (2.44) give $0 \leq \Delta_{1/2} \leq N^{-\gamma}$ in the notation of Proposition 2.8. For large N this is at most cB^{-6} . The proposition bounds every whitened quartic moment error by $CB^{4/3}N^{-\gamma/18}$. Composing a physical coordinate monomial with $\mathcal{L}_\lambda$ gives a homogeneous quartic with uniformly bounded coefficients: each of its four linear factors is a row of $\mathcal{L}_\lambda$, whose coefficients are bounded on K^k . The dimension is bounded because $k \leq 4$. Dividing by a fixed coefficient bound reduces to the normalization in Proposition 2.8, and multiplying back changes only C . Therefore any fixed $0 < \chi < \gamma/18$ gives (1.2) after absorbing the polylogarithmic factor, on the same graph event for all choices. $\square$

Proof of Corollary 1.3. Work first on the event of Theorem 1.2. Its scalar case applies simultaneously to every eigenvector, so it controls each coordinate of a tuple of any fixed size k . Taking four copies of that coordinate in (1.2) gives, for all large N ,

$$\mathbb{E}_*|X_{a,j}|^4 \leq 4, \quad \mathbb{E}_*|\mathbf{X}|^4 \leq 4(k(d+1))^2.$$

These are empirical moments under $\mathbb{E}_*$: their only randomness is the uniform vertex and neighbor ordering on the fixed graph. Convergence in probability in (1.3) instead refers to the random graph. The second inequality follows from $(\sum_{a,j} X_{a,j}^2)^2 \leq k(d+1) \sum_{a,j} X_{a,j}^4$. Thus the squares and quadratic products of all star coordinates are uniformly integrable, uniformly over the selected vectors. For example,

$$\mathbb{E}_*[\|\mathbf{X}\|^2 \mathbf{1}_{|\mathbf{X}| > R}] \leq R^{-2} \mathbb{E}_*|\mathbf{X}|^4 \leq C_{d,k} R^{-2}.$$

A product of two coordinates is bounded in absolute value by the sum of their squares divided by two, so it has the same tail control. This is the exact step that rules out mass escaping to rare, very large coordinates; convergence in distribution alone would not preserve variance.

We now apply the qualitative joint neighborhood conclusion of [1]. It is uniform over fixed orthonormal tuples of almost eigenvectors at a common energy. In the present setting, every unit linear combination $\sum_a c_a \mathbf{u}_a$ satisfies

$$\left\| (A - \lambda_1) \sum_a c_a \mathbf{u}_a \right\|_2^2 = \sum_a c_a^2 (\lambda_a - \lambda_1)^2 \leq \delta_N^2, \quad \sum_a c_a^2 = 1.$$

Hence the shrinking eigenvalue diameter is precisely the almost-eigenvector condition in that result. Its joint neighborhood conclusion applies in particular to all channels of the sampled star.

To see both the normalization and uniformity, argue by contradiction. If (1.3) fails, there are a subsequence and selections on graphs in the fourth-moment and Backhausz–Szegedy events for which the Wasserstein distance stays bounded away from zero. The uniform formulation of their theorem permits these graph-dependent choices; a diagonal choice of its accuracy parameters makes all of them tend to zero along this subsequence. Compactness of K and the diameter assumption allow a further subsequence with $\lambda_a \rightarrow \lambda$ for every a . The displayed fourth-moment bound gives tightness of the empirical star laws. Backhausz–Szegedy identifies each subsequential limit as a centered joint Gaussian wave, possibly with a loss of covariance. Uniform integrability and (2.28) now give its covariance exactly:

$$\lim_{N \rightarrow \infty} \mathbb{E}_* X_{a,i} X_{b,j} = \mathbf{1}_{a=b} (\Sigma_\lambda)_{ij}.$$

In particular, each root has variance one, and distinct channels are independent because the joint limit is Gaussian. The limiting star law is therefore $\nu_{\mathbf{Z}_{(\lambda, \dots, \lambda)}}$.

Weak convergence together with uniform integrability of the squared norm implies convergence in W_2 . One way to see this implication is to couple a weakly convergent subsequence almost surely in the fixed finite-dimensional Euclidean space. The squared distance is bounded by twice the sum of the two squared norms, whose tails are uniformly integrable. Truncation followed by dominated convergence then gives a coupling with mean squared distance tending to zero. The Gaussian comparison laws $\nu_{\mathbf{Z}_\lambda}$ also converge in W_2 by continuity of $\mathcal{L}_\lambda$. The triangle inequality contradicts the positive discrepancy. The fourth-moment event has polynomially small complement, while the qualitative Backhausz–Szegedy events have probability tending to one; no rate for their intersection is required. For a graph-measurable tuple converging in probability to a common energy, choose a deterministic $\delta_N \downarrow 0$ that bounds its eigenvalue diameter with probability tending to one, and apply (1.3). $\square$

Remark 2.14. The scalar fourth-moment bound alone prevents loss of variance in the qualitative step. Mixed fourth moments are needed later to control covariances between quadratic marks at different eigenvectors. Those estimates retain their polynomial rate for arbitrary bulk energies. Restricting the coupling in Corollary 1.3 to edge coordinates gives conditional bounded-Lipschitz convergence for every fixed graph-measurable selection at nearby energies. All uses of this convergence keep the number of selected vectors fixed while taking $N \rightarrow \infty$.

3 Point-process convergence and loop equations

We prove the main point-process theorem. Switching identities first give microscopic tightness and a Gaussian marked limit. The stationary marked flow then yields the full loop hierarchy, which identifies every subsequential point-process limit. The moment bounds needed to pass to correlation measures are proved separately in Appendix A; they are not inputs to this section.

3.1. Switching and microscopic tightness

Fix a bulk energy E and set $z_N(w) = E + w/(N\sigma)$. The polynomial mixed fourth-moment estimate controls the conditional variance of the switching compression. Reversibility then yields scalar tightness and capped moments.

3.1.1. The exact switching identities. Conditional on the graph, sample two independent uniform oriented edges (l, a) and (b, c) . If their four endpoints are distinct and neither $\{l, b\}$ nor $\{a, c\}$ is an edge, replace $\{l, a\}, \{b, c\}$ by $\{l, b\}, \{a, c\}$. Otherwise retain the graph. Write $\tilde{A}$ for the resulting adjacency matrix and call a proposal accepted when it meets these conditions. Set

$$D = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad L = PD = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

Let $\mathbf{e}_v$ denote the coordinate unit vector at v in $\mathbb{R}^N$. For every proposal, define

$$\mathbf{w}_+ = \frac{\mathbf{e}_l + \mathbf{e}_b - \mathbf{e}_a - \mathbf{e}_c}{2}, \quad \mathbf{w}_- = \frac{\mathbf{e}_l + \mathbf{e}_a - \mathbf{e}_b - \mathbf{e}_c}{2}, \quad W = (\mathbf{w}_+, \mathbf{w}_-).$$

The notation $\mathbb{E}_*$ denotes conditional expectation over this independent proposal, before imposing acceptance. Put

$$K(z) = W^T G(z) W.$$

Lemma 3.1 (Exact rank-two switching identities). *The graph pair $(A, \tilde{A})$ is exchangeable. The rejection probability is $O_d(N^{-1})$, uniformly in the initial simple regular graph. On accepted proposals, $W^T W = I_2$, the reverse proposal has $\tilde{W} = WP$, and*

$$\tilde{A} - A = 2WDW^T, \tag{3.1}$$

$$\tilde{G}(z) = G(z) - G(z)W(K(z) + D/2)^{-1}W^T G(z), \tag{3.2}$$

$$\mathcal{D}(z) := \frac{\det(\tilde{A} - z)}{\det(A - z)} = \det(I_2 + 2DK(z)), \tag{3.3}$$

$$\tilde{m}_N(z) - m_N(z) = -\frac{1}{N} \partial_z \log \mathcal{D}(z), \tag{3.4}$$

$$\tilde{K}(z) = P \left[D/2 - \frac{1}{4} D(K(z) + D/2)^{-1} D \right] P. \tag{3.5}$$

Here tildes indicate the switched graph and its reverse proposal; thus $\tilde{G}(z) = (\tilde{A} - z)^{-1}$, $\tilde{m}_N(z) = N^{-1} \text{Tr } \tilde{G}(z)$, and $\tilde{K}(z) = \tilde{W}^T \tilde{G}(z) \tilde{W}$. The logarithmic derivative is independent of the logarithm branch. For every nonreal z ,

$$|\tilde{m}_N(z) - m_N(z)| \leq \frac{2\pi}{N |\text{Im } z|}. \tag{3.6}$$

Proof. The accepted proposal $(l, a), (b, c)$ has inverse $(l, b), (a, c)$; both have conditional probability $(Nd)^{-2}$. This correspondence is an involution on the graph-proposal space, with rejected moves as self-loops, and therefore proves detailed balance for the uniform graph law. Each endpoint of a uniform oriented edge is a uniform vertex, independently of the other sampled edge. Each forbidden equality has probability N^{-1} and each forbidden cross adjacency has probability d/N . A union bound proves the rejection estimate.

On the four coordinates ordered as (l, a, b, c) , the increment is

$$\tilde{A} - A = \begin{pmatrix} 0 & -1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 1 & 0 & 0 & -1 \\ 0 & 1 & -1 & 0 \end{pmatrix},$$

with all other entries zero. Expansion of $2\mathbf{w}_+ \mathbf{w}_+^T - 2\mathbf{w}_- \mathbf{w}_-^T$ gives this matrix. The two displayed columns have norm one and zero inner product when the endpoints are distinct.

The reverse proposal interchanges these columns. This proves (3.1) and identifies the reverse compression basis. The determinant lemma gives (3.3). Its determinant is nonzero for nonreal z , since both adjacency matrices are real symmetric. Woodbury's identity gives (3.2). Compressing it and using

$$K - K(K + D/2)^{-1}K = D/2 - \frac{1}{4}D(K + D/2)^{-1}D,$$

proves (3.5). For the sign in the trace identity, recall that $\partial_z \log \det(A - z) = -\text{Tr}(A - z)^{-1}$. Differentiating the logarithm of the determinant ratio therefore gives $-N\tilde{m}_N + Nm_N$, proving (3.4). This matrix identity does not require commutativity: substitute $K = (K + D/2) - D/2$ and expand in the indicated order. We used $D^2 = I_2$.

The increment has rank at most two. The two eigenvalue counting functions therefore differ by at most two. Integration by parts against $(t - z)^{-1}$, followed by $\int_{\mathbb{R}} |t - z|^{-2} dt = \pi/|\text{Im } z|$, proves (3.6). $\square$

For $t \in \mathbb{C}$, write

$$V(t) = \text{diag}(1 - t/d, 1 + t/d).$$

If $\mathbf{u}_\alpha$ is a basis eigenvector orthogonal to the constant vector, put $\mathbf{g}_\alpha = \sqrt{N}W^T\mathbf{u}_\alpha$. Orthogonality and oriented-edge averaging give

$$\mathbb{E}_*\mathbf{g}_\alpha = 0, \quad \mathbb{E}_*\mathbf{g}_\alpha\mathbf{g}_\beta^T = \mathbf{1}_{\alpha=\beta}V(\lambda_\alpha). \quad (3.7)$$

The two components are half the sum of the edge differences and half the difference of the edge sums. Thus (3.7) follows from independence of the edges and their exact covariance; edge reversal makes each difference orthogonal to each sum. Also $W^T\mathbf{1} = 0$.

Consequently, with

$$\hat{m}_N(z) = m_N(z) - \frac{1}{N(d - z)},$$

the exact conditional mean is

$$\overline{K}(z) := \mathbb{E}_*K(z) = V(z)\hat{m}_N(z) - (1 - N^{-1})D/d. \quad (3.8)$$

This formula remains valid for a disconnected graph: take the constant vector as one basis vector and all other eigenvectors orthogonal to it.

Here is also a direct derivation of the mean formula. The spectral representation and (3.7) give

$$\mathbb{E}_*K(z) = \frac{1}{N} \sum_{i \neq 1} \frac{V(\lambda_i)}{\lambda_i - z}.$$

Since $V(\lambda_i) = V(z) - (\lambda_i - z)D/d$, the sum is exactly $V(z)\hat{m}_N(z) - (1 - N^{-1})D/d$. All these conditional expectations use the proposal *before* acceptance. Conditioning first on acceptance would slightly change the sampling law and lose these exact covariance identities. We instead keep the rejection error separate in each reversibility argument.

3.1.2. A conditional variance estimate. Fix compact bulk intervals $J \Subset K$ with E in the interior of J . Let $\chi > 0$ be the mixed fourth-moment exponent supplied by Theorem 1.2 on K . Choose

$$0 < \varepsilon < \min(\chi/4, 1/4).$$

Let Ω_N be the intersection of its graph event with the spectral event of Proposition 2.10 at the scale $N^{-1+\varepsilon}$ and the local tree event in that proposition. After reducing $\beta > 0$,

$$\mathbb{P}(\Omega_N^c) \leq CN^{-\beta}.$$

The intervals and ε remain fixed throughout this subsection.

Lemma 3.2 (Conditional variance of the compressed resolvent). *Fix $c_* > 0$ and a sufficiently small $\eta_* > 0$. Uniformly for $z = E + i\eta$ with $c_*/N \leq \eta \leq \eta_*$, on Ω_N ,*

$$\frac{1}{N} \sum_{\lambda_i \in J} |\lambda_i - z|^{-1} \leq CN^\varepsilon, \quad (3.9)$$

$$\mathbb{E}_* \|K(z) - \bar{K}(z)\|^2 \leq C \left(1 + \frac{\operatorname{Im} m_N(z)}{N\eta} \right) + N^{-\tau} \quad (3.10)$$

for some $\tau > 0$, with J the inner bulk interval fixed above. The absolute-sum estimate also holds, with a compact-dependent constant, at $z = z_N(w)$ uniformly for w in a fixed compact subset of $\mathbb{C}^+$.

Proof. This estimate uses fourth moments, not convergence of the full mark law. The diagonal terms in the variance sum are controlled by the Ward identity. For distinct eigenvectors, the polynomial error from Theorem 1.2 must be small enough to survive an absolute sum over all bulk eigenvalues. The choice $2\varepsilon < \chi$ is made for precisely this reason.

The averaged local law in Proposition 2.10 gives

$$\frac{1}{N} \#\{i : |\lambda_i - x| \leq h\} \leq Ch, \quad h \geq N^{-1+\varepsilon},$$

for the real energies needed here. At $z = E + ic_*/N$, the central interval of width $N^{-1+\varepsilon}$ contributes $O(N^\varepsilon)$ and the dyadic annuli contribute $O(\log N)$. Increasing η decreases the weights; denominator comparison gives the compact-uniform assertion for $z_N(w)$.

The part of $G(z)$ from eigenvalues outside J has uniformly bounded operator norm. Since $\|W\|$ is uniformly bounded on all proposals, its centered conditional second moment is bounded by a constant. The part inside J is

$$\frac{1}{N} \sum_{\lambda_i \in J} \frac{\mathbf{g}_i \mathbf{g}_i^T}{\lambda_i - z}.$$

For entries indexed by $a, b, c, e \in \{1, 2\}$, combining the mixed fourth-moment estimate Theorem 1.2 with the independent two-edge sampling gives

$$|\operatorname{Cov}_*((\mathbf{g}_i)_a(\mathbf{g}_i)_b, (\mathbf{g}_j)_c(\mathbf{g}_j)_e)| \leq C\mathbf{1}_{i=j} + CN^{-\chi}.$$

In the four-factor expansion, the distributions $(4, 0)$, $(2, 2)$, and $(3, 1)$ between the two edges use, respectively, the fourth-moment theorem, exact second moments, and vanishing first moments. Gaussian quadratic marks have zero covariance at distinct eigenvectors and bounded covariance at the same eigenvector.

Summing entrywise and using the equivalence of matrix norms in dimension two, we bound the centered variance of the bulk part by

$$\frac{C}{N^2} \sum_i |\lambda_i - z|^{-2} + CN^{-\chi} \left(\frac{1}{N} \sum_{\lambda_i \in J} |\lambda_i - z|^{-1} \right)^2 \leq \frac{C \operatorname{Im} m_N(z)}{N\eta} + CN^{-\chi+2\varepsilon}.$$

The first term is the exact Ward sum. Combining the inside and outside parts proves (3.10). More explicitly, the central interval contains at most $CN \cdot N^{-1+\varepsilon} = CN^\varepsilon$ eigenvalues, each contributing at most C to the normalized reciprocal-distance sum when $\eta \geq c_*/N$. On an annulus of radii $h, 2h$ with $h \geq N^{-1+\varepsilon}$, there are at most CNh eigenvalues and each contributes at most $(Nh)^{-1}$, giving a constant per annulus. This explains why the absolute sum is only a small power of N , rather than the potentially much larger bound η^{-1} . Since $\chi - 2\varepsilon > 0$, it can be absorbed into the exponent τ in the stated result. $\square$

3.1.3. Reversibility and scalar moments.

Proposition 3.3 (Microscopic scalar tightness at a fixed energy). *With the event Ω_N above, there are $M_0, C, \kappa > 0$ such that*

$$\mathbb{P}(\Omega_N \cap \{|m_N(E + i\eta)| > M\}) \leq C \exp(-\kappa M \log(2 + M)) + CN^{-\beta}, \quad M \geq M_0, \quad (3.11)$$

uniformly for $c_/N \leq \eta \leq \eta_*$. Define*

$$\Gamma_{N,\eta} = \Omega_N \cap \{|m_N(E + i\eta)| \leq \log N\}.$$

Then, uniformly in this range of heights,

$$\mathbb{P}(\Gamma_{N,\eta}^c) \leq CN^{-\beta}, \quad \mathbb{E}[\mathbf{1}_{\Gamma_{N,\eta}} |m_N(E + i\eta)|^r] \leq C_r \quad (0 < r < \infty). \quad (3.12)$$

There is also a single graph event Γ_N of probability at least $1 - CN^{-\beta}$ such that, for every compact $B \subset \mathbb{C} \setminus \mathbb{R}$ and fixed $0 < r < \infty$,

$$\mathbb{E}[\mathbf{1}_{\Gamma_N} \sup_{w \in B} |s_N(w)|^r] \leq C_{B,r}. \quad (3.13)$$

In particular s_N is tight for locally uniform analytic convergence on both half-planes, related by Schwarz reflection.

Proof. The mechanism is a contradiction between a large forward determinant and a large reverse determinant. A large scalar transform makes each determinant typically quadratic in that transform, but the two determinants must be reciprocal. Thus a large scalar transform forces an atypical compression in at least one direction. Iterating this observation gives a much stronger tail than a single variance bound.

Write $T_A = |m_N(z)|$. By (3.10),

$$\mathbb{E}_* \|K - \overline{K}\|^2 \leq C(1 + T_A)$$

throughout the indicated height range. The coefficient of $\hat{m}_N^2$ in $\det(I_2 + 2D\overline{K})$ is $4(z^2/d^2 - 1)$; its modulus is uniformly bounded below. All remaining coefficients are bounded, and $|\hat{m}_N - m_N| \leq C/N$. It follows that, for fixed small $\delta > 0$ and sufficiently large M_0 ,

$$T_A \geq M_0, \quad \|K - \overline{K}\| \leq \delta T_A \implies |\mathcal{D}(z)| \geq \delta T_A^2.$$

Here the determinant perturbation is at most $C\delta T_A^2 + C\delta T_A$. Chebyshev's inequality bounds the probability that the compression fails this condition:

$$\mathbb{P}_*(\|K - \overline{K}\| > \delta T_A) \leq C/T_A \quad \text{on } \Omega_N \cap \{T_A \geq M_0\}. \quad (3.14)$$

The rank shift in (3.6) is at most $A_0 = 2\pi/c_*$. On an accepted proposal the reverse determinant is $\mathcal{D}^{-1}$. If both graphs lie in Ω_N , both compressions satisfy their favorable events, and $T_A > M$ for large M , then

$$1 = |\mathcal{D}\mathcal{D}_{\text{rev}}| \geq \delta^2 T_A^2 \tilde{T}_A^2 > 1,$$

since $\tilde{T}_A \geq T_A - A_0$. Thus, when the scalar transform is large and the forward compression is favorable, the reverse proposal must fail the concentration estimate.

Set $F_N(M) = \mathbb{P}(\Omega_N \cap \{T_A > M\})$. Rejected proposals contribute at most $(C/N)F_N(M)$. The initial concentration failure contributes at most $(C/M)F_N(M)$ by (3.14). The reversed failure contributes at most $C(M - A_0)^{-1}F_N(M - A_0)$, using the exact reversal on accepted

proposals. The probability that either needed graph event fails is at most $CN^{-\beta}$ by stationarity. Therefore, after absorption and an increase of M_0 ,

$$F_N(M) \leq \frac{C}{M} F_N(M - A_0) + CN^{-\beta}.$$

To see the gain from iteration, set $n = \lfloor (M - M_0)/A_0 \rfloor$. The homogeneous coefficient after n steps is bounded by

$$\prod_{j=0}^{n-1} \frac{C}{M - jA_0} \leq \frac{(C/A_0)^n}{n!} \leq C \exp\{-\kappa M \log(2 + M)\},$$

after changing constants for bounded M . Once $C/M_0 \leq 1/2$, the inhomogeneous products sum to a bounded geometric series. This proves (3.11) uniformly in η . Integrating the tail only up to $\log N$ gives (3.12), because $N^{-\beta}(\log N)^r$ tends to zero for every fixed r .

For (3.13), use $\Gamma_N = \Gamma_{N,1/(N\sigma)}$, choosing $c_* \leq 1/\sigma$. The spectral representation with unit masses at x_i gives, on each compact $B \subset \mathbb{C}^+$,

$$|s_N(w) - s_N(i)| \leq |w - i| \sum_i \frac{1}{|x_i - w| |x_i - i|} \leq C_B \operatorname{Im} s_N(i).$$

Here $|x - i|/|x - w|$ is uniformly bounded for real x and $w \in B$. Thus $\sup_B |s_N| \leq C_B |s_N(i)|$, with the same conclusion on the lower half-plane by reflection. The height moment estimate, together with the fixed real subtraction, now proves (3.13) and hence analytic tightness by normal-family compactness. Cauchy's formula also gives the corresponding capped moments for each fixed spectral derivative. The cap in the moment statement is essential at this stage. The additive error $N^{-\beta}$ in the tail estimate cannot be integrated to infinity with a uniform bound, since the trivial finite- N bound on the transform is of order N at microscopic height. It can be integrated up to $\log N$, because $N^{-\beta}(\log N)^r \rightarrow 0$. For tightness of the analytic field, caps whose probability tends to one are already enough, and this is all that the point-process proof needs. Only for the correlation upgrade do we remove the cap on counting observables; this is done in Lemmas A.1–A.3 of Appendix A. $\square$

3.2. The Gaussian marked limit

We identify finitely many eigenvector marks using Corollary 1.3 and control the remaining spectral sum using Theorem 1.2. The number of retained marks stays fixed while $N \rightarrow \infty$.

Define the fixed matrices and constants

$$V = V(E), \quad H = a_E V - D/d, \quad q = \frac{d-2}{\sqrt{4(d-1) - E^2}}, \quad c = \sqrt{1 + q^2},$$

and the normalized marked transform

$$S_N(w) = \sigma^{-1} V^{-1/2} (K(z_N(w)) - H) V^{-1/2}. \quad (3.15)$$

Its imaginary part is positive semidefinite on $\mathbb{C}^+$. Here V, H, q, c depend only on the fixed energy E and degree d ; K, S_N also depend on the edge proposal. For a basis eigenvector orthogonal to the constant vector, with eigenvalue in the chosen bulk interval, set

$$\xi_i = V(\lambda_i)^{-1/2} g_i.$$

These finite- N marks come from the independent edge proposal. By (3.8),

$$\mathbb{E}_* S_N(w) = s_N(w) I_2 + O_B(N^{-1}(1 + |m_N(z_N(w))|)), \quad (3.16)$$

uniformly on each fixed compact $B \subset \mathbb{C}^+$.

3.2.1. The centered spectral tail.

Lemma 3.4 (Removal of the remote eigenvector marks). *For every compact $B \subset \mathbb{C}^+$ and sufficiently large fixed R , there are graph events $\Gamma_{N,R}$ of probability tending to one such that*

$$\limsup_{N \rightarrow \infty} \mathbb{E} \left[\mathbf{1}_{\Gamma_{N,R}} \sup_{w \in B} \left\| S_N(w) - s_N(w)I_2 - \sum_{|x_i| \leq R} \frac{\xi_i \xi_i^T - I_2}{x_i - w} \right\|^2 \right] \leq \frac{C_B}{R}. \quad (3.17)$$

The constant does not depend on R . The events are graph-measurable; the expectation in (3.17) is over both the graph and the independent edge proposal. Equivalently it is the graph expectation of the conditional proposal average $\mathbb{E}_*$.

Proof. We split the spectrum twice. A smooth macroscopic cutoff separates energies away from E , where the resolvent is a continuous function of A and local tree geometry is enough. Inside the bulk cutoff, we retain the microscopic window $[-R, R]$ and use the fourth-moment estimate for the remaining centered sum. Centering is crucial: the uncentered tail need not tend to zero, because it carries the macroscopic background of the Stieltjes transform.

Take $\Gamma_{N,R}$ to be the intersection of Ω_N , Γ_N , and $\Gamma_{N,R/(N\sigma)}$. For every fixed $R \geq 1$, the last event is defined for all sufficiently large N . Its height-moment constants in (3.12) are independent of R . Finite-prefix Gaussianity will be applied separately through Corollary 1.3, whose error tends to zero in probability.

Choose a smooth real function ϑ , equal to one near E and supported in a compact bulk interval J . The outer part of the compression has scalar weight

$$f_{N,w}(t) = \frac{1 - \vartheta(t)}{t - z_N(w)}.$$

Uniformly for $t \in [-d, d]$ and w in a fixed compact set, this converges to the continuous real function $f(t) = (1 - \vartheta(t))/(t - E)$, extended across E . Given $\delta > 0$, choose a real polynomial Q_δ with $\sup_{[-d,d]} |f - Q_\delta| \leq \delta$. Functional calculus bounds the compression error by $C\delta + o(1)$.

For the fixed degree of Q_δ , the local tree event in Proposition 2.10 ensures that all but a vanishing fraction of roots have the needed tree neighborhood. Two independently sampled edge neighborhoods of this fixed radius are disjoint, with no path of length at most $\deg Q_\delta$ between them, except with probability $O(N^{-1})$. On the remaining proposals, the compression of $Q_\delta(A)$ is the same deterministic two-edge tree compression: its local entries count the same walks and its cross-edge entries vanish. Since its norm is bounded on every proposal, its centered conditional second moment tends to zero. The outer compression thus has centered variance at most $C\delta^2 + o(1)$. First let $N \rightarrow \infty$, then $\delta \rightarrow 0$.

For the bulk part, the covariance calculation in Lemma 3.2 gives the following bound for its centered contribution from $|x_i| > R$:

$$C_B \sum_{|x_i| > R} |x_i - w|^{-2} + CN^{-\chi} \left(\sum_{\lambda_i \in J} |x_i - w|^{-1} \right)^2.$$

The second term is $O_B(N^{-\chi+2\varepsilon})$ on Ω_N , by (3.9). If $R \geq 2 \sup_{w \in B} |w| + 1$, then

$$\sum_{|x_i| > R} |x_i - w|^{-2} \leq C_B \sum_i \frac{1}{x_i^2 + R^2} = \frac{C_B}{R} \operatorname{Im} s_N(iR). \quad (3.18)$$

The expectation of the final imaginary part, restricted to $\Gamma_{N,R}$, is uniformly bounded by (3.12). The centered bulk tail therefore satisfies the required C_B/R estimate.

For the retained indices, $\vartheta(\lambda_i) = 1$ for large N , and

$$V^{-1/2}(\mathbf{g}_i \mathbf{g}_i^T - V(\lambda_i))V^{-1/2} = \boldsymbol{\xi}_i \boldsymbol{\xi}_i^T - I_2 + O((R/N)(1 + |\boldsymbol{\xi}_i|^2)).$$

Theorem 1.2 bounds $\mathbb{E}_*[(1 + |\boldsymbol{\xi}_i|^2)(1 + |\boldsymbol{\xi}_j|^2)]$ uniformly. The squared retained error is therefore at most $C_{R,B}N^{-2}\mu_N([-R, R])^2$ in conditional expectation. Since

$$\mu_N([-R, R]) \leq 2R \operatorname{Im} s_N(iR),$$

its expectation on $\Gamma_{N,R}$ tends to zero. The mean error in (3.16) also tends to zero there by (3.13).

We explain the last analytic step, which will recur below. Cover B by finitely many disks D_j whose concentric disks of twice the radius remain off the real axis. For an analytic scalar entry F of the error field, Cauchy's formula on the larger circle gives

$$\sup_{w \in D_j} |F(w)|^2 \leq C_{B,j} \int_{\partial(2D_j)} |F(z)|^2 |dz|.$$

The same bound applies entrywise to matrices. The preceding pointwise second-moment estimates are uniform on these circles; Tonelli's theorem then gives the squared-supremum estimate (3.17). The constants may depend on the fixed compact set but not on R once R exceeds that set. $\square$

3.2.2. The scalar measure and the finite prefixes.

Lemma 3.5 (Scalar Nevanlinna subsequential limits). *Every sequence has a subsequence on which (μ_N, s_N) converges in distribution, vaguely in the first coordinate and locally uniformly in the second. Each limit has*

$$s(w) = a + \int_{\mathbb{R}} \left(\frac{1}{x-w} - \frac{x}{1+x^2} \right) d\mu(x), \quad a \in \mathbb{R}, \quad \int_{\mathbb{R}} \frac{d\mu(x)}{1+x^2} < \infty. \quad (3.19)$$

The measure μ is a locally finite integer counting measure. For every compact $B \subset \mathbb{C} \setminus \mathbb{R}$ and fixed $0 < r < \infty$,

$$\mathbb{E} \sup_{w \in B} |s(w)|^r < \infty, \quad \sup_{Y \geq 1} \mathbb{E} |s(iY)|^r < \infty. \quad (3.20)$$

Proof. The representing point measure and the analytic function must be extracted jointly. Vague convergence of point measures records only particles in fixed bounded intervals; an analytic limit could in principle retain an additional contribution from particles escaping to infinity. In the Nevanlinna representation this contribution is a nonnegative linear term. The uniform bound at large imaginary height is what rules it out.

Analytic tightness follows from Proposition 3.3. The bound

$$\mu_N([-R, R]) \leq (1 + R^2) \operatorname{Im} s_N(i)$$

gives tightness of the point measures on each compact interval. A joint subsequence therefore exists. The Nevanlinna representation and continuity of its representing measure on finite intervals give

$$s(w) = a + \alpha w + \int_{\mathbb{R}} \left(\frac{1}{x-w} - \frac{x}{1+x^2} \right) d\mu(x), \quad \alpha \geq 0.$$

Indeed $\int (1+x^2)^{-1} d\mu_N(x) = \operatorname{Im} s_N(i)$ is tight. Weighted-measure convergence identifies the finite-interval measure with μ ; mass escaping to infinity contributes to α . Locally bounded integer masses remain integer under vague convergence.

By bounded truncation and Fatou's lemma, (3.13) holds in the limit, since the omitted events have vanishing probability. Applying (3.12) at $\eta = Y/(N\sigma)$ gives the bound uniform in $Y \geq 1$. Since $\alpha Y \leq \operatorname{Im} s(iY)$, letting $Y \rightarrow \infty$ in expectation gives $\alpha = 0$. For the passage from capped to limiting moments, fix a bounded continuous truncation of the relevant supremum and let $N \rightarrow \infty$. Its contribution on the complement of the cap is at most its bound times $\mathbb{P}(\Gamma_N^c)$, which tends to zero. Increasing the truncation and applying monotone convergence gives the desired limiting moment bound. Thus no unproved uniform integrability of the uncapped finite- N transform is being invoked.

The representing measure can also be recovered locally from $\pi^{-1} \operatorname{Im} s(x + i\eta) \, dx$ as $\eta \downarrow 0$. This inversion formula identifies it with the vague limit of the particle measures on intervals whose endpoints carry no atoms. A locally bounded sequence of integer counting measures has an integer counting limit: isolate each limiting point in a sufficiently small interval and use that the number of approximating points is an integer. Multiplicities may merge at this stage; simplicity will follow from identification or, in the tagged setting, from a separate argument. $\square$

Lemma 3.6 (Conditional Gaussian representation of the marks). *The triples (μ_N, s_N, S_N) are jointly tight. Each subsequential limit can be realized with marks ξ_x , independent standard real two-dimensional Gaussians conditionally on (μ, s) , such that*

$$S(w) = s(w)I_2 + \sum_{x \in \mu} \frac{\xi_x \xi_x^T - I_2}{x - w}. \quad (3.21)$$

Points with multiplicity receive separate marks. The series converges conditionally in L^2 , locally uniformly off the real axis. Its tail outside $[-R, R]$ has squared unconditional L^2 supremum on a fixed compact set bounded by C_B/R , for sufficiently large R . Moreover,

$$\mathbb{E} \sup_{w \in B} (|s(w)| + \|S(w)\|)^r < \infty \quad (3.22)$$

for every compact $B \subset \mathbb{C} \setminus \mathbb{R}$ and fixed $0 < r < \infty$, and the same holds for each fixed spectral derivative.

Proof. There are two logically separate claims: conditional Gaussianity of any fixed finite prefix, and control of the infinite remainder. The first uses only Corollary 1.3. The second uses the polynomial fourth-moment estimates through Lemma 3.4. We never apply qualitative convergence to a number of eigenvectors that grows with N .

Fix a subsequential scalar limit and a cutoff R such that $\mu(\{-R, R\}) = 0$ almost surely. Such cutoffs can be chosen to tend to infinity: (3.20) gives finite expected counts on compact intervals, so only countably many deterministic points can have positive probability of being atoms.

On the event of at most k eigenvalues in $[-R, R]$, select their eigenvectors in spectral order and pad with nearby orthonormal eigenvectors so that all k selected eigenvalues tend to E . For example, if $a > 0$ is the coarse-rigidity exponent, the window $[E - N^{-a/2}, E + N^{-a/2}]$ contains at least k bulk eigenvalues for all large N and accommodates every retained vector. For fixed k , Corollary 1.3 gives conditional joint edge convergence in probability, uniformly over these graph-measurable selections. The two edges are independent; the bounded linear maps defining $\mathbf{g}_i$ and $V(\lambda_i)^{-1/2}$ therefore give independent standard Gaussian limiting marks, by (3.7).

The conditional bounded-Lipschitz error is bounded and tends to zero in probability, hence also in expectation. This remains true after multiplication by any bounded graph-measurable function. For clarity, let b_N be a bounded graph-measurable function and let f be a bounded

Lipschitz test of the retained marks. If ϵ_N is the conditional Gaussian approximation error, then

$$|\mathbb{E}\{b_N[\mathbb{E}_*f(\boldsymbol{\xi}_1, \dots, \boldsymbol{\xi}_k) - \mathbb{E}_{\mathbf{G}}f(\mathbf{G}_1, \dots, \mathbf{G}_k)]\}| \leq \|b_N\|_\infty \mathbb{E}\epsilon_N \rightarrow 0.$$

Here $\mathbf{G}_1, \dots, \mathbf{G}_k$ are independent standard Gaussian vectors, and $\mathbb{E}_{\mathbf{G}}$ refers only to them. The outer expectation is over the graph, since the proposal was averaged by $\mathbb{E}_*$. Choose b_N among bounded tests of the scalar spectral state and pass to its joint limit. A monotone-class argument then gives the product Gaussian conditional law given the full state (μ, s) , including the real constant in its Nevanlinna representation, not merely given the particle locations. Counting tightness removes the cap k . Changing the order at repeated limiting points only permutes independent identically distributed marks. A consistent exhaustion by cutoffs therefore defines the Gaussian family on all particles.

Finite marked sums are analytically tight. Lemma 3.4 approximates $S_N - s_N I_2$ by these sums on every compact set, taking $N \rightarrow \infty$, then $k \rightarrow \infty$, then $R \rightarrow \infty$. This proves joint tightness. In a subsequential coupling, Fatou's lemma transfers the truncated bounds at each fixed cutoff to the limit. The indicators tend to one almost surely along a further subsequence simultaneously at the countable cutoffs, since their failure probabilities vanish.

Conditionally on (μ, s) , the centered summands are independent quadratic Gaussian chaoses. Their variance sum is bounded by

$$C \sum_{x \in \mu} |x - w|^{-2} = C \frac{\operatorname{Im} s(w)}{\operatorname{Im} w} < \infty.$$

Thus the centered Gaussian series converges conditionally in L^2 . Cauchy's formula on surrounding circles proves its locally uniform version. For R large relative to a compact set, the limiting version of (3.18), together with $\sup_{Y \geq 1} \mathbb{E} \operatorname{Im} s(iY) < \infty$, gives the unconditional squared L^2 tail bound C_B/R . The finite-prefix identification and the vanishing tail identify the limit with (3.21).

The required higher moments are obtained only after taking the Gaussian limit. For a centered scalar quadratic Gaussian chaos Z , hypercontractivity gives $\|Z\|_{L^r} \leq (r-1)\|Z\|_{L^2}$ for $r \geq 2$. Apply this conditionally to each entry of a finite centered marked sum. The bound is independent of the number of summands, so it passes to the L^2 limit. For $0 < r < 2$, conditional Cauchy-Schwarz is enough. This bounds the conditional moments in terms of the variance. Spectral derivatives introduce fixed powers of $|\operatorname{Im} w|^{-1}$. Nevanlinna bounds control these variances on surrounding circles by $C_B \operatorname{Im} s(i)$. Thus (3.20) and Cauchy's formula give (3.22), including derivatives, without higher eigenvector moments. One should distinguish the two series in this construction. The scalar Nevanlinna sum is compensated by $x/(1+x^2)$. The additional matrix sum is instead centered in the Gaussian marks and converges in conditional L^2 . It need not be absolutely convergent term by term. Its imaginary part, combined with the scalar part, is

$$\operatorname{Im} S(w) = \sum_{x \in \mu} \frac{\operatorname{Im} w}{|x - w|^2} \boldsymbol{\xi}_x \boldsymbol{\xi}_x^T \succeq 0.$$

This positive series converges by the same weighted second-moment bound. The distinction between compensation in space and centering in marks is useful when the switching map is applied. $\square$

3.2.3. The stationary marked state.

Proposition 3.7 (The limiting switching involution). *Every subsequential limit (μ, s, S) has the representation (3.19), the conditional Gaussian marks (3.21), and all moments in (3.22). Its law is invariant under*

$$(\mu, s, S) \mapsto (\mu, s, OSO^T), \quad O \in O(2). \quad (3.23)$$

Almost surely $S(w) + qD$ is invertible for every nonreal w . The map

$$\tilde{S} = -qD - c^2 L(S + qD)^{-1} L^T, \quad \tilde{s} = s - \partial_w \log \det(S + qD) \quad (3.24)$$

together with the representing point measure $\tilde{\mu}$ of $\tilde{s}$ is a law-preserving involution. The joint law of a state and its image is invariant under exchange of the two states. Furthermore,

$$|\tilde{s}(w) - s(w)| \leq \frac{2\pi}{|\operatorname{Im} w|}. \quad (3.25)$$

For nonreal w, v , the conditional Gaussian identities are

$$\mathbb{E}[S_{22}(w) - S_{11}(w) \mid \mu, s] = 0, \quad (3.26)$$

$$\mathbb{E}[S_{11}(w)S_{22}(w) \mid \mu, s] = s(w)^2, \quad (3.27)$$

$$\mathbb{E}[S_{12}(w)S_{12}(v) \mid \mu, s] = \frac{s(w) - s(v)}{w - v}. \quad (3.28)$$

These conditional expectations average only the Gaussian marks, with the full scalar state fixed. The products are bilinear: no complex conjugation is intended. Conjugates are obtained by taking v in the opposite half-plane. At coincident arguments the divided difference means $s'(w)$.

Proof. The finite- N Woodbury identity is converted into the normalized map by a deterministic change of coordinates. Its coefficients are of order one at fixed d ; no expansion in $1/d$ is used. There is also a genuine continuity issue: convergence of S_N does not automatically imply convergence of $(S_N + qD)^{-1}$. We resolve it by expressing the inverse in terms of the reversed field, which is tight by reversibility.

The Gaussian representation and moment assertions were proved above. Replacing each ξ_x by $O\xi_x$ preserves its conditional product law and changes the centered series to $O(S - sI_2)O^T$. This proves (3.23).

We first compute the marked switching exactly at finite N . Substitution of a_E and σ gives

$$H + D/2 = \sigma q V^{1/2} D V^{1/2}, \quad \sigma^{-1} \frac{1}{2} V^{-1/2} P D V^{-1/2} = cL.$$

Inserting these identities into (3.5) and (3.15) yields, on accepted proposals,

$$\tilde{S}_N = -qD - c^2 L(S_N + qD)^{-1} L^T. \quad (3.29)$$

The determinant identity becomes

$$\mathcal{D}(z_N(w)) = -c^{-2} \det(S_N(w) + qD).$$

Equations (3.4) and $z'_N(w) = 1/(N\sigma)$ therefore give the exact normalized identities

$$\tilde{s}_N = s_N - \partial_w \log \det(S_N + qD), \quad |\tilde{s}_N(w) - s_N(w)| \leq \frac{2\pi}{|\operatorname{Im} w|}. \quad (3.30)$$

The accepted graph-proposal measure is invariant under reversal. Its total mass is $1 - O(N^{-1})$. Conditioning this joint measure on acceptance changes the original law by $O(N^{-1})$ in total variation and preserves the reversal identity. In particular the original and reversed triples have identical marginal laws under this conditioning and are both tight. To control inversion in the limit, extract a joint analytic subsequential limit of the two triples. Put $M_N = S_N + qD$. Equation (3.29) writes the whole inverse field as

$$M_N^{-1} = -c^{-2} L^T (\tilde{S}_N + qD) L.$$

The right side is tight in the analytic topology. Passing to the limit in $M_N M_N^{-1} = I_2$ gives directly

$$(S + qD)[-c^{-2}L^T(\tilde{S} + qD)L] = I_2.$$

Thus the bracketed matrix is the inverse. The identity holds analytically throughout both half-planes on one probability-one set.

Cauchy's formula gives convergence of spectral derivatives, while the last display gives convergence of the inverse. Thus (3.29) and (3.30) give (3.24) and (3.25). The reversed scalar function has the representing point measure given by the vague limit of the reversed eigenvalues, by Lemma 3.5. It is therefore $\tilde{\mu}$. The symmetry under exchange of the finite state and its image passes to the limiting joint law.

To check the involution directly, write $M = S + qD$. Then

$$\tilde{M} = -c^2 L M^{-1} L^T, \quad \tilde{M}^{-1} = -c^{-2} L M L^T, \quad \det \tilde{M} = \frac{c^4}{\det M}.$$

Since $L^2 = -I_2$, the matrix transformation applied twice returns M . The two scalar logarithmic derivatives cancel by the determinant identity. Thus the map is an involution on the full state, including its representing point measure.

Finally the two Gaussian coordinates are independent. Their centered quadratic series therefore give (3.26) and (3.27), while

$$\mathbb{E}[S_{12}(w)S_{12}(v) \mid \mu, s] = \sum_{x \in \mu} \frac{1}{(x - w)(x - v)}.$$

The absence of a linear term in (3.19) identifies this sum with the divided difference in (3.28). Cauchy–Schwarz gives absolute convergence of the sums. The conditional and unconditional moment bounds above also allow differentiation and passage to coincident arguments in these identities. For reference, the Gaussian contractions can be read directly from one particle. If $\xi = (\xi_1, \xi_2)^T \sim N(0, I_2)$, then

$$\mathbb{E}(\xi_1^2 - 1)(\xi_2^2 - 1) = 0, \quad \mathbb{E}(\xi_1 \xi_2)^2 = 1, \quad \mathbb{E}(\xi_1^2 - 1) = \mathbb{E}(\xi_2^2 - 1) = 0.$$

Different particles have independent marks. Thus diagonal centered series in different coordinates have zero covariance, while the covariance of the off-diagonal series is the sum of the products of its scalar kernels. The algebraic identity

$$\frac{1}{x - w} - \frac{1}{x - v} = \frac{w - v}{(x - w)(x - v)}$$

gives the divided difference. If the scalar representation had contained a term αw , the divided difference would also have contained α , with no corresponding mark contraction. This explains why removal of the linear term precedes the loop-equation calculation. $\square$

3.3. The marked switching flow

We derive the bulk loop equations from the marked state of Proposition 3.7, retaining its matrices D, P, L and constants $q > 0$, $c^2 = 1 + q^2$. The marked switching map is

$$\mathcal{T}(s, S) = \left(s - \partial_w \log \det(S + qD), -qD - c^2 L(S + qD)^{-1} L^T \right).$$

Here and below a logarithmic derivative means $\partial_w \log \det M = \text{Tr}(M^{-1} \partial_w M)$, independently of a logarithm branch.

The scalar representation in Proposition 3.7 gives

$$\sum_i \frac{1}{(x_i - w)(x_i - z)} = \frac{s(w) - s(z)}{w - z}, \quad \sum_i \frac{1}{|x_i - w|^2} = \frac{\operatorname{Im} s(w)}{\operatorname{Im} w}.$$

Both sums converge absolutely; the first is interpreted by a derivative at coincident arguments. In applying the flow below, we will use the finiteness of all polynomial moments of marked values and spectral derivatives, established in (3.22).

Lemma 3.8 (The stationary conjugated-rotation flow). *Let*

$$O_\theta = e^{\theta L}, \quad \mathcal{R}_\theta(s, S) = (s, O_\theta S O_\theta^T), \quad \Phi_\theta = \mathcal{T} \mathcal{R}_\theta \mathcal{T}.$$

For each fixed real θ , Φ_θ is defined almost surely and preserves the joint marked law. Its infinitesimal vector field is

$$\mathcal{V}S = \alpha[L, S] + \kappa(P + SPS), \quad \mathcal{V}s = 2\kappa \partial_w S_{12}, \quad \alpha = \frac{1 - q^2}{1 + q^2}, \quad \kappa = \frac{2q}{1 + q^2}. \quad (3.31)$$

For every polynomial F in finitely many scalar and marked values and their spectral derivatives,

$$\mathbb{E} \mathcal{V}F = 0. \quad (3.32)$$

Proof. The switching is a discrete involution, not a small perturbation. The continuous parameter comes from rotating the Gaussian marks. Conjugating that rotation by switching preserves the law and creates a nontrivial scalar motion. We first compute its derivative algebraically and then justify differentiating expectations; these are separate steps.

Put $M = S + qD$ and write $T(S)$ for the matrix component of $\mathcal{T}(s, S)$. Since $L^T = -L$,

$$T(S) + qD = -c^2 L M^{-1} L^T, \quad (T(S) + qD)^{-1} = -c^{-2} L M L^T.$$

Proposition 3.7 gives the involution and its invariance, while (3.23) gives invariance under $\mathcal{R}_\theta$. For each fixed angle the successive inputs have the original marked law and lie in the domain of $\mathcal{T}$ almost surely. Hence Φ_θ preserves that law, and $\Phi_{\theta+\varphi} = \Phi_\theta \Phi_\varphi$ wherever defined.

The vector field follows by differentiating the conjugated rotation. For a symmetric matrix increment $\mathcal{H}$,

$$dT_Z[\mathcal{H}] = c^2 L(Z + qD)^{-1} \mathcal{H}(Z + qD)^{-1} L^T.$$

At $Z = T(S)$, take $\mathcal{H} = [L, T(S)]$. The preceding inverse identity gives

$$dT_{T(S)}[\mathcal{H}] = -c^{-2} M \mathcal{H} M.$$

Indeed $\mathcal{H}$ is symmetric and traceless, so $L \mathcal{H} L = \mathcal{H}$. Also,

$$\mathcal{H} = -2qP + c^2(-M^{-1}L + LM^{-1}),$$

and hence

$$\mathcal{V}M = [L, M] + \kappa M P M.$$

To track the constants, expand

$$[L, M] = [L, S] + 2qP, \quad M P M = S P S - q[L, S] - q^2 P.$$

Indeed $PD = L$, $DP = -L$, and $DPD = -P$. The coefficient of $[L, S]$ is $1 - \kappa q = \alpha$, and that of P is $2q - \kappa q^2 = \kappa$. Therefore

$$[L, M] + \kappa M P M = \alpha[L, S] + \kappa P + \kappa S P S,$$

which is the matrix equation in (3.31).

Let $M_\theta = S_\theta + qD$ be the matrix component along the flow. Composing the scalar determinant identities gives

$$s_\theta - s = \partial_w \log \frac{\det M_\theta}{\det M}.$$

Since $\text{Tr}(M^{-1}[L, M]) = 0$,

$$\mathcal{V}s = \partial_w \text{Tr}(M^{-1}\mathcal{V}M) = \kappa \partial_w \text{Tr}(PM) = 2\kappa \partial_w S_{12}.$$

The group identity gives (3.31) at every regular angle of an orbit.

For a polynomial observable F , its orbit and derivative are rational in $\sin \theta, \cos \theta$, with nonzero denominators at zero. Invariance at each deterministic angle and (3.22) give

$$\mathbb{E} \int_0^T |\mathcal{V}F(\Phi_\theta(s, S))| d\theta = T \mathbb{E} |\mathcal{V}F(s, S)| < \infty.$$

A nonremovable real pole of the orbit would make this integral infinite. Its finitely many real singularities on $[0, T]$ are therefore removable; after filling them, the orbit is absolutely continuous, even when $\text{Im } S$ is only semidefinite. The fundamental theorem of calculus yields

$$0 = \mathbb{E}[F(\Phi_T(s, S)) - F(s, S)] = \int_0^T \mathbb{E} \mathcal{V}F(\Phi_\theta(s, S)) d\theta = T \mathbb{E} \mathcal{V}F.$$

This proves (3.32). The integration argument is preferable to an unsupported exchange of a derivative and an expectation at zero. It uses stationarity at each deterministic time and integrability of the derivative along the orbit. Tonelli's theorem first gives an almost surely integrable derivative on every bounded time interval. Any nonremovable pole of a rational orbit would contradict that integrability. Filling the removable singularities then permits the fundamental theorem of calculus. No deterministic lower bound on $\det(S + qD)$ is required. $\square$

3.4. Identification of the point-process limit

We now close the main-text argument. Averaging the stationary flow conditionally on its scalar data gives the full microscopic loop hierarchy. The characterization theorem of [9] then identifies the limit, and tightness turns subsequential identification into convergence of the full sequence.

Proposition 3.9 (The full microscopic loop hierarchy). *Let (μ, s, S) be any subsequential marked limit in Proposition 3.7. Then, for every integer $p \geq 1$ and every $w_1, \dots, w_p \in \mathbb{C} \setminus \mathbb{R}$,*

$$\begin{aligned} 0 = \mathbb{E} & \left[(s'(w_1) + s(w_1)^2 + 1) \prod_{j=2}^p s(w_j) \right] \\ & + 2 \sum_{j=2}^p \mathbb{E} \left[\partial_{w_j} \frac{s(w_1) - s(w_j)}{w_1 - w_j} \prod_{\substack{2 \leq \ell \leq p \\ \ell \neq j}} s(w_\ell) \right]. \end{aligned} \tag{3.33}$$

At coincident arguments the divided difference is continued holomorphically. Consequently μ has the law of Θ_1 .

Proof of Proposition 3.9. For $p = 1$, the desired identity resembles a Stein identity: it says that the expectation of a differential expression in the resolvent vanishes. That single equation

is not the characterization theorem. We derive all p simultaneously, including mixed upper- and lower-half-plane arguments, so that the full cited hierarchy applies.

By (3.26)–(3.28), conditionally on (μ, s) ,

$$\begin{aligned}\mathbb{E}[S_{11}(w) - S_{22}(w) \mid \mu, s] &= 0, & \mathbb{E}[S_{11}(w)S_{22}(w) \mid \mu, s] &= s(w)^2, \\ \mathbb{E}[S_{12}(w)S_{12}(v) \mid \mu, s] &= \frac{s(w) - s(v)}{w - v}.\end{aligned}$$

At $v = w$, the last expression is $s'(w)$. The moment bounds justify spectral differentiation of these identities. Since $[L, S]_{12} = S_{11} - S_{22}$ and $(SPS)_{12} = S_{11}S_{22} + S_{12}^2$, equation (3.31) gives

$$\mathcal{V}S_{12} = \alpha(S_{11} - S_{22}) + \kappa(1 + S_{11}S_{22} + S_{12}^2), \quad \mathcal{V}s = 2\kappa S'_{12}.$$

Apply (3.32) to $F = S_{12}(w_1) \prod_{j=2}^p s(w_j)$. The product rule gives

$$\begin{aligned}\mathcal{V}F &= [\alpha(S_{11}(w_1) - S_{22}(w_1)) + \kappa(1 + S_{11}(w_1)S_{22}(w_1) + S_{12}(w_1)^2)] \prod_{j=2}^p s(w_j) \\ &\quad + 2\kappa \sum_{j=2}^p S_{12}(w_1)S'_{12}(w_j) \prod_{\substack{2 \leq \ell \leq p \\ \ell \neq j}} s(w_\ell).\end{aligned}$$

First average the marks conditionally on (μ, s) . The α term vanishes, and the three displayed contractions evaluate the remaining terms. Then average the scalar state and divide by $\kappa > 0$. This is the tower property $\mathbb{E}\mathcal{V}F = \mathbb{E}[\mathbb{E}(\mathcal{V}F \mid \mu, s)]$, and it gives (3.33). The same real marks are used in both half-planes, so mixed arguments require no separate construction. The moment bounds allow holomorphic continuation to coincident arguments.

In the terminology of [9, Definition 1.3], particle-generated means that the representing measure of the Nevanlinna function is a locally finite integer atomic measure, with multiplicities retained. Equation (3.33) is exactly [9, Assumption 1.4] at $\beta = 1$: the derivative coefficient is $(2 - \beta)/\beta = 1$ and the interaction coefficient is $2/\beta = 2$. All its expectations are finite by the compact moment bounds. Since $\beta = 1$ is rational, [9, Theorem 1.5] identifies μ as Θ_1 in the intensity- $1/\pi$ normalization fixed in the introduction. $\square$

Proposition 3.10 (Fixed-energy point-process convergence). *For each fixed bulk energy E , the point measure $\mu_N = \sum_j \delta_{N\sigma(\lambda_j - E)}$, where $\sigma = \pi\rho_d(E)$, converges in law for the vague topology to Θ_1 , the Sine₁ process with intensity $1/\pi$. This is precisely the main point-process theorem.*

Proof of Proposition 3.10 and the main theorem. Proposition 3.7 gives a joint marked limit along a further subsequence of every subsequence. Proposition 3.9 identifies its configuration law as that of Θ_1 . Thus every subsequential limit of μ_N has the same law.

For completeness, if the full sequence failed to converge to this law, some bounded continuous test on the space of point measures would have expectations separated from the limiting value along a subsequence. Joint tightness would provide a further convergent subsequence; the preceding identification would force its expectation to converge to that same value, a contradiction.

Since $\sigma = \pi\rho_d(E)$, this is already the asserted convergence

$$\sum_{j=1}^N \delta_{N\pi\rho_d(E)(\lambda_j - E)} \implies \Theta_1.$$

The eigenvalue d leaves every compact microscopic interval, so it may be included or omitted. This proves the main theorem. Only convergence in law has been used; no uniform integrability of uncapped finite- N counting observables is asserted here. $\square$

The stronger conclusions. The main-text argument is now complete. Theorem 1.1 is proved in the appendices. Appendix A adds uniform integrability and obtains the fixed-energy correlation measures. Appendix B treats a deterministic eigenvalue label, derives conditional Gibbs laws for its tagged limit, and identifies the consecutive gaps using Appendix C. These are additional arguments, not formal consequences of vague point-process convergence.

A Fixed-energy correlation universality

We prove Theorem 1.1(ii). The main theorem already gives convergence in law at each fixed energy. The additional issue is uniform integrability: the k -point observable is bounded by a constant times the k th power of a local particle count, and is not a bounded point-process test. We combine the scalar tail from Section 3.1 with a high-probability counting cap, then pass to expectations. Throughout this appendix, E is fixed and all graph probabilities and expectations refer to the original uniform graph law, unless a different graph is explicitly indicated.

Set

$$n_N(I) = \#\{j : N\pi\rho_d(E)(\lambda_j - E) \in I\}$$

for a fixed bounded interval I . Thus $n_N(I) = \mu_N(I)$ in the same coordinates as the main theorem and Theorem 1.1. Proposition 3.3 gives, for some $b > 0$,

$$\mathbb{P}(n_N(I) > M) \leq C_I e^{-c_I M \log(2+M)} + C_I N^{-b}, \quad M \geq M_I. \quad (\text{A.1})$$

Indeed the Poisson kernel at a fixed imaginary parameter bounds the count on I by $C_I \text{Im } s_N$. The graph-event failure contributes $O(N^{-b})$, with b independent of the moment order.

Lemma A.1 (Repair of finitely many geometric defects). *Fix $0 < c_0 < 1/100$, $D_0 > 0$, and $R = \lfloor (c_0/4) \log_{d-1} N \rfloor$. Outside an event of probability $O(N^{-D_0})$, a uniform simple regular graph can be transformed by at most K_{D_0} valid switchings into a simple regular graph such that every radius- R ball has excess at most one and fewer than N^{c_0} of these balls are not trees. The number K_{D_0} is independent of N .*

Proof. For a finite graph H , its excess is $|E(H)| - |V(H)| +$ the number of connected components. It counts the edges that must be removed to obtain a spanning forest. In a connected ball, excess zero means a tree and excess one permits a single independent cycle. The goal is not to make the entire graph a tree, but to eliminate the finitely many clusters containing two or more short independent cycles. We also need a margin in the number of non-tree balls, since a repair can modify some nearby neighborhoods.

We first obtain a margin in the number of non-tree balls. If C_ℓ counts cycles of lengths at most $\ell = O(\log N)$, then for each fixed integer $r \geq 1$,

$$\mathbb{E} C_\ell^r \leq (C_r(\ell + 1))^{C_r} (d - 1)^{r\ell}. \quad (\text{A.2})$$

To see this, expand over ordered r -tuples of cycles, allowing repetitions, and consider their union. It has minimum degree at least two. Write $h = e - v \geq 0$. Suppress degree-two vertices in every component that is not an isolated cycle. The resulting kernel has minimum degree at least three. If it has v_0 vertices and e_0 edges, then $2e_0 \geq 3v_0$ and $e_0 - v_0 \leq h$. Thus $v_0 \leq 2h$ and $e_0 \leq 3h$. Suppression records a length for each kernel edge; these lengths are at most $r\ell$. The kernel, subdivisions, isolated cyclic components, and the r cycle markings have at most $(C_r(\ell + 1))^{C_r(h+1)}$ possibilities. For each union the expected number of embeddings in the configuration model is at most

$$\frac{N^v \prod_x (d)_{\deg x}}{(Nd - 1)(Nd - 3) \cdots (Nd - 2e + 1)} \leq 2N^{-h} (d - 1)^e.$$

Here $(d)_j = d(d-1)\cdots(d-j+1)$ is the falling factorial, $(d)_j \leq d(d-1)^{j-1}$ for $j \geq 1$, and $e = O(\log N)$; degree greater than d gives zero embeddings. The sum over $h \leq r\ell$ is bounded by a geometric series, since a fixed power of $\log N$ divided by N tends to zero. Conditioning on simplicity costs a bounded factor. This proves (A.2).

Take $\ell = 2R + 1$ and $r > 8(D_0 + 3)/c_0$. Markov's inequality gives $C_\ell \leq N^{5c_0/8}$ outside probability $O(N^{-D_0-2})$. Every non-tree radius- R ball contains a cycle of length at most $2R + 1$. Each such cycle can belong to at most $C_d \ell (d-1)^R$ of these balls. Thus their number is at most

$$C_d \ell (d-1)^R C_\ell \leq N^{7c_0/8+o(1)} < \frac{1}{2} N^{c_0}. \quad (\text{A.3})$$

Call a vertex bad if its radius- $2R$ ball has excess at least two. For fixed prescribed roots, expose their radius- $2R$ balls sequentially in the configuration model, including all half-edges at depth $2R$ and therefore discovering vertices at depth $2R + 1$. Stop if the exposed vertex sets meet. Each exploration makes at most $M = C_d(d-1)^{2R+1} = N^{c_0/2+o(1)}$ pairing queries. Before stopping, each query has conditional probability at most $C_d M/N$ of a collision within the current exploration. Each pairing to a previously discovered vertex increases the excess by at most one; a pairing to a new vertex adds one edge and one vertex and does not increase it. Starting from a root, excess at least two therefore requires two collision queries. For one exploration there are at most M^2 choices of their locations and each pair has conditional probability at most $(C_d M/N)^2$, giving the bound $C_d M^4/N^2$. If the prescribed roots are at distance greater than $4R + 2$, stopping cannot occur. Choosing two query locations for each root and summing over ordered tuples therefore gives

$$\mathbb{P}(\text{there are } t \text{ separated bad roots}) \leq N^t (C_d M^4/N^2)^t = N^{-(1-2c_0)t+o(1)}.$$

Choose the fixed integer t with $(1 - 2c_0)t > D_0 + 3$. A maximal such collection has fewer than t vertices and covers every bad vertex within distance $4R + 2$.

By [23, Proposition 2.1], applied with parameter $8c_0$, every radius- $8R$ ball has excess at most a fixed K_0 , outside probability $O(N^{-D_0-2})$, provided $(1 - 8c_0)K_0 > D_0 + 3$. For each chosen center, select a spanning tree of its radius- $8R$ ball. Delete the union $\mathcal{F}$ of the omitted edges; its size is at most tK_0 . Each originally bad radius- $2R$ ball is contained in one of these radius- $8R$ balls, since $6R + 2 \leq 8R$ for large N . The remaining graph $\mathcal{G} \setminus \mathcal{F}$, of maximum degree d , consequently satisfies

$$\text{excess } B_{2R}(v, \mathcal{G} \setminus \mathcal{F}) \leq 1 \quad \text{for every } v. \quad (\text{A.4})$$

For vertices that were not bad, this also follows by monotonicity of excess under taking subgraphs.

Write the deleted edges as $\{l_j, a_j\}$. For each one choose an oriented edge (b_j, c_j) of the original graph such that $B_{10R}(b_j)$ is a tree and $\{b_j, c_j\}$ is at distance greater than $20R$ from all deletion endpoints and previously chosen remote edges. Such choices exist: the structural estimate with parameter $10c_0$ excludes at most $N^{10c_0} = o(N)$ roots, with arbitrarily large polynomial probability, and the distance exclusions cover at most $C_d t K_0 (d-1)^{20R} = O_{D_0}(N^{5c_0}) = o(N)$ vertices. Perform the switches

$$\{l_j, a_j\}, \{b_j, c_j\} \mapsto \{l_j, b_j\}, \{a_j, c_j\}.$$

The four endpoints of each switch are distinct, and its added edges were absent by the distance restrictions. Thus the switches are valid and restore regularity.

Deleting $\{b_j, c_j\}$ separates its two branches inside the remote tree ball. A path of length at most $2R$ entering either branch through a new edge cannot leave through its distant boundary or through the other branch. It also cannot encounter another repair site. Hence

the new branches attach trees and give no new short connection between the old vertices. The cyclic part of a final radius- R ball is contained either in an old radius- R ball or in the old radius- $2R$ ball of its first attachment vertex. Equation (A.4) bounds its excess by one. Only $O_{D_0}((d-1)^R)$ roots can have their radius- R ball changed: each is within distance R of a switching endpoint in the original or final graph. Together with (A.3), this gives fewer than N^{c_0} non-tree balls. Choose all centers, spanning trees, and remote edges in a fixed label order to obtain a deterministic repair map. The order of the repair choices matters only for defining the map, not for the estimates. The list of deleted edges is fixed first. Remote edges are chosen away from *all* deletion endpoints, so an earlier switching cannot remove a later deletion edge or create a forbidden edge for it. Removing all deletion and remote edges first, and then adding the new edges, describes the same final graph. This viewpoint makes (A.4) directly applicable to the old part of every final short ball. The number of repair sites is bounded independently of N , which is essential for the counting cap in the next lemma. $\square$

Lemma A.2 (A high-probability microscopic counting cap). *For every $D_0 > 0$ and bounded interval I , there exist $B, C < \infty$ such that*

$$\mathbb{P}(n_N(I) > C(\log N)^B) \leq C_{D_0, I} N^{-D_0}. \quad (\text{A.5})$$

Proof. The repaired graph is not uniformly distributed in its geometric class. We therefore cannot simply apply a uniform-graph local law to its distribution. Instead we bound how many original graphs can have the same repaired image, and choose the failure exponent of the local law large enough to absorb this multiplicity.

Let Ψ be the map in Lemma A.1, using at most K switches. Its image belongs to the geometric class $\overline{\Omega}(c_0)$ of [23, Definition 1.1], because $\omega_d \geq 1$. Each simple regular graph has at most $C_d N^2$ switching neighbors. By reversibility every fiber of Ψ has size at most $(K+1)(C_d N^2)^K$. If $\mathcal{A} \subset \overline{\Omega}(c_0)$, then

$$\mathbb{P}(\Psi(\mathcal{G}) \in \mathcal{A}, \mathcal{G} \text{ is repairable}) \leq (K+1)(C_d N^2)^K \mathbb{P}(\mathcal{G} \in \mathcal{A}).$$

The local law [23, Theorem 4.2] holds with arbitrarily large polynomial probability under the uniform law on $\overline{\Omega}(c_0)$. Take its failure exponent greater than $2K + D_0 + 2$. At height $\eta = (\log N)^B/N$, with B sufficiently large, it gives $\text{Im } m_{\Psi(\mathcal{G})}(E + i\eta) \leq C$ and hence at most $C(\log N)^B$ eigenvalues in $E + I/(N\pi\rho_d(E))$. The fiber bound transfers this event to the repaired graph with failure $O(N^{-D_0})$. Finally

$$\text{rank}(A_{\mathcal{G}} - A_{\Psi(\mathcal{G})}) \leq 2K,$$

so rank interlacing at the two interval endpoints bounds the change in the eigenvalue count by $4K$, proving (A.5). The fiber estimate follows by reversing at most K switchings: a fixed image has at most $\sum_{j=0}^K (C_d N^2)^j \leq (K+1)(C_d N^2)^K$ possible preimages. For a bad subset $\mathcal{A}$ of the geometric class, its relative size under all regular graphs is no larger than its relative size under that class. Thus the local-law failure probability multiplied by the fiber bound is still $O(N^{-D_0})$ with the chosen exponent. Finally, an interval count is a difference of two half-line counts. A rank- $2K$ perturbation changes either half-line count by at most $2K$, explaining the factor $4K$ rather than $2K$ for the interval. $\square$

Lemma A.3 (Uniform integrability of microscopic counts). *For each fixed integer $p \geq 1$, the family $n_N(I)^p$ is uniformly integrable.*

Proof. Uniform integrability means that the expected contribution from large values tends to zero uniformly in N , not merely that those values have small probability. We combine two estimates with different purposes: the strong scalar tail controls moderate counts, while the

arbitrarily high-probability cap handles the rare graphs on which that tail estimate has only a polynomial error.

Choose $D_0 > p + 1$ in Lemma A.2, and let $L_N = C(\log N)^B$ be its bound. Integrating (A.1) up to L_N , and using $n_N(I) \leq N$ beyond it, gives

$$\limsup_{N \rightarrow \infty} \mathbb{E}[n_N(I)^p \mathbf{1}_{n_N(I) > M}] \leq C_p M^p e^{-c_I M \log(2+M)} + C_p \int_M^\infty t^{p-1} e^{-c_I t \log(2+t)} dt.$$

The discarded terms are at most $CN^{-b}L_N^p + CN^{p-D_0}$ and tend to zero. The right-hand side tends to zero as $M \rightarrow \infty$, which proves uniform integrability. The tail-integration identity behind the estimate is

$$\mathbb{E}[X^p \mathbf{1}_{M < X \leq L}] \leq M^p \mathbb{P}(X > M) + \int_M^L p t^{p-1} \mathbb{P}(X > t) dt \quad (X \geq 0).$$

Here $X = n_N(I)$ and $L = L_N$. The remaining event $X > L_N$ contributes at most $N^p \mathbb{P}(X > L_N) \leq CN^{p-D_0}$. For the finitely many N below any chosen threshold, the variable is bounded by N , so its tail eventually vanishes. The displayed limsup estimate therefore proves uniform integrability for the whole family, not just along a subsequence. $\square$

Proof of Theorem 1.1(ii). For a point measure ζ , let $\mathcal{C}_\varphi(\zeta)$ be the sum of φ over its ordered distinct particles. Take a bounded interval I containing all coordinate projections of $\text{supp } \varphi$ in its interior. Then

$$|\mathcal{C}_\varphi(\zeta)| \leq \|\varphi\|_\infty \zeta(I)^k.$$

At a simple locally finite configuration with no particles at the boundary of I , this functional is continuous for vague convergence: match the finitely many particles in a compact neighborhood of $\text{supp } \varphi$ and pass their coordinates to the limit. The main point-process theorem gives convergence of the graph point measure to Θ_1 in these same coordinates. This process is simple. Its intensity measure is $\pi^{-1} dx$, so it almost surely has no particle at either fixed endpoint of I . Lemma A.3 gives convergence of the expectation of the first sum in (1.1) to $\mathbb{E}\mathcal{C}_\varphi(\Theta_1)$. The GOE expectation has the same limit by the classical bulk correlation asymptotics, see [8, (2.3)], after multiplying the usual density-normalized coordinates by π on both sides. Equivalently, one applies that result to the test $\mathbf{v} \mapsto \varphi(\pi \mathbf{v})$. This proves (1.1). To justify the expectation passage explicitly, compose $\mathcal{C}_\varphi$ with a continuous truncation of the real line. The resulting observable is bounded and continuous at almost every configuration under the limiting law. The almost-sure-continuity version of the continuous mapping theorem therefore gives convergence of its expectation. Then remove the truncation using $|\mathcal{C}_\varphi| \leq \|\varphi\|_\infty n_N(I)^k$ and Lemma A.3. Repeated labels are excluded in the factorial sum, even if two finite- N eigenvalues happen to coincide; at the simple limiting configuration the matched-particle description is unambiguous. $\square$

B Gap universality at a deterministic label

We prove Theorem 1.1(i). A deterministic eigenvalue label cannot be recovered by simply centering the fixed-energy limit at one of its particles. We therefore adapt the marked construction of Section 3 while retaining the chosen label. The rooted loop equation gives rigidity; coordinate integration by parts gives conditional Gibbs laws. The comparison in Appendix C then identifies the gap law. This argument uses the eigenvector and switching inputs from the main text, but does not require the counting upgrade in Appendix A.

B.1. Tagged marked limits

For a deterministic bulk label $i = i_N$, pass to a subsequence with $\gamma_i \rightarrow E$ and, in the descending eigenvalue order, put

$$\sigma = \pi \rho_d(E), \quad x_{N,j} = N\sigma(\lambda_{i-j} - \lambda_i), \quad j \in \{i - N, \dots, i - 1\}.$$

The relative labels increase, $x_{N,0} = 0$, and every fixed offset stays in the bulk. Throughout this appendix the point measure is centered at the tag:

$$\mu_N = \sum_{j=i-N}^{i-1} \delta_{x_{N,j}}.$$

Thus μ_N now denotes the tagged measure, not the fixed-energy measure from Section 3. Its limit μ is likewise rooted at label zero; it is not identified with the unrooted law of Θ_1 . The proof retains the limiting labelled array and identifies its gaps separately. All estimates are uniform in the original bulk label.

The change of sign in the index is deliberate: although the original spectrum is decreasing, the relative coordinates $(x_{N,j})$ are increasing. For example, $x_{N,-1} = N\sigma(\lambda_{i+1} - \lambda_i) \leq 0$, while $x_{N,1} = N\sigma(\lambda_{i-1} - \lambda_i) \geq 0$. A switching can move the chosen eigenvalue by more than a prescribed microscopic amount, so fixed-energy tightness cannot simply be reused after substituting $E = \lambda_i$. The next lemma supplies the needed random-center estimate.

Retaining V, H from (3.15) and writing $z_i(w) = \lambda_i + w/(N\sigma)$, define

$$s_{N,i}(w) = \sigma^{-1}(m_N(z_i(w)) - a_E), \quad S_{N,i}(w) = \sigma^{-1}V^{-1/2}(K(z_i(w)) - H)V^{-1/2}.$$

Lemma B.1 (Scalar tightness around a deterministic label). *Fix $c_* > 0$ and a sufficiently small $\eta_* > 0$. There are $C, c, \beta > 0$ and a graph event Ω_N with $\mathbb{P}(\Omega_N^c) \leq CN^{-\beta}$ such that, for $T_i = |m_N(\lambda_i + i\eta)|$,*

$$\mathbb{P}(\Omega_N \cap \{T_i > M\}) \leq Ce^{-c(\log M)^2} + CN^{-\beta}, \quad c_*/N \leq \eta \leq \eta_*, \quad M \geq C. \quad (\text{B.1})$$

Consequently the events $\Gamma_{N,i,\eta} = \Omega_N \cap \{T_i \leq \log N\}$ have probability tending to one and $\sup_{N,i,\eta} \mathbb{E}[\mathbf{1}_{\Gamma_{N,i,\eta}} T_i^r] < \infty$ for every fixed $0 < r < \infty$.

Proof. The proof differs from the fixed-energy argument in one essential way: after switching, the old center need not be the new eigenvalue with the same label. We first find a nearby reversed eigenvalue with one of five possible labels, then control the reversed compression uniformly over all possible small shifts of that center. Only after this uniform estimate is available do we select the nearby label.

Choose three nested compact bulk intervals around E and include coarse rigidity in the event of Lemma 3.2. The uniform counting bound and separation from the outside spectrum make (3.10) uniform in the middle interval, including all centers below and all labels within $O(\log N)$ of i .

Write $z = \lambda_i + i\eta$ and $T = T_i$. Since $D^2 = I_2$, (3.8) gives $K + D/2 = m_N(z)V(z) + O(1) + (K - \bar{K})$. The least singular value of $V(z)$ is bounded below. For fixed small $\delta > 0$ and large T ,

$$\|K - \bar{K}\| \leq \delta T \implies \|(K + D/2)^{-1}\| \leq C/T.$$

The conditional failure probability is C/T . Let $\mathbf{g}_i = \sqrt{N}W^T \mathbf{u}_i$. By (3.7), $\mathbb{E}_*|\mathbf{g}_i|^2 = 2$, so $|\mathbf{g}_i|^2 \leq c_1 N \eta T$ also fails with probability at most C/T . On an accepted proposal satisfying both bounds, (3.2) gives

$$\langle \mathbf{u}_i, \tilde{G}(z) \mathbf{u}_i \rangle = \frac{i}{\eta} + \frac{\mathbf{g}_i^T (K + D/2)^{-1} \mathbf{g}_i}{N \eta^2}, \quad \text{Im} \langle \mathbf{u}_i, \tilde{G}(z) \mathbf{u}_i \rangle \geq \frac{3}{4\eta},$$

for small c_1 . Let ν be the spectral measure of $\tilde{A}$ at the unit vector $\mathbf{u}_i$, and write $p = \nu([\lambda_i - \eta, \lambda_i + \eta])$. The Poisson kernel is at most $1/\eta$ everywhere and at most $1/(2\eta)$ outside this interval. Consequently

$$\frac{3}{4\eta} \leq \int \frac{\eta}{(t - \lambda_i)^2 + \eta^2} d\nu(t) \leq \frac{p}{\eta} + \frac{1-p}{2\eta},$$

so $p \geq 1/2$. In particular, the interval contains a reversed eigenvalue. Interlacing supplies a nearby eigenvalue with $|j - i| \leq 2$: either extreme label can replace a nearby eigenvalue beyond this range.

For a scalar Herglotz transform, parameters of equal height whose real parts differ by at most that height satisfy

$$\operatorname{Im} m(z) \asymp \operatorname{Im} m(z_0), \quad |m(z) - m(z_0)| \leq C \operatorname{Im} m(z_0), \quad |m(z)| \asymp |m(z_0)|.$$

These bounds follow by comparing denominators and remain valid on fixed scaled compact subsets of $\mathbb{C}^+$. Together with (3.6) they give $\tilde{T}_j := |\tilde{m}_N(\tilde{\lambda}_j + i\eta)| \asymp T$. By (3.5), $\tilde{K}(z)$ is bounded, whereas its conditional mean has size comparable with $\tilde{T}_j$.

To estimate this deviation at the old, proposal-dependent center, set

$$W_j(w) = \tilde{K}(\tilde{\lambda}_j + \eta w) - \widetilde{\tilde{K}}(\tilde{\lambda}_j + \eta w).$$

Cauchy's formula on the circle $|w - 2i| = 7/4$ and (3.10) give

$$\mathbb{E}_* \sup_{|u| \leq 1} \|W_j(u + i)\|^2 \leq C(1 + \tilde{T}_j). \quad (\text{B.2})$$

The circle has heights in $[1/4, 15/4]$ and surrounds $[-1, 1] + i$, so we use the variance lemma with its height range adjusted accordingly. The centers used here correspond to fixed labels of the reversed graph. In this display $\mathbb{E}_*$ averages proposals with the reversed graph held fixed, and $\widetilde{\tilde{K}}$ is that conditional proposal mean. The old center is selected only after taking the supremum in (B.2).

Put $F_i(M) = \mathbb{P}(\Omega_N \cap \{T_i > M\})$. Exact proposal reversal, (B.2), and conditional Markov give, after absorbing the initial failures and rejection,

$$F_i(M) \leq \frac{C}{M} \sum_{|j-i| \leq 2} F_j(c_0 M) + CN^{-\beta}, \quad 0 < c_0 < 1.$$

At depth n , every label lies within $2n$ of the starting label. The sum over the at most five choices at each step can be absorbed by replacing C with $5C$. If $n \asymp \log M / |\log c_0|$ is the first time that $c_0^n M$ reaches a sufficiently large constant, the homogeneous product is bounded by

$$\prod_{j=0}^{n-1} \frac{5C}{c_0^j M}.$$

Its logarithm is $n \log(5C) - n \log M - \frac{n(n-1)}{2} \log c_0 = -c(\log M)^2 + O(\log M)$, with a positive c after reducing constants. Choosing the terminal threshold large makes all successive coefficients at most $1/2$, so the accumulated exceptional terms are bounded by a geometric sum. Since $T_i \leq N/c_*$, all labels remain within $O(\log N)$ of i . This proves (B.1); integration of the tail bound up to $\log N$ gives the capped moments. This tail gives all capped moments because $\int_C^\infty M^{r-1} e^{-c(\log M)^2} dM < \infty$ for each fixed r . At $M = \log N$, however, the leading failure bound is $e^{-c(\log \log N)^2}$. It tends to zero but is not asserted to be a negative power of N . In the tagged subsequence arguments below, only a vanishing cap failure probability is needed. $\square$

Proposition B.2 (Construction and normalization of a tagged limit). *Every sequence has a joint subsequential limit of all fixed relative labels and the scalar and marked transforms centered at λ_i . It has $x_0 = 0$, infinitely many particles in both directions, and the representations (3.19) and (3.21). The marks are independent standard real two-dimensional Gaussians conditionally on the full scalar state. The bounds (3.20) and (3.22) hold at every fixed label, with uniform constants, and*

$$s(iY) \longrightarrow i \quad \text{in probability,} \quad \mathbb{E}\|S(iY) - s(iY)I_2\|^2 \leq C/Y. \quad (\text{B.3})$$

Its law is invariant under Gaussian rotations and under $\mathcal{T}$ followed by recentering at the same ordered label.

Proof. We separate construction of a rooted point measure from tightness of its labels. Initially a fixed neighboring label is allowed to escape to infinity. We first prove the normalization at imaginary infinity and show that infinitely many particles remain on both sides. Only then can fixed neighboring labels be recovered without circular reasoning.

1. *Construct the scalar field and Gaussian marks.* First apply Lemmas 3.4–3.6 at λ_i , using Lemma B.1 for scalar moments. The uniform variance estimate above supplies the centered tail bound, and Corollary 1.3 applies to graph-measurable finite prefixes. All retained eigenvalues converge to E by coarse rigidity. For padding, use the window $[\lambda_i - N^{-a/2}, \lambda_i + N^{-a/2}]$, where a is the coarse-rigidity exponent; it contains enough eigenvalues for every fixed prefix. This allows γ_i to approach E arbitrarily slowly. The number of selected vectors is fixed while N tends to infinity. At the fixed constants E, σ, V, H , the mean error is

$$O((|\lambda_i - E| + N^{-1})(1 + |m_N|)) = o_{\mathbb{P}}(1)$$

on compact sets by coarse rigidity and capped moments. The squared conditional whitening error is bounded by $C_{R,B}(|\lambda_i - E| + R/N)^2 \mu_N([-R, R])^2$ and vanishes on the two height caps. The outside polynomial approximation still converges to $(1 - \vartheta(t))/(t - E)$. We obtain a rooted marked limit, initially allowing escaping neighbors. Uniform height moments remove its scalar linear coefficient and give the second bound in (B.3).

2. *Define switching without changing the origin.* The old-origin reverse $(s^\circ, S^\circ) = \mathcal{T}(s, S)$ exists analytically. Two particles make $\text{Im } S$ positive definite. With only the root, $S(w) = aI_2 - \xi\xi^T/w$; a kernel of $S + qD$ would lie in $\xi^\perp \cap \ker(aI_2 + qD)$, an event of probability zero since $q > 0$. Passing the inverse identities to the limit gives

$$|s^\circ(w) - s(w)| \leq \frac{2\pi}{|\text{Im } w|}, \quad (S^\circ + qD)L(S + qD)L^T = -c^2I_2. \quad (\text{B.4})$$

3. *Rule out a reverse matrix linear term.* If $u \leq e \leq v$ are consecutive eigenvalues, every spectral point t lies outside (u, v) . For $t \leq u$, its distance from e is at least its distance from u ; for $t \geq v$, use v instead. Comparing each spectral projection's nonnegative Poisson weight gives

$$\text{Im } G(e + i\eta) \preceq \text{Im } G(u + i\eta) + \text{Im } G(v + i\eta).$$

Let S_N° denote the reverse field at the old origin. Fix $Y \geq 1$, put $\eta = Y/(N\sigma)$, and intersect acceptance with the five reversed caps $\tilde{\Gamma}_{N,i+k,\eta}$ to obtain $\Theta_{N,Y}$. Interlacing, reversal and $\text{Tr } \bar{K} = 2\hat{m}_N$ give

$$\mathbb{E}[\mathbf{1}_{\Theta_{N,Y}} \text{Tr } \text{Im } S_N^\circ(iY)] \leq C \sum_{|k| \leq 2} \mathbb{E}[\mathbf{1}_{\tilde{\Gamma}_{N,i+k,\eta}} \text{Im } \hat{m}_N(\tilde{\lambda}_{i+k} + i\eta)] \leq C.$$

Since $\mathbb{P}(\Theta_{N,Y}) \rightarrow 1$, Fatou yields

$$\sup_{Y \geq 1} \mathbb{E} \operatorname{Tr} \operatorname{Im} S^\circ(iY) < \infty. \quad (\text{B.5})$$

Thus S° has no matrix linear coefficient.

4. *Show that the reverse process is nonempty.* An empty reverse scalar measure would make the old measure finite by the rank interval bound. Absorb its compensators into a and, writing $n \geq 1$ for its multiplicity, set

$$Q(w) = \prod_{j=1}^n (x_j - w) \det \left(aI_2 + qD + \sum_{j=1}^n \frac{\xi_j \xi_j^T}{x_j - w} \right).$$

The polynomial Q satisfies $s^\circ = a - Q'/Q$. Its degree is n unless $a = \pm q$, when it is $n - 1$ almost surely: the leading coefficient is a nonzero multiple of a sum of Gaussian squares in the null direction of $aI_2 + qD$. Its zeros are real by the Herglotz property. Emptiness therefore forces $n = 1$, $a = \pm q$. Direct inversion gives the positive linear coefficient $c^2 \xi_2^{-2} \mathbf{e}_1 \mathbf{e}_1^T$ or $c^2 \xi_1^{-2} \mathbf{e}_2 \mathbf{e}_2^T$, contradicting (B.5).

5. *Identify the value at imaginary infinity.* Extract jointly the five reverse anchored states and their shifts $\Delta_k = N\sigma(\lambda_{i+k} - \lambda_i)$ in the extended real line. A finite reverse pole and interlacing imply that some Δ_k is finite. At each fixed anchor the centered Gaussian field $Z_k = S_k - s_k I_2$ satisfies $\mathbb{E} \sup_{|u| \leq R} \|Z_k(u + iY)\|^2 \leq C_R/Y$ for large Y , by Cauchy's formula and its Gaussian series. Hence

$$\mathbb{P}(\|S^\circ(iY) - s^\circ(iY)I_2\| > \varepsilon) \leq \mathbb{P}(\min_k |\Delta_k| > R) + \frac{C_R}{\varepsilon^2 Y}.$$

Let $Y \rightarrow \infty$ and then $R \rightarrow \infty$. Here the supremum over possible shifts is taken before the anchor is selected. Write $S = sI_2 + Z$ and $S^\circ = s^\circ I_2 + Z^\circ$. Both centered fields tend to zero in probability at iY , and both scalar fields are tight uniformly in Y by the height moments and the rank bound. Since $LDL^T = -D$, the leading part of the matrix product in (B.4) is

$$(s^\circ I_2 + qD)(sI_2 - qD) = (s^\circ s - q^2)I_2 + q(s - s^\circ)D.$$

Taking half its trace and using $c^2 = 1 + q^2$ gives $s^\circ(iY)s(iY) + 1 \rightarrow 0$ in probability. The scalar rank bound then gives $s(iY)^2 + 1 \rightarrow 0$, and the upper half-plane selects i .

6. *Prevent labels from escaping.* If only finitely many particles lie to the left, remove their terms. The remaining real summands $x/(x^2 + Y^2) - x/(1 + x^2)$ decrease with Y . An almost surely convergent subsequence in (B.3) makes their limit finite, so monotone convergence gives $\sum_{x \geq 1} x^{-1} < \infty$. The bound $Y/(x^2 + Y^2) \leq (2x)^{-1}$ would then make the imaginary part tend to zero. Reflection excludes a finite right side. For any fixed positive integer j , the limiting process has at least j particles to the right in $[0, R]$ with probability tending to one as $R \rightarrow \infty$. Choose R to avoid deterministic atoms and use convergence of the integer counts in that interval. The same is then true of the finite- N process for large N , which bounds its j th neighboring position. Apply the same argument on the left. This is the additional ingredient, beyond local counting tightness, that bounds each label. Rank interlacing then bounds the same-label reverse shift. Diagonal extraction gives the joint labelled state and the asserted moments and marking at each offset.

7. *Retain the deterministic label under reversal.* The label in the inverse map is fixed by the exact inertia identity

$$N_{>}(\tilde{A}, t) - N_{>}(A, t) = n_{-}(D/2 + K(t)) - 1 = n_{-}(S_{N,i}(w) + qD) - 1, \quad t = \lambda_i + \frac{w}{N\sigma}, \quad (\text{B.6})$$

where w is real and t lies in a common spectral gap. Here $N_>$ counts eigenvalues strictly above its argument, and n_- counts negative eigenvalues. For completeness, apply Sylvester's law of inertia to the block matrix

$$\begin{pmatrix} A - t & W \\ W^T & -D/2 \end{pmatrix}.$$

Taking the Schur complement of $-D/2$ gives $\tilde{A} - t$, whereas taking that of $A - t$ gives $-(D/2 + K(t))$. For a common real gap, all the relevant matrices are invertible. Since $-D/2$ has one positive and one negative eigenvalue, equality of the positive inertias gives the first equality in (B.6). The second follows from the positive congruence $D/2 + K = \sigma V^{1/2}(S_{N,i} + qD)V^{1/2}$. Integer-valued measure convergence excludes poles on compact subsets of common gaps; Nevanlinna bounds then give analytic convergence there. Locally constant inertia therefore fixes the limiting counting offset and label zero. Exact proposal reversal preserves the recentered law; Gaussian rotations preserve it with all labels fixed. The seven steps also show the roles of the two normalizations. Keeping the old origin makes the rank-two algebra exact. Recentering at the same ordered label is postponed until tightness and the inertia identity specify that label in the limit. At every other fixed relative label the original index is still deterministic and bulk, so the same construction gives the same moment bounds. No translation invariance or label-stationarity of the limiting point process is assumed in this argument. $\square$

Lemma B.3 (Simplicity of the tagged limiting process). *The labelled scalar configuration in Proposition B.2 is almost surely simple.*

Proof. We compare the multiplicities before and after one switching. The matrix field can cancel at most two scalar particles at an old location, while a new location is generated by a zero of a 2×2 determinant. Independent Gaussian marks make the exceptional degeneracies have probability zero. Stationarity then transfers simplicity of the image back to the original configuration.

At an old pole of multiplicity $m \geq 2$, its Gaussian matrix pole weight R is positive definite. With $t = w - x$, $S + qD = -R/t + O(1)$, so $(S + qD)^{-1} = -tR^{-1} + O(t^2)$. The matrix image is analytic. Moreover $\det(S + qD) = t^{-2} \det R (1 + O(t))$, so $\partial_w \log \det(S + qD) = -2/t + O(1)$. The scalar pole is $-m/t$ in the variable $t = w - x$; after subtracting this logarithmic derivative its coefficient is $-(m - 2)/t$. Thus its multiplicity is $m - 2$. Stationary Gaussian marking excludes $m \geq 3$, since every scalar atom has a nonzero matrix pole weight.

An old double pole disappears. At an old simple pole write $S + qD = -\xi\xi^T/t + A_0 + O(t)$. The determinant pole coefficient is $-\xi^T \text{adj}(A_0)\xi$. The matrix A_0 is independent of ξ and is nonzero almost surely: after conditioning on all but one other mark, it is a fixed matrix plus a nonzero multiple of a Gaussian rank-one matrix. Thus this coefficient is nonzero almost surely, and the scalar logarithmic derivative cancels the simple pole.

In an old spectral gap, $S'(x)$ is positive definite. If one eigenvalue of $S(x) + qD$ vanishes and the other does not, the derivative of the vanishing eigenvalue is $\mathbf{v}^T S'(x) \mathbf{v} > 0$ for its unit eigenvector $\mathbf{v}$. The determinant therefore has a simple zero. A multiple zero can only occur if both eigenvalues vanish, namely $S(x) + qD = 0$. To exclude the latter, condition on all but one Gaussian mark at a particle x_p outside the gap and write $S(x) + qD = A_0(x) + \xi\xi^T/(x_p - x)$. Since $A'_0(x) > 0$, its eigenvalues strictly increase and $\det A_0(x)$ has at most two zeros in the gap. At either conditionally fixed point, $\xi\xi^T = -(x_p - x)A_0(x)$ has probability zero. A countable union over gaps and stationarity prove simplicity. The strict inequalities used here hold almost surely simultaneously in every gap: after removing any one fixed mark, there remain infinitely many independent Gaussian marks spanning $\mathbb{R}^2$, and each contributes a

positive coefficient to $S'(x)$. There are only countably many particles and gaps, so all the zero-probability exceptional events can be discarded at once. Each old scalar pole then disappears, and every newly created pole is simple. By the law-preserving recentered switching, the original scalar configuration is simple as well. $\square$

Lemma B.4 (The recentered stationary flow). *Apply $\Phi_\theta = \mathcal{T}\mathcal{R}_\theta\mathcal{T}$ with the label convention (B.6), and recenter at its label zero. This is a law-preserving flow with locally Lipschitz ordered particle trajectories. Put $R_j = \xi_j \xi_j^T$, $\zeta_j = (R_j)_{12}$, and $H_j = \lim_{w \rightarrow x_j} (S(w) - R_j/(x_j - w))$. Thus R_j is the matrix pole weight and H_j the regular part; the complex-analytic residue at $w = x_j$ is $-R_j$. Its moving-frame vector field is*

$$\begin{aligned} \mathcal{V}_0 x_j &= -2\kappa(\zeta_j - \zeta_0), & \mathcal{V}_0 R_j &= \alpha[L, R_j] + \kappa(H_j P R_j + R_j P H_j), \\ \mathcal{V}_0 s &= 2\kappa(S'_{12} - \zeta_0 s'), & \mathcal{V}_0 S &= \alpha[L, S] + \kappa(P + S P S) - 2\kappa\zeta_0 S', \end{aligned}$$

with α, κ from (3.31). Invariant differentiation $\mathbb{E}\mathcal{V}_0 F = 0$ holds for smooth functions of finitely many off-real fields whose function and derivatives have polynomial growth. It also holds for finite-coordinate tests involving pole weights, with compact position cutoffs and collision cutoffs at each pole whose regular part occurs in the derivative.

Proof. Off the real axis the formulas follow from Lemma 3.8. For coordinate integration by parts we need more: particle paths must remain controlled when they approach a collision, and the chosen label must not jump. We establish these facts by a finite symmetric matrix linearization on each bounded real interval before matching pole coefficients.

For the intermediate unrooted field $S^1 = \mathcal{T}S$, the total counting shift at a common real gap is

$$[n_-(S + qD) - 1] + [n_-(O_\theta S^1 O_\theta^T + qD) - 1] \in [-2, 2].$$

Its changes are precisely crossings of final poles. Thus each ordered pole stays between the original labels offset by two, and the count at $\theta = 0$ fixes its continuous branch.

To handle possible collisions, we use a finite matrix representation. On a bounded real interval, remove the finitely many intermediate poles $t_1, \dots, t_m$ and write

$$O_\theta S^1(w) O_\theta^T + qD = H_\theta(w) + V_\theta \text{diag}(t_\ell - w)^{-1} V_\theta^T.$$

The regular part is analytic across the interval, and $H'_\theta(x)$ is positive definite, since the infinitely many outside Gaussian pole weights span $\mathbb{R}^2$. The finite symmetric matrix

$$Q_\theta(w) = \begin{pmatrix} H_\theta(w) & V_\theta \\ V_\theta^T & \text{diag}(w - t_\ell) \end{pmatrix}$$

satisfies

$$\det Q_\theta(w) = \prod_{\ell=1}^m (w - t_\ell) \det(O_\theta S^1(w) O_\theta^T + qD), \quad \partial_w Q_\theta = \text{diag}(H'_\theta, I_m) > 0.$$

The product cancels the intermediate scalar poles in the logarithmic derivative. Thus the determinant zeros are exactly the final scalar poles, with multiplicities. On compact parameter intervals this derivative has a strictly positive lower bound. Weyl's inequality controls the change of each ordered matrix eigenvalue by the norm of the matrix increment, which is locally Lipschitz in θ . Increasing w increases every eigenvalue at a rate bounded below by the smallest eigenvalue of $\partial_w Q_\theta$. Dividing the first bound by this positive lower bound gives a finite random Lipschitz constant for each ordered zero. This proves local Lipschitz continuity even at multiple roots. At simple poles the implicit function theorem makes them analytic.

Near a simple pole, write $S(w) = R_j/(x_j - w) + H_j + O(w - x_j)$. The rank-one identity $R_j P R_j = (\xi_j^T P \xi_j) R_j = 2\zeta_j R_j$ identifies the double pole of $\kappa S P S$. On the other hand,

$$\mathcal{V}\left(\frac{R_j}{x_j - w}\right) = \frac{\mathcal{V}R_j}{x_j - w} - \frac{R_j \mathcal{V}x_j}{(x_j - w)^2}.$$

Matching double poles yields $\mathcal{V}x_j = -2\kappa\zeta_j$. Matching simple poles yields $\mathcal{V}R_j = \alpha[L, R_j] + \kappa(H_j P R_j + R_j P H_j)$. The minus sign in the double-pole differentiation is important. Subtracting the root velocity gives the moving-frame formulas. By stationarity, the configuration is almost surely simple at each fixed time. Fubini's theorem therefore implies that collision times form a set of Lebesgue measure zero almost surely. Off-real derivatives are integrable by the marked moments and Gaussian moments. If $|x_j| \leq R$ and its two neighboring gaps are at least δ , the matrix Nevanlinna representation gives

$$\|H_j\| \leq C_{R,\delta}(\|S(i)\| + |\xi_j|^2).$$

Indeed the compensated kernel of every other particle is bounded by $[R + (1 + R^2)/\delta]/(1 + x_\ell^2)$. This bounds the regular parts on the supports of the collision cutoffs. The locally Lipschitz trajectories and these integrable derivatives justify the fundamental theorem of calculus. Stationarity at both endpoints then gives $\mathbb{E}\mathcal{V}_0 F = 0$ for the stated tests. A collision cutoff is a smooth function that vanishes whenever a specified neighboring gap is smaller than δ , and is one when that gap is larger than 2δ . It is needed only when the observable involves a regular part H_j or a pole-weight derivative. On its support all the bounds above are uniform; near a collision the observable is identically zero. Thus the local absolutely continuous pieces fit together, and the exceptional collision times of Lebesgue measure zero do not contribute to the integrated stationarity identity. $\square$

B.2. Rooted loop equation and rigidity

Let $\mathbf{x} = (x_j)_{j \in \mathbb{Z}}$ be the increasing configuration of Proposition B.2, which is simple by Lemma B.3, with $x_0 = 0$. Retain $\zeta_j = \xi_{j1}\xi_{j2}$. The unrooted vector field (3.31) moves the pole x_j at velocity $-2\kappa\zeta_j$. Consequently its expression $\mathcal{V}_0$ in the frame following label zero is

$$\mathcal{V}_0 s = 2\kappa(S'_{12} - \zeta_0 s'), \quad \mathcal{V}_0 S_{12} = \alpha(S_{11} - S_{22}) + \kappa(1 + S_{11}S_{22} + S_{12}^2) - 2\kappa\zeta_0 S'_{12}.$$

Deleting the tagged atom leaves the Herglotz function $r(w) = s(w) + 1/w$. For finitely many nonreal arguments v and a differentiable function F of the corresponding values $r(v)$, invariant differentiation of $S_{12}(w)F$ gives

$$0 = \mathbb{E}[(1 + r(w)^2 + r'(w) - 2r(w)/w)F] + 2 \sum_v \mathbb{E}[C(w, v)D_v F], \quad (\text{B.7})$$

Here $D_v F$ denotes differentiation with respect to the *value* $r(v)$ in the test function, whereas ∂_v in the next display differentiates the *spectral parameter*. These two derivatives should not be confused. When conjugate values occur, they are interpreted as Wirtinger derivatives: for $z = x + iy$, $\partial_z = (\partial_x - i\partial_y)/2$ and $\partial_{\bar{z}} = (\partial_x + i\partial_y)/2$. We have

$$C(w, v) = \partial_v \frac{r(w) - r(v)}{w - v} + \frac{r'(v)}{w}, \quad C(w, w) = \frac{1}{2}r''(w) + \frac{r'(w)}{w}.$$

Indeed, conditional Gaussian averaging gives

$$\mathbb{E}[S_{12}(w)\zeta_0 \mid s] = -\frac{1}{w}, \quad \mathbb{E}[S'_{12}(w)\zeta_0 \mid s] = \frac{1}{w^2}, \quad \mathbb{E}[S_{12}(w)S_{12}(v) \mid s] = \frac{s(w) - s(v)}{w - v}.$$

Together with (3.27), these identities give (B.7). Here is the cancellation explicitly. Since $s = r - 1/w$,

$$1 + s(w)^2 + s'(w) - \frac{2}{w^2} = 1 + r(w)^2 + r'(w) - \frac{2r(w)}{w}.$$

For the cross term, Gaussian averaging before deleting the root gives

$$\partial_v \frac{s(w) - s(v)}{w - v} + \frac{s'(v)}{w}.$$

The identity $(s(w) - s(v))/(w - v) = (r(w) - r(v))/(w - v) + 1/(wv)$ shows that its additional $-1/(wv^2)$ cancels the $1/(wv^2)$ from $s'(v)/w$, leaving exactly $C(w, v)$. At $v = w$, Taylor expansion of the divided difference gives $\frac{1}{2}r''(w)$, accounting for the factor one half.

We use (B.7) for functions whose first derivatives have polynomial growth in the fixed off-real fields and whose denominators are separated from zero on the state space. Their derivatives along $\mathcal{V}_0$ are integrable by the marked moment bounds. The labelled flow of Lemma B.4, stationarity at each fixed time, and Fubini justify differentiation of the expectation. The displayed derivatives involve only off-real fields and the root mark, so this argument requires no derivative of ζ_0 .

Lemma B.5 (Local law for the tagged limit). *There is an absolute constant C such that, for every integer $p \geq 1$ and every $w \in \mathbb{C}^+$,*

$$\|s(w) - \mathbf{i}\|_{2p} \leq \frac{C\sqrt{p}}{\operatorname{Im} w}. \quad (\text{B.8})$$

The constants are uniform over all deterministic tag marginals. Moreover,

$$\mathbb{P}(x_1 > R) + \mathbb{P}(-x_{-1} > R) \leq Ce^{-cR^2}, \quad R \geq 1. \quad (\text{B.9})$$

Proof. The test is chosen so that the leading nonlinear term becomes a positive even moment. The factor $z + \mathbf{i}$ in the denominator never vanishes for a Herglotz value z , and retaining it in the estimates cancels the otherwise troublesome factor $\operatorname{Im} z$ in the derivative bounds. This avoids any prior quantitative local law for the limiting process.

Put $Y = \operatorname{Im} w$, $z = r(w)$, and $\Delta = z - \mathbf{i}$. The sums below omit the root label $j = 0$. For $w = u + \mathbf{i}Y$,

$$\operatorname{Im} r(w) = \sum_{j \neq 0} \frac{Y}{|x_j - w|^2}, \quad r'(w) = \sum_{j \neq 0} \frac{1}{(x_j - w)^2}, \quad r''(w) = 2 \sum_{j \neq 0} \frac{1}{(x_j - w)^3}.$$

Use $|x_j - w| \geq Y$ to compare these absolutely convergent sums. The same comparison for the divided difference, at $v = w$ and $v = \bar{w}$, gives

$$|r'(w)| \leq \frac{\operatorname{Im} z}{Y}, \quad |r''(w)| \leq \frac{2\operatorname{Im} z}{Y^2}, \quad |C(w, w)| + |C(w, \bar{w})| \leq \frac{4\operatorname{Im} z}{Y^2}.$$

In (B.7) take the two scalar arguments $z = r(w)$ and $\bar{z} = r(\bar{w})$, and set

$$F(z, \bar{z}) = \frac{(z - \mathbf{i})^{p-1}(\bar{z} + \mathbf{i})^p}{z + \mathbf{i}}.$$

Since $\operatorname{Im} z \geq 0$, the denominator has modulus at least one, and $|\Delta|$, $|z|$, and $\operatorname{Im} z$ are each at most $|z + \mathbf{i}|$. Thus

$$(1 + z^2)^F = |\Delta|^{2p}, \quad |F_z| + |F_{\bar{z}}| \leq \frac{2p|\Delta|^{2p-2}}{|z + \mathbf{i}|}.$$

The term with coefficient $p - 1$ is omitted when $p = 1$. Keeping the denominator in this bound cancels the factor $\operatorname{Im} z$ from the derivative estimates. If $M = \|\Delta\|_{2p}$, (B.7) and Hölder yield

$$M^{2p} \leq \frac{3}{Y} M^{2p-1} + \frac{8p}{Y^2} M^{2p-2}.$$

For $M > 0$, division by M^{2p-2} gives a quadratic inequality, so

$$M \leq \frac{3 + \sqrt{9 + 32p}}{2Y}.$$

The case $M = 0$ is immediate. Adding $|1/w| \leq 1/Y$ proves (B.8). In particular, for a sufficiently small $c > 0$,

$$\sup_{w \in \mathbb{C}^+} \mathbb{E} \exp(c(\operatorname{Im} w)^2 |s(w) - i|^2) < \infty.$$

On $x_1 > R$, every atom is at distance at least $R/2$ from the midpoint $R/2$. For $\eta = R/100$, termwise comparison of the Poisson kernels gives

$$\operatorname{Im} s(R/2 + i\eta) \leq \frac{1}{20} \operatorname{Im} s(R/2 + iR).$$

If both imaginary parts were in $[3/4, 5/4]$, the displayed comparison would imply $3/4 \leq (5/4)/20$, a contradiction. Therefore at least one of the two values of $|s - i|$ is at least $1/4$. Their imaginary heights are fixed positive multiples of R , so exponential Markov gives a bound Ce^{-cR^2} . This gives the right-gap estimate in (B.9); the left-gap estimate follows in the same way on $(-R, 0)$. The exponential moment assertion follows directly from the even moment bounds: expand the exponential and use $\mathbb{E}[Y^{2p}|s(w) - i|^{2p}] \leq (C\sqrt{p})^{2p}$. Stirling's estimate $p! \geq (p/e)^p$ makes the resulting series geometric when c is sufficiently small. All constants therefore remain uniform in the original deterministic tag. $\square$

Lemma B.6 (Rigidity and principal-value normalization). *For every $\epsilon, D > 0$ there is $C_{\epsilon, D}$ such that, for every integer $K \geq 2$,*

$$\mathbb{P} \left(\max_{|j| \leq 2K} |x_j - \pi j| > K^\epsilon \right) \leq C_{\epsilon, D} K^{-D}. \quad (\text{B.10})$$

For every fixed $1 \leq p < \infty$, the maximum on the left has L^p norm at most $C_{p, \epsilon} K^\epsilon$. Almost surely,

$$s(w) = \lim_{M \rightarrow \infty} \sum_{j=-M}^M \frac{1}{x_j - w}, \quad (\text{B.11})$$

locally uniformly off the poles.

Proof. The local law is a bound for a smoothed counting measure. To turn it into labelled rigidity, we first make it uniform on a collection of complex rectangles, then integrate along a contour to estimate interval counts, and finally invert the count starting at $x_0 = 0$. The principal-value identity is obtained only after this rigidity has been proved.

For a dyadic height $h \geq 1$, Cauchy's formula on disks of radius comparable to h and (B.8) give

$$\left\| \sup_{|u-u_0| \leq h/4, h \leq v \leq 2h} v |s(u + iv) - i| \right\|_{2p} \leq C\sqrt{p}.$$

The surrounding circles have heights comparable to h . Covering $|u| \leq 4R$, $1 \leq v \leq 8R$ requires $O(R + \log R)$ such rectangles. At a threshold $A_D \sqrt{\log R}$, the probability for one

rectangle is at most $(C\sqrt{p}/(A_D\sqrt{\log R}))^{2p}$. Choose p to be a sufficiently large multiple of $\log R$ and then A_D large enough. This probability is $O(R^{-D-2})$. A union bound over the $O(R + \log R)$ rectangles gives, with probability at least $1 - C_D R^{-D}$,

$$\sup_{|u| \leq 4R, 1 \leq v \leq 8R} v |s(u + iv) - i| \leq C_D \sqrt{\log R}. \quad (\text{B.12})$$

Here we work throughout with the infinite labelled limit constructed above.

Fix $a, b \in [-R, R]$, $a \leq b$, and let

$$f(t) = \frac{1}{\pi} (\arctan(b - t) - \arctan(a - t)).$$

Positivity and the scalar representation imply

$$\int f \, d\mu - \frac{b - a}{\pi} = \frac{1}{\pi} \operatorname{Im} \int_a^b (s(u + i) - i) \, du.$$

Deform the last integral to height $2R$. Equation (B.12) bounds the top by $C_D \sqrt{\log R}$ and the two vertical sides by $C_D (\log R)^{3/2}$.

For $c \in [-R, R]$ and $1 \leq r \leq 4R$, the same event gives

$$\mu([c - r, c + r]) \leq 2r \operatorname{Im} s(c + ir) \leq C(r + \sqrt{\log R}).$$

The difference between f and the interval indicator is bounded by one within distance one of an endpoint, by C/r on an endpoint annulus of radius $r \in [1, 4R]$, and by CR/r^2 outside $[-4R, 4R]$. Dyadic summation bounds the near contribution by $C_D (\log R + \sqrt{\log R})$. For the far annuli, (B.8) at $i2^n R$ ensures simultaneously $\mu([-2^n R, 2^n R]) \leq C2^n R$, $n \geq 0$, outside an event of probability Ce^{-cR^2} . Their contributions sum to a constant. Thus

$$\sup_{a, b \in [-R, R], a \leq b} \left| \mu((a, b)) - \frac{b - a}{\pi} \right| \leq C_D (\log R)^{3/2}$$

with probability at least $1 - C_D R^{-D}$. Endpoint atoms contribute at most two. With $R = 10\pi K$, the estimate first confines all particles with $|j| \leq 2K$ to the interval; counting from $x_0 = 0$ then gives (B.10).

For the L^p conclusion, let $H_K = \max_{|j| \leq 2K} |x_j - \pi j|$. The nearest-gap tail at every fixed tag gives a uniform bound for the L^{2p} norm of each adjacent gap. Expressing a position as a sum of at most $2K$ gaps and applying Minkowski gives $\|H_K\|_{2p} \leq C_p K$. On the good rigidity event, $H_K \leq K^\epsilon$; on its complement, Cauchy–Schwarz gives

$$\mathbb{E} H_K^p \mathbf{1}_{H_K > K^\epsilon} \leq \|H_K\|_{2p}^p \mathbb{P}(H_K > K^\epsilon)^{1/2} \leq C K^{p-D/2}.$$

The failure exponent D can be chosen arbitrarily large. This proves the stated moment bound, which will later be used for conditional means, not only for typical boundary locations.

Finally, apply (B.10) on dyadic scales with a fixed $\epsilon < 1$ and summable failure probabilities. Almost surely, $x_j = \pi j + O(|j|^\epsilon)$ for all sufficiently large $|j|$. The paired sum in (B.11) therefore converges locally, and its difference from the equally spaced sum tends to zero at imaginary infinity by domination with $C|j|^{\epsilon-2}$. The equally spaced paired sum is

$$\lim_{M \rightarrow \infty} \sum_{j=-M}^M \frac{1}{\pi j - w} = -\cot w,$$

which tends to i at $w = iY$, because $-\cot(iY) = i \coth Y$. Moreover, for large $|j|$,

$$\left| \frac{1}{x_j - iY} - \frac{1}{\pi j - iY} \right| \leq C \frac{|j|^\epsilon}{j^2 + Y^2} \leq C |j|^{\epsilon-2}.$$

The majorant is summable when $\epsilon < 1$, and each fixed term tends to zero as $Y \rightarrow \infty$. Dominated convergence proves the normalization of the paired sum. The paired sum and s have the same representing measure and zero linear coefficient. Their difference is therefore a real constant, which vanishes by the normalization at infinity. $\square$

B.3. Conditional Gibbs measures

B.3.1. Coordinate integration by parts. Let $\mathbf{P}$ denote the law of the simple rooted array $\mathbf{x} = (x_j)_{j \in \mathbb{Z}}$ constructed above. Its transform s is determined by the principal-value formula (B.11). Retain $\zeta_j = \xi_{j1}\xi_{j2}$ and write the regular scalar part as

$$h_j = \lim_{w \rightarrow x_j} \left(s(w) - \frac{1}{x_j - w} \right).$$

The expectation in the integration-by-parts identities below is under this rooted state and, when marks occur, their conditional Gaussian law. Since s is determined by $\mathbf{x}$, conditioning on $\mathbf{x}$ already fixes the entire scalar state. Write

$$\psi(t) = \frac{t}{1+t^2}, \quad a = \operatorname{Re} s(i), \quad s(w) = a + \sum_{\ell \in \mathbb{Z}} \left(\frac{1}{x_\ell - w} - \psi(x_\ell) \right).$$

The series converges locally off the poles, and $\sum_{\ell} (1+x_\ell^2)^{-1} = \operatorname{Im} s(i) < \infty$. The moving-frame pole-coefficient calculation gives

$$\mathcal{V}_0 x_\ell = -2\kappa(\zeta_\ell - \zeta_0), \quad \mathcal{V}_0 a = -2\kappa \sum_{\ell \neq 0} (\zeta_\ell - \zeta_0) \psi'(x_\ell), \quad \mathbb{E}(\mathcal{V}_0 \zeta_j \mid \mathbf{x}) = 2\kappa h_j.$$

Taking real parts of $\mathcal{V}_0 s$ at i gives the middle identity; $|\psi'(t)| \leq (1+t^2)^{-1}$ makes its series integrable.

For $j \neq 0$, define $D_j = \partial_{x_j} + \psi'(x_j) \partial_a$. Gaussian contraction in the invariant derivatives of $\zeta_j F$ and $\zeta_0 F$ gives

$$\mathbb{E} D_j F = \mathbb{E} h_j F, \quad \mathbb{E} \sum_{\ell \neq 0} D_\ell F = -\mathbb{E} h_0 F. \quad (\text{B.13})$$

We use smooth cylinder tests compactly supported in a and finitely many positions, with neighboring gaps at least δ at each differentiated pole weight. For $0 < \eta < \delta/2$,

$$|h_j - \operatorname{Re} s(x_j + i\eta)| \leq \frac{\eta}{\delta} \operatorname{Im} s(x_j + i\eta).$$

With $\eta = \delta/4$, the fixed-tag moments bound the drift. For the second identity, we separate the root; its scalar-derivative coefficient $\sum_{\ell \neq 0} \psi'(x_\ell)$ converges absolutely. All other derivatives involve Gaussian mark products.

Here is the Gaussian contraction yielding (B.13). For a scalar cylinder test F , the chain rule gives

$$\mathcal{V}_0 F = -2\kappa \sum_{\ell \neq 0} (\zeta_\ell - \zeta_0) D_\ell F.$$

The sum is finite except for the absolutely convergent coefficient of $\partial_a F$. The conditional Gaussian law gives $\mathbb{E}[\zeta_j \mid \mathbf{x}] = 0$ and $\mathbb{E}[\zeta_j \zeta_\ell \mid \mathbf{x}] = \mathbf{1}_{j=\ell}$. Thus, for $j \neq 0$,

$$0 = \mathbb{E} \mathcal{V}_0(\zeta_j F) = 2\kappa \mathbb{E} h_j F - 2\kappa \mathbb{E} D_j F.$$

For $j = 0$, the contraction $\mathbb{E}[\zeta_0(\zeta_\ell - \zeta_0) \mid \mathbf{x}] = -1$ for $\ell \neq 0$ changes the sign and gives the second identity. To check the remaining mean, H_j is independent of the mark ξ_j conditionally on the scalar state, and $\mathbb{E}[H_j \mid \mathbf{x}] = h_j I_2$. Taking the $(1, 2)$ entry of the pole-weight equation and using $\mathbb{E}[R_j \mid \mathbf{x}] = I_2$ gives $\mathbb{E}(\mathcal{V}_0 \zeta_j \mid \mathbf{x}) = 2\kappa h_j$. The commutator term has zero conditional expectation.

The correction in D_j is not optional. Moving a single particle while holding the regular background fixed changes its compensator $\psi(x_j)$, so the scalar coordinate a must change by $\psi'(x_j)$ times that displacement. For a finite $I \subset \mathbb{Z} \setminus \{0\}$ and $j \in I$,

$$D_j \left(a - \sum_{k \in I} \psi(x_k) \right) = \psi'(x_j) - \psi'(x_j) = 0.$$

Every outside position is also annihilated by D_j . Thus, if we use $a - \sum_{k \in I} \psi(x_k)$ and the outside positions as the fixed coordinates, D_j is simply the ordinary partial derivative in the inside position x_j . Smooth exterior-cylinder factors may depend on this corrected scalar. This is the coordinate change used to disintegrate the identity below.

B.3.2. Conditional law of a finite block. To free the root, translate by $\tau \in \mathbb{R}$: set $z_j = x_j + \tau$, $s^\tau(w) = s(w - \tau)$, and

$$a^\tau = a + \sum_{\ell} (\psi(x_\ell + \tau) - \psi(x_\ell)). \quad (\text{B.14})$$

This sum converges absolutely. For $|\tau| \leq R$,

$$\sum_{\ell} |\psi'(x_\ell + \tau)| \leq C_R \sum_{\ell} (1 + x_\ell^2)^{-1} = C_R \operatorname{Im} s(i).$$

This bound and Gaussian moments dominate the truncated flow derivatives, extending the identities to bounded smooth tests of (B.14).

Integrate the translated law against $d\tau$ to obtain the sigma-finite measure $\mathbf{Q}$. It is not a probability measure: since $z_0 = \tau$, its mass on $\{|z_0| \leq R\}$ is $2R$. The normalized probability measure is introduced below. The derivative $\tilde{D}_j = \partial_{z_j} + \psi'(z_j) \partial_{a^\tau}$ is the pushforward of D_j for $j \neq 0$, and of $\partial_\tau - \sum_{\ell \neq 0} D_\ell$ for $j = 0$. Lebesgue integration and (B.13) give, for every label,

$$\int \tilde{D}_j F d\mathbf{Q} = \int h_j F d\mathbf{Q}. \quad (\text{B.15})$$

The regular parts are translation invariant, and we retain their collision cutoffs.

Fix a finite consecutive block I , which may contain zero. Choose $J \notin I$ and a nonnegative smooth compactly supported φ with integral one. Then

$$d\mathbf{Q}_\varphi = \varphi(z_J) d\mathbf{Q}, \quad \int_{\mathbb{R}} \varphi(x_J + \tau) d\tau = 1. \quad (\text{B.16})$$

This probability law preserves every relative observable.

More precisely, for any bounded measurable function F of the relative array,

$$\int F((z_j - z_0)_j) d\mathbf{Q}_\varphi = \mathbb{E}^{\mathbf{P}} \left[F((x_j)_j) \int_{\mathbb{R}} \varphi(x_J + \tau) d\tau \right] = \mathbb{E}^{\mathbf{P}} F((x_j)_j).$$

The infinite translation measure $\mathbf{Q}$ has thus been normalized without biasing any gap statistic. It is important that $J \notin I$: using a weight attached to an inside particle would produce an extra derivative of that weight in the conditional integration-by-parts formula. Since the normalizing label lies outside the block, inside derivatives annihilate the weight. Removing a translation cutoff with derivative $O(R^{-1})$ costs $O(R^{-1})$ by (B.16), so (B.15) holds under $\mathbf{Q}_\varphi$ for inside labels. Translation invariance preserves the localized drift bounds.

Lemma B.7 (Conditional Gibbs laws from integration by parts). *Under $\mathbf{Q}_\varphi$, condition on the outside positions $\mathbf{z} = (z_\ell)_{\ell \notin I}$. The inside variables are $\mathbf{t} = (t_j)_{j \in I}$, and $\mathcal{D}_I(\mathbf{z})$ is the chamber*

$$z_{\min I-1} < t_{\min I} < \cdots < t_{\max I} < z_{\max I+1}.$$

There is an exterior-measurable real scalar b such that the conditional law of $(z_j)_{j \in I}$ is

$$\frac{\mathbf{1}_{\mathcal{D}_I(\mathbf{z})}(\mathbf{t})}{Z_I(\mathbf{z}, b)} \prod_{j < k, j, k \in I} |t_j - t_k| \exp \left(\sum_{j \in I} \mathcal{U}_{\mathbf{z}, b}(t_j) \right) \prod_{j \in I} dt_j, \quad (\text{B.17})$$

where

$$\mathcal{U}_{\mathbf{z}, b}(t) = -bt + \sum_{\ell \notin I} \left[\log \frac{|t - z_\ell|}{\sqrt{1 + z_\ell^2}} + t\psi(z_\ell) \right].$$

The combined summands converge absolutely at infinity. All particles of I , including the former root, are ordinary variables in (B.17).

Proof. We do not assume beforehand that a conditional density exists. The integration-by-parts identities first identify the logarithmic derivative of the conditional measure in the distributional sense. An elementary argument on the connected ordered chamber then proves absolute continuity and identifies the density. Finally we show that the auxiliary scalar being conditioned on is already determined by the exterior.

1. *Disintegrate with the correct scalar fixed.* First condition on the outside positions and also on $b = a^\tau - \sum_{j \in I} \psi(z_j)$. Let ν denote this conditional probability measure on the inside variables $\mathbf{t}$; at this point we do not yet assume it has a density. The inside derivatives annihilate b and all outside coordinates. Moreover

$$h_j = \sum_{k \in I \setminus \{j\}} \frac{1}{t_k - t_j} + b + \sum_{\ell \notin I} \left(\frac{1}{z_\ell - t_j} - \psi(z_\ell) \right).$$

We localize to rational compact boxes inside the chamber, whose exterior separation is determined by its two neighbors. A monotone-class argument extends (B.15) to bounded exterior factors. Countable dense test families on these boxes give the conditional identities on one full-measure set. Conditional probabilities and identities in this proof are understood for almost every exterior and scalar b , not for every arbitrary boundary configuration.

2. *Recover the density from its logarithmic derivative.* For the candidate unnormalized density w_I in (B.17),

$$\partial_j \log w_I = \sum_{k \in I \setminus \{j\}} \frac{1}{t_j - t_k} - b + \sum_{\ell \notin I} \left(\frac{1}{t_j - z_\ell} + \psi(z_\ell) \right) = -h_j.$$

The conditional identities say $\int \partial_j f \, d\nu = \int h_j f \, d\nu$ for compactly supported smooth f in the chamber. Hence, in distributions, $\partial_j \nu = -h_j \nu = (\partial_j \log w_I) \nu$. On a compact subset of the chamber w_I is strictly positive and smooth, so the distributional product rule gives

$$\partial_j (w_I^{-1} \nu) = 0 \quad \text{for every } j \in I.$$

To see why this identifies a density, convolve the locally finite measure $w_I^{-1} \nu$ with a smooth approximate identity on a ball compactly contained in the chamber. Every derivative of the convolution is zero on a slightly smaller ball, so that convolution is constant there. Passing to the limit shows that the measure is a constant multiple of Lebesgue measure on that ball.

Overlapping balls have the same constant; connectedness of the ordered chamber therefore gives $\nu = cw_I \, d\mathbf{t}$ throughout it.

3. *Normalize the density.* For bounded t and large $|z|$, Taylor expansion gives

$$\log \frac{|t-z|}{\sqrt{1+z^2}} + t\psi(z) = -\frac{1+t^2}{2z^2} + O_R(|z|^{-3}), \quad |t| \leq R.$$

Thus the tail, and its derivatives on compact subintervals, converge uniformly because $\sum_\ell (1+z_\ell^2)^{-1} < \infty$. There are only finitely many nearby exterior factors. Those at the two boundary particles vanish as an inside particle approaches the boundary; the Vandermonde factors vanish at inside collisions. There is no positive blow-up at the chamber boundary. Thus $0 < Z_I < \infty$; simplicity gives mass one in the chamber and $c = Z_I^{-1}$.

4. *Remove the extra scalar conditioning.* For $|\tau| \leq R$ and $Y \geq 2R+1$, kernel comparison gives

$$|s(iY - \tau) - s(iY)| \leq \frac{C_R}{Y} \operatorname{Im} s(iY).$$

The height moments and (B.3) therefore imply $s^\tau(iY) \rightarrow i$ in probability uniformly for bounded τ . Under the probability law $\mathbf{Q}_\varphi$, the translation variable is finite almost surely, so its tail can be removed by taking $R \rightarrow \infty$ after $Y \rightarrow \infty$. Taking real parts yields

$$b = \lim_{Y \rightarrow \infty} \sum_{\ell \notin I} \left(\psi(z_\ell) - \frac{z_\ell}{z_\ell^2 + Y^2} \right) \quad \text{in probability.}$$

The finite inside resolvent sum vanishes. An almost surely convergent subsequence makes b measurable in the completed exterior sigma-field, so the additional conditioning can be removed. This final measurability step matters: conditioning only on the exterior would otherwise produce a mixture of densities with different linear potentials $-bt$. The normalization at imaginary infinity shows that there is a single exterior-determined b , so the conditional law is exactly the Gibbs density stated in the lemma. The former root is now one of the ordinary inside variables; there is no hidden atom constrained to stay at zero. $\square$

B.4. Identification of the gap law

The preceding argument supplies rigidity and a conditional log-gas structure. The following proposition isolates the consequence for gaps. Let Θ_1^0 be the Palm process of Θ_1 , with increasing points $(y_j)_{j \in \mathbb{Z}}$ and $y_0 = 0$, and write $G_j = y_{j+1} - y_j$ for its consecutive gaps. These gaps have mean π ; they are not divided by π .

The Palm law describes the configuration seen from a typical particle, rather than from a deterministic point in space. For a bounded interval I of positive length and a bounded measurable rooted test F , it is characterized by Campbell's identity

$$\mathbb{E} \sum_{x \in \Theta_1 \cap I} F(\Theta_1 - x) = \frac{|I|}{\pi} \mathbb{E}^0 F(\Theta_1^0).$$

Here $\mathbb{E}^0$ is expectation under the Palm probability law. Taking $F = 1$ recovers $\mathbb{E}\Theta_1(I) = |I|/\pi$; in particular, the prefactor is one for $I = [0, \pi]$. This avoids interpreting Palm conditioning as conditioning on the probability-zero event of an atom at one prescribed location.

Proposition B.8 (Gap identification from rigidity and conditional Gibbs laws). *Let $\mathbf{P}$ be the law of a strictly increasing array $(x_j)_{j \in \mathbb{Z}}$ with $x_0 = 0$. Suppose the following hold.*

- (i) For every $\epsilon, D > 0$ and $1 \leq p < \infty$, there are constants such that, for every integer $K \geq 2$,

$$\begin{aligned} \mathbb{P} \left(\max_{|j| \leq 2K} |x_j - \pi j| > K^\epsilon \right) &\leq C_{\epsilon, D} K^{-D}, \\ \left\| \max_{|j| \leq 2K} |x_j - \pi j| \right\|_{L^p(\mathbb{P})} &\leq C_{p, \epsilon} K^\epsilon. \end{aligned} \quad (\text{B.18})$$

- (ii) For every sufficiently large even K , there is a probability law $\mathbb{Q}_K$ on strictly increasing arrays $(z_j)_{j \in \mathbb{Z}}$ such that

$$\text{Law}_{\mathbb{Q}_K}((z_j - z_0)_{j \in \mathbb{Z}}) = \mathbb{P}.$$

For $I = \{-K, \dots, K\}$, conditionally on all outside coordinates, the inside coordinates have density

$$\frac{\mathbf{1}_{\{z_{-K-1} < t_{-K} < \dots < t_K < z_{K+1}\}}}{Z_I} \prod_{\substack{j < k \\ j, k \in I}} |t_j - t_k| \exp \left(- \sum_{j \in I} W_I(t_j) \right) \prod_{j \in I} dt_j, \quad (\text{B.19})$$

where $0 < Z_I < \infty$ and W_I is an exterior-measurable C^1 function on (z_{-K-1}, z_{K+1}) satisfying

$$W'_I(t) = \lim_{R \rightarrow \infty} \sum_{K < |\ell| \leq R} \frac{1}{z_\ell - t}. \quad (\text{B.20})$$

The principal value pairs opposite labels and converges locally uniformly on this interval. The auxiliary laws may depend on K .

Then, for every fixed $r \in \mathbb{Z}$ and $k \geq 1$,

$$(x_{r+1} - x_r, \dots, x_{r+k} - x_{r+k-1}) \stackrel{d}{=} (G_0, \dots, G_{k-1}).$$

The auxiliary laws in (ii) need not be the same for different K , and they need not be invariant under translations. What is required is that each preserves the same relative array law and allows all particles of the chosen block to move. This is exactly what (B.16) provides. The proposition does not assume that the original rooted process is already stationary under changing the tag.

The proof is given in Appendix C. It compares large conditional blocks with GOE blocks using the local log-gas universality theorem of Erdős and Yau [16]. Rigidity controls the effect of completing the distant exterior with Gaussian classical locations. All particles in the conditional block, including label zero, are ordinary variables in (B.19).

Proof of Theorem 1.1(i). We verify the two hypotheses of the abstract gap-identification proposition for every subsequential tagged law, and only then return to the finite- N normalization and decreasing eigenvalue order. Consider any deterministic bulk indices i_N . Pass to a subsequence with $\gamma_{i_N} \rightarrow E$ in the strict bulk and then to a jointly labelled tagged limit with law $\mathbb{P}$. Proposition B.2 and Lemma B.3 give a strictly increasing array with $x_0 = 0$. Lemma B.6 gives the first hypothesis of Proposition B.8.

For the second hypothesis, take $I = \{-K, \dots, K\}$ and use $\mathbb{Q}_K = \mathbb{Q}_\varphi$ from (B.16), with normalizing label $J = K + 2$. Its relative array has law $\mathbb{P}$. Lemma B.7 gives (B.19) with $W_I = -\mathcal{U}_{z, b}$. The translated principal-value identity (B.11) gives

$$a^\tau = \text{PV} \sum_{\ell \in \mathbb{Z}} \psi(z_\ell), \quad b = \text{PV} \sum_{\ell \notin I} \psi(z_\ell).$$

Differentiating the conditional potential therefore cancels the compensators and yields (B.20). Rigidity gives local uniform convergence of the paired sum. The proposition now identifies every fixed consecutive gap vector of the tagged limit.

Since $\rho_d(\gamma_{i_N})/\rho_d(E) \rightarrow 1$, replacing $N\sigma$ by $N\pi\rho_d(\gamma_{i_N})$ gives the theorem's normalization, without any further dilation. Take $r = -k$ and reverse the k coordinates to obtain the descending graph gaps. Finite GOE reflection maps the corresponding reversed block to right gaps at label $M/2 + 1$, whose limit is $(G_0, \dots, G_{k-1})$ by Lemma C.2. The bulk-label comparison in [16, Theorem 2.3] gives the same limit for the GOE gaps at the original indices, with the factor $N\pi\rho_{sc}$ used in the theorem. The fixed factor π is absorbed by composing the test function in the cited comparison with this dilation. Every fixed neighboring gap vector is tight by Proposition B.2; thus smooth compactly supported tests identify the weak limit and convergence extends to $F \in C_b(\mathbb{R}^k)$. If uniformity over the bulk indices failed, there would be $\varepsilon_0 > 0$ and deterministic indices i_N along a subsequence for which the displayed difference in the theorem is at least ε_0 . Compactness of the bulk quantile interval gives a further subsequence with $\gamma_{i_N} \rightarrow E$. The tagged construction and identification above force both the graph and GOE expectations along that subsequence to have the same limit, a contradiction. Thus the convergence is uniform over the full bulk index range, not merely pointwise for a fixed proportion i/N . $\square$

C Gap identification by Gaussian completion

We prove Proposition B.8, which is invoked at the end of Appendix B. Gaussian completion, exterior rigidity, and conditional mean bounds reduce its abstract gap law to the GOE reference law. The fixed-energy point-process theorem and the correlation upgrade do not use this comparison. Throughout this appendix, $\mathbf{P}$ and $\mathbf{Q}_K$ satisfy the two hypotheses of that proposition. In the conditional comparison, probabilities of exterior events and unsubscripted expectations are under $\mathbf{Q}_K$. Superscripts μ_∞ , μ_M and GOE_M specify, respectively, the two fixed-exterior conditional laws and the unconditional GOE law. The GOE identification proof states its own expectation convention. Only in this appendix, GOE eigenvalues are ordered increasingly, $\mu_1 \leq \dots \leq \mu_M$, to agree with the labelled arrays.

Gaussian completion here means replacing only the *distant exterior particle locations* of a limiting conditional log gas by suitable GOE classical locations. It does not mean adding a Gaussian matrix perturbation to the graph. The nearby boundary particles and the whole inside chamber are kept unchanged. A force estimate then compares the two conditional densities, after which the cited local log-gas theorem compares their gaps. The use of that theorem is isolated in this appendix.

C.1. Exterior rigidity and conditional means

Set $\delta_* = 1/32$, choose $\xi > 0$ sufficiently small for [16, Proposition 5.3], and put

$$\xi_* = \frac{\xi^2}{16}, \quad \nu = \frac{\xi^2}{64}, \quad 0 < \epsilon < \min(\xi^2/1000, 1/5), \quad M = K^{16}, \quad L = K^5.$$

Here $K \rightarrow \infty$ through even integers.

These powers provide comfortable separation rather than optimized rates. The inside block has order K particles, the original exterior is kept out to order $L = K^5$, and the auxiliary GOE has size $M = K^{16}$. Thus $K \ll L \ll M$, while $K^3/L = K^{-2}$ will control the full conditional Hamiltonian error. The small exponents are chosen before K tends to infinity. Work under the auxiliary law $\mathbf{Q}_K$ in Proposition B.8, put $I = \{-K, \dots, K\}$, and condition

on the exterior sigma-field $\mathcal{E}$. Define the exterior-measurable midpoint and interval length by

$$m = \frac{z_{-K-1} + z_{K+1}}{2}, \quad \ell = z_{K+1} - z_{-K-1}, \quad J = (-\ell/2, \ell/2).$$

Use the single centered array $u_j = z_j - m$ for every label j . After conditioning on $\mathcal{E}$, the coordinates with $j \notin I$ are fixed. Let μ_∞ be the conditional law of the inside array $(u_j)_{j \in I}$, and put $\alpha_j = j\ell/(2K+2)$ for its equally spaced reference locations. The rigidity assumption (B.18) and a dyadic union give, outside an event of probability $O(K^{-D})$,

$$\ell = 2\pi(K+1) + O(K^\epsilon), \quad |u_j - \pi j| \leq C(K+|j|)^\epsilon \quad (|j| > K). \quad (\text{C.1})$$

Indeed centering changes each rooted position by $O(K^\epsilon)$. Moreover $H = \max_{j \in I} |u_j - \alpha_j|$ is translation invariant and $\|H\|_p \leq C_{p,\epsilon} K^\epsilon$. Conditional Jensen's inequality and Markov's inequality give

$$\mathbb{P} \left(\max_{j \in I} |\mathbb{E}^{\mu_\infty} u_j - \alpha_j| > K^{\xi_*} \right) \leq C_{p,\epsilon} K^{-p(\xi_* - \epsilon)}.$$

Thus, for every D , an exterior event of probability $1 - O(K^{-D})$ satisfies (C.1) and

$$\max_{j \in I} |\mathbb{E}^{\mu_\infty} u_j - \alpha_j| \leq K^{\xi_*}. \quad (\text{C.2})$$

To check that the event of good conditional means is an exterior event, observe that $\max_j |\mathbb{E}[u_j | \mathcal{E}] - \alpha_j| \leq \mathbb{E}[H | \mathcal{E}]$. Conditional Jensen gives $\mathbb{E}(\mathbb{E}[H | \mathcal{E}]^p) \leq \mathbb{E}H^p$. This proves the displayed probability estimate without assuming independence between the inside and outside coordinates. Since $\epsilon < \xi_*$, taking p sufficiently large gives any prescribed power K^{-D} . The L^p part of the rigidity hypothesis is used here; a high-probability position estimate alone would not directly control these conditional means.

Fix an integer r and $k \geq 1$. The two gap vectors are

$$\begin{aligned} \mathbf{g}_r &= (u_{r+1} - u_r, \dots, u_{r+k} - u_{r+k-1}), \\ \mathbf{g}_{M,r} &= M(\mu_{M/2+r+1} - \mu_{M/2+r}, \dots, \mu_{M/2+r+k} - \mu_{M/2+r+k-1}). \end{aligned}$$

The first is unaffected by centering at m . For fixed r, k , all its labels lie in the inside block once K is large enough. The GOE factor is M , because $\pi\rho_{\text{sc}}(0) = 1$; both gap vectors therefore have the mean-spacing- π convention.

C.2. Comparison by Gaussian completion

Lemma C.1 (Gaussian completion and conditional comparison). *For $O \in C_c^\infty(\mathbb{R}^k)$, uniformly over exteriors satisfying (C.1) and (C.2),*

$$\left| \mathbb{E}^{\mu_\infty} O(\mathbf{g}_r) - \mathbb{E}^{\text{GOE}_M} O(\mathbf{g}_{M,r}) \right| \leq C_O (K^{-2} + K^{-c_*} + e^{-cM^{c_2}})$$

for some positive c_*, c, c_2 .

Proof. 1. Keep the near boundary and complete only the far boundary. The comparison measure must have the same ordered chamber as μ_∞ . Keeping all exterior locations with $K < |j| \leq L$ ensures this. The small outward shift in the farther Gaussian locations prevents an ordering defect where the two descriptions meet.

Use GOE coordinates $u = M\lambda$ and relative classical positions

$$\mathcal{J}_M = \{-M/2 + 1, \dots, M/2\}, \quad \int_{-2}^{\Gamma_j/M} \rho_{\text{sc}}(v) dv = \frac{1}{2} + \frac{j}{M}.$$

Then $\Gamma_{-j} = -\Gamma_j$ for $1 \leq j < M/2$, the unpaired position is $\Gamma_{M/2} = 2M$, and

$$|\Gamma_j| \geq \pi|j|, \quad \Gamma_j = \pi j + O(j^3/M^2) \quad (|j| \leq 2L).$$

With $\Delta = L^{2\epsilon}$, complete the exterior by

$$y_j = \begin{cases} u_j, & K < |j| \leq L, \\ \Gamma_j + \text{sgn}(j)\Delta, & j \in \mathcal{J}_M, |j| > L. \end{cases}$$

The outward shift dominates both junction errors. Hence the array is ordered, its configuration interval is J , and

$$\max_{j \in \mathcal{J}_M \setminus I} |y_j - \Gamma_j| \leq CK^{10\epsilon}. \quad (\text{C.3})$$

Its conditional GOE law μ_M has one-particle Hamiltonian

$$W_M(v) = \frac{v^2}{4M} - \sum_{j \in \mathcal{J}_M \setminus I} \log |v - y_j|.$$

2. Compare the forces and then the whole Hamiltonians. By the conditional-force assumption (B.20), the corresponding force for μ_∞ is $W'_\infty(v) = \text{PV} \sum_{|j| > K} (u_j - v)^{-1}$. The forces agree for $K < |j| \leq L$. Their remaining terms are

$$R_\infty(v) = \text{PV} \sum_{|j| > L} \frac{1}{u_j - v}, \quad R_M(v) = \frac{v}{2M} + \sum_{j \in \mathcal{J}_M, |j| > L} \frac{1}{y_j - v}.$$

Opposite-label pairing and the inverse-square tail give

$$|R_\infty(0)| \leq C \sum_{j > L} j^{\epsilon-2} \leq CL^{\epsilon-1}, \quad R_M(0) = \frac{1}{2M + \Delta}, \quad \sup_J (|R'_\infty| + |R'_M|) \leq C/L.$$

Since $L^\epsilon \leq K$, the force difference is at most CK/L . Choose additive constants so that $W_M(0) = W_\infty(0)$; this does not change either conditional probability law. The force bound and $|v| \leq CK$ imply

$$\sup_{v \in J} |W_M(v) - W_\infty(v)| \leq CK^2/L.$$

There are $2K + 1$ inside particles. Since their Vandermonde factors are identical, the *full* Hamiltonians therefore differ by at most $CK^3/L = CK^{-2}$ uniformly throughout the chamber. If two unnormalized densities differ by a multiplicative factor between $e^{-\delta}$ and e^δ , their normalizing constants satisfy the same bound, and the ratio of normalized densities lies between $e^{-2\delta}$ and $e^{2\delta}$. Hence their total variation distance is at most $e^{2\delta} - 1$. Also $|u_j| \leq CK$ on this common chamber, so a coordinate mean changes by at most CK times the total variation distance. These observations give

$$\|\mu_\infty - \mu_M\|_{\text{TV}} \leq CK^{-2}, \quad \max_{j \in I} |\mathbb{E}^{\mu_\infty} u_j - \mathbb{E}^{\mu_M} u_j| \leq CK^{-1}. \quad (\text{C.4})$$

In particular, (C.2) transfers to μ_M with an adjusted constant.

3. Verify the good-boundary and conditional-mean hypotheses. We verify the hypotheses of [16] with size M and block center $M/2$. The relevant powers are

$$M^{\delta_*} \leq K \leq M^{1/4}, \quad M^\nu = K^{\xi^2/4}, \quad M^{\xi_*\delta_*/2} = K^{\xi^2/64}.$$

Both tolerances dominate the bound in (C.3). After division by M , this gives all three bounds in [16, (4.7)], with exponent $\nu = \xi^2\delta_*/2$ or $\xi_*\delta_*/2$ and a fixed bulk margin; the

nonbulk bounds are weaker. Equation (C.1) gives [16, (4.9)], absorbing its $O(1)$ particle-count difference in K^ξ .

Apply [16, Theorem 4.2] with exponent ξ_* . The stronger boundary condition and transferred means give Gaussian rigidity on scale K^{ξ_*} . In the units of [16], the particle coordinates are u_j/M ; thus a tolerance $M^{-1}K^{\xi_*}$ there is K^{ξ_*} here. The target threshold is $M^\nu = K^{\xi^2/4}$, whereas the conditional fluctuation scale is $K^{\xi_*} = K^{\xi^2/16}$. Their ratio is $K^{3\xi^2/16}$, and squaring it in the Gaussian tail gives $K^{3\xi^2/8}$. Since $\max_{j \in I} |\alpha_j - \Gamma_j| \leq CK^\epsilon$,

$$\mathbb{P}^{\mu_M}(|u_j - \Gamma_j| > M^\nu) \leq C \exp(-cK^{3\xi^2/8}), \quad j \in I. \quad (\text{C.5})$$

Decrease the exponent c_2 in the global Gaussian estimates [16, (5.1)–(5.4)] so that $16c_2 < 3\xi^2/8$. Then (C.5) is at most $\exp(-c_1 M^{c_2}/2)$ for large K . This is exactly the conditional bound defining R^* in [16, (5.8)]; the weaker global constants remain valid in its Lemma 5.2 and Proposition 5.3.

4. Compare with a typical Gaussian boundary. Apply that proposition with both ensembles Gaussian and integrate the second exterior over the R^* event of its Lemma 5.2, whose complement has probability $O(e^{-cM^{c_2}})$. For the fixed anchor r , take the two block centers to be $M/2$ and $M/2 - r$, respectively; the comparison gap then has label $M/2$, as required in the proposition. The cited theorem uses anchored differences, which for an increasing array are $d_j = u_r - u_{r+j}$, $1 \leq j \leq k$. Our consecutive positive gaps are recovered by $u_{r+1} - u_r = -d_1$ and $u_{r+j} - u_{r+j-1} = d_{j-1} - d_j$ for $j \geq 2$. This invertible linear change of variables takes a smooth compactly supported gap test to a smooth compactly supported anchored test, with controlled derivative norm. Where the cited comparison uses mean-spacing-one gaps, apply it to $\mathbf{v} \mapsto O(\pi \mathbf{v})$; this fixed dilation preserves smoothness, compact support, and the error bounds up to constants depending on O . The factor π times the semicircle density at label $M/2 + r$ is $1 + O(M^{-2})$, and Theorem 2.3 of [16] moves the Gaussian comparison from label $M/2$ to $M/2 + r$. These errors and (C.4) prove the lemma. $\square$

C.3. Identification of the GOE gap limit

Lemma C.2 (Identification of the GOE reference gaps). *For each fixed r, k and $O \in C_c^\infty(\mathbb{R}^k)$, $\mathbb{E}^{\text{GOE}_M} O(\mathbf{g}_{M,r}) \rightarrow \mathbb{E} O(G_0, \dots, G_{k-1})$ as $M \rightarrow \infty$ through even integers.*

Proof. A fixed-label gap law is not literally a fixed-energy point-process law. We first use the GOE gap comparison to replace the fixed label by a short label average. Rigidity turns that average into a spatial average over actual particles. Campbell's formula then identifies it with a Palm expectation, to which the GOE kernel asymptotics apply.

In this proof, $\mathbb{E}$ averages the GOE eigenvalues, except for an expectation explicitly involving the limiting Palm gaps $(G_0, \dots, G_{k-1})$. The notation $\mathbb{E}_{M,t}^0$ below denotes the finite-GOE Palm expectation, not the graph or exterior law. Using the increasing GOE eigenvalues μ_j fixed above, put $\Xi_M = \sum_j \delta_{M\mu_j}$, $n_M = \lfloor M^{1/4} \rfloor$, and $\mathcal{A}_M = \{j \in \mathbb{Z} : |j| \leq n_M\}$. Theorem 2.3 of [16] gives

$$\mathbb{E} O(\mathbf{g}_{M,r}) - \frac{1}{|\mathcal{A}_M|} \sum_{j \in \mathcal{A}_M} \mathbb{E} O(\mathbf{g}_{M,j}) \rightarrow 0.$$

On these labels, π times the local semicircle density is $1 + O(M^{-3/2})$, a negligible change for O . Gaussian rigidity replaces this label set by the points of Ξ_M in $[-\pi n_M, \pi n_M]$, with only $O(M^\eta)$ endpoint errors for any $0 < \eta < 1/4$, outside an exponentially rare event. Let F_O apply O to the first k right gaps of a rooted configuration, with value zero if fewer neighbors

exist. Write $\rho_{M,q}$ for the correlation densities of Ξ_M and $\mathbb{E}_{M,t}^0$ for its Palm expectation at t , translated to zero. The label average consequently differs by $o(1)$ from

$$\frac{1}{|\mathcal{A}_M|} \mathbb{E} \sum_{u \in \Xi_M \cap [-\pi n_M, \pi n_M]} F_O(\Xi_M - u) = \frac{1}{|\mathcal{A}_M|} \int_{-\pi n_M}^{\pi n_M} \rho_{M,1}(t) \mathbb{E}_{M,t}^0 F_O dt.$$

Under this Palm law the nonroot correlation densities are

$$\frac{\rho_{M,q+1}(t, t + v_1, \dots, t + v_q)}{\rho_{M,1}(t)}.$$

Apply [11, arXiv v3, Theorem 1.1 and Corollary 1.2] with $V(x) = x^2/2$, so its GOE weight is $e^{-x^2/4}$ and $x = \sqrt{M}\mu$. The centers $t/\sqrt{M} = O(M^{-1/4})$ remain in a compact set. The resulting kernel convergence, expressed in the coordinates $M\mu$, is therefore uniform for $|t| \leq \pi n_M$ and bounded $\mathbf{v}$, and $\rho_{M,1}(t) = 1/\pi + o(1)$. The bounded matrix entries, including the sign entry, give the Pfaffian bound $C^q(2q)^{q/2}$ by Hadamard's inequality. After fixing the root and dividing by $q!$, these bounds are summable in every bounded interval. For example, for a nonnegative continuous f of compact support, the reduced Palm Laplace functional is

$$\sum_{q=0}^{\infty} \frac{(-1)^q}{q!} \int \prod_{j=1}^q (1 - e^{-f(v_j)}) \frac{\rho_{M,q+1}(t, t + v_1, \dots, t + v_q)}{\rho_{M,1}(t)} d\mathbf{v}.$$

For finite M the series terminates; we can regard the remaining terms as zero. The Pfaffian bounds, divided by $q!$ and multiplied by the fixed support volume to the power q , give a summable majorant independent of M, t . Dominated convergence proves uniform convergence of these local Laplace functionals. This identifies uniform local convergence of the reduced Palm laws; adding the fixed root gives the Palm laws Θ_1^0 . If $\text{supp } O \subset [-R_O, R_O]^k$, then F_O depends only on $[0, kR_O + 1]$ and is almost surely continuous under the simple Palm law. Consequently $\mathbb{E}_{M,t}^0 F_O \rightarrow \mathbb{E}O(G_0, \dots, G_{k-1})$ uniformly. The prefactor in the Campbell identity satisfies

$$\frac{1}{|\mathcal{A}_M|} \int_{-\pi n_M}^{\pi n_M} \rho_{M,1}(t) dt = \frac{2\pi n_M}{|\mathcal{A}_M|} \left(\frac{1}{\pi} + o(1) \right) \rightarrow 1.$$

Thus the spatial-window length and the intensity cancel, giving the claimed Palm expectation. For the endpoint replacement, divide the number $O(M^\eta)$ of possible mismatched labels by $|\mathcal{A}_M| \asymp M^{1/4}$; it vanishes for $\eta < 1/4$. The boundedness of O controls the rare exceptional event as well. For locality of F_O , a nonzero value requires all first k gaps to be at most R_O , so the required neighbors lie within $[0, kR_O]$. If a required next gap is larger, its absence is detected within $[0, kR_O + 1]$ and the test is zero. Under a simple Palm process with no particle at the fixed outer boundary, this is a bounded, almost surely continuous local observable. Thus local Palm convergence suffices; no assertion about an unbounded search for neighbors is needed. $\square$

Proof of Proposition B.8. At this stage the microscopic graph limit has already been taken. The only large parameters are the conditional block size K and the auxiliary GOE size $M = K^{16}$. In particular, none of the qualitative eigenvector limits is being used uniformly in K . For $O \in C_c^\infty(\mathbb{R}^k)$, integrate Lemma C.1 over the exterior. Equations (C.1) and (C.2) hold outside an event of probability $O(K^{-D})$, so

$$\left| \mathbb{E}^{\mathbf{Q}_K} O(\mathbf{g}_r) - \mathbb{E}^{\text{GOE}_M} O(\mathbf{g}_{M,r}) \right| \leq C_O (K^{-2} + K^{-D} + K^{-c_*} + e^{-cM^{c_2}}).$$

Relative-gap expectations under $\mathbf{Q}_K$ equal those under $\mathbf{P}$ by hypothesis, even though $\mathbf{Q}_K$ depends on K . Let $K \rightarrow \infty$ through even integers, with $M = K^{16}$, and apply Lemma C.2.

Equality for smooth compactly supported tests determines the claimed distribution of each finite gap vector. Indeed such tests determine finite Borel measures on $\mathbb{R}^k$ by approximation of continuous compactly supported functions. Both sides are probability laws. The left side is independent of K because each $\mathbf{Q}_K$ preserves the same relative law $\mathbf{P}$, so passing to larger conditional blocks does not change the object being identified. $\square$

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